

PART 1

1. A dynamometer shows that the average pull required for a certain machine is 20KN. If the tractor travels at 5kph, what is the power developed?

Given: $F = 20\text{KN}$, $S = 5\text{kph}$

Required: power developed

Solution:

$$P = \frac{(20,000 \text{ N} \times 5000 \frac{\text{m}}{\text{s}})}{3600\text{sec}} = 27.78 \text{ kW}$$

2. A dynamometer test shows that an engine delivers an average output of 50hp at a speed of 1500rpm. What torque is exerted?

Given: $P = 50\text{hp}$, $N = 1500\text{rpm}$

Required: torque

Solution:

$$P = 2\pi TN; T = \frac{P}{2\pi N} = \frac{(50\text{hp}) (76.2\text{kg}\cdot\text{m/s}) (9.8\text{N})}{2\pi(1500/60\text{min})} = 237.7 \text{ N}\cdot\text{m}$$

3. What is the piston displacement in cubic inches of a 4.5 in by 6 in cylinder engine?

Given: $D = 4.5 \text{ in}$, $H = 6 \text{ in}$

Solution:

$$PD = \frac{\pi D^2 L}{4} = \frac{\pi(4.5\text{in})^2(6\text{in})}{4} = 95.42 \text{ in}^3$$

4. An engine with 60mm x 60mm engine runs 1500rpm. What is the maximum piston speed inside the cylinder?

Given: Engine dimension: 60mm x 60mm

Solution:

$$V = \pi DN = \frac{\pi(0.06\text{m}) (1500)}{60 \text{ sec}} = 4.71 \text{ m/s}$$

5. A tractor pulls a 2500kg of loads at a speed of 5kph on a flat ground. If the coefficient of rolling resistance is 0.25, what is the drawbar power that the tractor generates?

Given: $F = 2500\text{kg}$, $RS = 0.25$, $V = 5\text{kph}$

Solution:

$$DHP = FV = \frac{(2500\text{kg}) (5000\text{m})}{3600 \text{ sec}} = \frac{3472.20\text{kg}\cdot\frac{\text{m}}{\text{s}}}{76.2\text{kg}\cdot\frac{\text{m}}{\text{s}}} = 45.57\text{hp} \times 0.25 = 11.40\text{hp}$$

6. A gasoline engine was tested for continuous 5hrs of operation. It consumed a total of 9 liters of fuels and developed a constant power of 3 kW. If the fuel contains 34,636MJ/m³, what is the approximate thermal efficiency?

Solution:

$$\eta_{th} = \frac{W}{Q} = \frac{3\text{kW}}{17.318\text{kW}} = 17.32\%$$

$$Q = \frac{(9 \text{ liters}/5\text{hrs}) (34636\text{MJ}/\text{m}^3)}{(1000 \text{ lit}) (3600 \text{ sec})} = 17.318 \text{ KW}$$

7. A pony brake test of an engine shows that the crankshaft rotates at 1530 rpm. The brake arm length is 0.75m and the net load on the scales is 20kgf. What if the power developed?

Given: $N = 1530 \text{ rpm}$, $a = 0.75 \text{ m}$, $F = 20\text{kgf}$

Solution:

$$\begin{aligned}
 P &= 2\pi FNa = \frac{2\pi(20\text{kgf})(1530\text{rpm})(0.75\text{m})}{60\text{sec}} = \left(2403.3\text{kgf} \cdot \frac{\text{m}}{\text{s}}\right) \left(\frac{9.8\text{N}}{\text{kgf}}\right) \\
 &= 23552.52 \text{ N} \cdot \frac{\text{m}}{\text{s}} \text{ or } 23552.52 \frac{\text{J}}{\text{s}} \text{ or } 23.55 \text{ kW} \\
 &= 23.55 \text{ kW} \times 1.34\text{hp} \\
 &= 31.56\text{hp}
 \end{aligned}$$

8. Determine that brake thermal efficiency of an engine if its mechanical efficiency is 87% and the indicated thermal efficiency is 45%

Given: ME = 87%, ITE = 45%

Solution:

$$\text{BTE} = 0.87 \times 0.45 \times 100 = 39\%$$

9. What is the rim speed of a 9 in diameter flywheel turning at 1800 rpm?

Given: D = 9 in, N = 1800rpm

Solution:

$$V = \pi DN = \pi \left(\frac{9\text{in}}{12\text{in}}\right) (1800\text{rpm}) = 4241.1\text{ft/min}$$

10. What is the speed ratio of the transmission unit to give the tractor a speed of 3.5kph? The engine speed is 2200 rpm and a rear wheel diameter of 55 cm.

Given: S = 3.5kph, N = 2200 rpm, D = 55cm

Solution:

$$\begin{aligned}
 V &= \pi DN = \pi(0.55\text{m})(2200\text{rpm}) \left(\frac{60\text{sec}}{1000}\right) = 228.08\text{kph} \\
 \text{SR} &= \frac{228.08\text{kph}}{3.5\text{kph}} = 65.2:1
 \end{aligned}$$

11. What is the efficiency of the transmission if the input speed is 2000 rpm, input torque of 220lb.ft and an output of torque at the final drive is 19lb-ft at 17,000rpm?

Solution:

$$\text{Efficiency} = \frac{(17,000\text{rpm})(19\text{lb} - \text{ft})}{(2000\text{rpm})(220\text{lb} - \text{ft})} = 73.4\%$$

12. How much power is actually received by a pump run by a 7.5hp engine running at 2800 rpm? If a 3% slip is allowed between belt and pulley.

Solution:

$$P = 7.5\text{hp} - (7.5\text{hp} \times 0.03) = 7.3\text{hp}$$

13. Determine the brake thermal efficiency of a 5hp gasoline engine consuming 2 liters of fuel in an hour?

Solution:

$$\text{BTE} = \frac{5 \text{ hp}}{\left(\frac{2\text{lit}}{\text{hr}}\right) \left(\frac{34330\text{MJ}}{\text{m}^3}\right) \left(\frac{1\text{m}}{31000\text{lit}}\right) \left(\frac{1\text{hr}}{3600\text{sec}}\right)} = \frac{5\text{hp}}{26\text{hp}} = 19\%$$

14. A gasoline engine has a thermal efficiency of 23%. Test shows it uses 1.2 liter of premium gasoline fuel per hour. What is the approximate power it generates?

Solution:

$$P = \frac{\left(1.2 \frac{\text{liter}}{\text{hr}}\right) \left(\frac{34330\text{MJ}}{\text{m}^3}\right)}{(1000\text{lit}) (3600\text{sec})} = 11443.30 \frac{\text{J}}{\text{s}} \times 0.23 = 2.632 \text{ kW}$$

15. If clearance volume of 3.5 x 4 in gasoline engine is 7 cubic inches, what is the compression ratio?

Solution:

$$CR = \frac{TV}{CV}; TV = PD + CV = \frac{\pi D^2 L}{4} + 7 \text{ in}^3 = \frac{\pi (3.5)^2 (4 \text{ in})}{4} + 7 \text{ in}^3 = 45.48 \text{ in}^3$$

$$CR = \frac{TV}{CV} = \frac{45.48 \text{ in}^3}{7 \text{ in}^3} = 6.5:1$$

16. A diesel engine produces 100kW of brake power while it is running at 2000rpm rated speed and consuming 37kg/hr. of fuel. Calculate the specific fuel consumption.

Solution:

$$SFC = \frac{\left(\frac{37 \text{ kg}}{\text{hr}}\right)}{(100 \text{ kW})} = 0.37 \text{ kg/kW-hr}$$

17. An engine is tested on a dynamometer and produced 450 N-m of torque at a speed of 2400 rpm. Calculate the power output.

Solution:

$$P = 2\pi NT = \frac{2\pi (2400 \text{ rpm}) (450 \text{ N-m})}{(1000) (60 \text{ sec})} = 113 \text{ kW}$$

18. The firing interval and firing order of a 4 stroke, 3 cylinder in-line engine?

Solution:

$$FI = \frac{(360) (2)}{3} = 240^\circ$$

19. What is the engine displacement of an $3\frac{3}{4} \times 4\frac{1}{4}$ in 4 cylinder engine?

Solution:

$$PD = \frac{\pi D^2 L}{4} = \frac{\pi (3.75 \text{ in})^2 (4.25 \text{ in}) (4)}{4} = 187.8 \text{ in}^3$$

20. What should be the number of teeth to be meshed with T3 to combine the linear speed of a 2 wheel tractor running at 2.175mph given the following data?

1 st reduction	2 nd reduction	3 rd reduction	Ground wheel diameter
N1 = 2000rpm D1 = 300mm D2 = 600mm	T3 = 13t T4 = ?	T5 = 30t T6 = 90t	D = 500mm

Solution:

$$\text{Forward speed} = \pi DN = 2.175 \text{ mph}, N \text{ or } N6 = 37.15 \text{ rpm}$$

$$N5T5 = N6T6, N5 = 111.44 \text{ rpm}, N5 = N4$$

$$N1D1 = N2D2, N2 = 1000 \text{ rpm}$$

$$T3N3 = T4N4,$$

$$T4 = 108 \text{ t}$$

21. Determine the time in hours to plow a 3 ha field using 200mm animal drawn plow with an average depth of cut of 100mm traveling at a speed of 1.554mph and efficiency of 85%. What is the total distance traveled in kilometer by the farmer upon completion of the plowing operation?

Given:

$$A = 3 \text{ ha}; d = 100 \text{ mm}; \text{Efficiency} = 85\%$$

$$W = 200 \text{ mm}; S = 1.554 \text{ mph}$$

Solution:

$$EFC = \frac{SWEff}{10} = \frac{(1.554\text{mph} \times 1.609\text{km})(0.2\text{m})(0.85)}{10} = 0.0425 \frac{\text{ha}}{\text{hr}}$$

$$T = \frac{3\text{ha}}{0.0425 \frac{\text{ha}}{\text{hr}}} = 70.5 \text{ hrs}$$

$$D = 1.554\text{mph} \times 1.609 \times 70.5 = 176.3\text{km}$$

22. The application rate of a certain chemical is recommended at 5 kg/ha. The calibration test of a 15 liter knapsack sprayer is as follows: Nozzle discharge = 1.5 lit/min, Swath = 2700mm and walking speed = 18m/min. Determine the total volume of water applied per hectare?

Given:

$$Q_{\text{recommended}} = 5\text{kg/ha} = 5\text{lit/ha}$$

$$\text{Capacity} = 15 \text{ liters}$$

$$Q_{\text{speed}} = 1.5 \text{ lit/min}$$

$$W = 2.7\text{m}$$

$$S = 18\text{m/min}$$

Solution:

$$V_t = \frac{(1\text{min})(10,000\text{m}^2)(1.5\text{lit})}{18\text{m} \times 2.7\text{m}} = 308.6 \text{ lit/ha}$$

23. Determine the efficiency of a 64V starting system if the rate for the battery when the engine crank is 190 ampere and the maximum power required for the cranking of the engine is 5hp.

Given:

$$V = 64 \text{ V} \quad I = 190\text{amp} \quad P = 5\text{hp}$$

Solution:

$$P = EI = (64\text{V})(190\text{amp}) = 12,160\text{W} = 16.29\text{hp}$$

$$\text{Eff} = \frac{5\text{hp}}{16.29\text{hp}} = 31\%$$

24. Compute the velocity of a windmill having 9ft diameter wheel and an efficiency of 25%. The actual power output is said to be 0.7235hp.

Solution:

$$P = \frac{\rho A V^3}{2}; \quad P = \frac{0.7235\text{hp}}{1.34\text{hp}} = 540\text{N} - \frac{\text{m}}{\text{s}} = 55.1\text{kg} - \frac{\text{m}}{\text{s}}$$

$$\left(55.1\text{kg} - \frac{\text{m}}{\text{s}}\right) = \frac{\left(\frac{1.2\text{kg}}{\text{m}^3}\right) \left(\frac{\pi \left(\frac{9\text{ft}}{3.28\text{ft}}\right)^2}{4}\right) V^3}{2}$$

$$V = \left(\frac{110.2}{(1.2)(5.91)}\right) = \frac{2.50}{0.25} = \frac{9.98\text{m}}{\text{s}}$$

25. A drawer dynamometer show that the average pull required for a certain machine is 2460lbs. if the tractor travels 1,670ft in 55 min, what is the rate of travel in mph?

Solution:

$$V = \frac{(1,670\text{ft})(60\text{min})(1\text{km})(1\text{mile})(1\text{m})}{(55\text{min})(3.28\text{ft})(1000\text{m})(1.609\text{km})(1\text{hr})} = 0.345\text{mph}$$

26. What would be the operating efficiency of an engine if its size to operate a 120V direct current electric generator is 18,91hp? Assume that it has maximum output of 65 amperes.

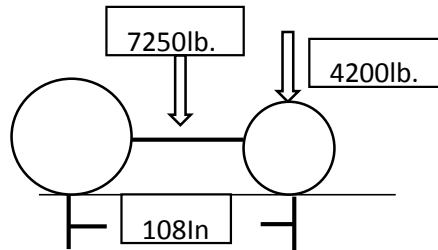
Solution:

$$P = 120 \times 65 = 10.452\text{hp}$$

$$\text{Efficiency} = \frac{10.452\text{hp}}{(18.91\text{hp})} = 55\%$$

27. A tractor has a wheel base of 108 in and weight of 7250lbs. the state weight on the front wheels is 4200lbs. calculate the location of the center of gravity longitudinally with respect to the rear axle.

Given:



Solution:

$$\text{Center of Gravity} = \frac{(4200\text{lbs})(108\text{in})}{7250\text{lbs}} = 62.57 \text{ in}$$

28. If a belt produces a tangential force of 132N on a pulley having a radius of 0.576m at a speed of 352rpm. What will be the power?

Solution:

$$P = 2\pi FN\alpha = \frac{2\pi(132\text{N})(352\text{rpm})(0.576\text{m})}{60\text{secs}} = 2.80\text{kW}$$

29. Calculate the developed horsepower required to pull a plow with a 5-16in bottom at a rate of 4.75mph if the draft is 7psi of furrow section and depth of cut is 4 inches.

Solution:

$$\text{HP} = \frac{(7\text{psi})(5 \times 16\text{inx}4\text{in})(4.75\text{mph} \times 1.609\text{km})}{(2.2\text{lb})(274)} = 28.40\text{hp}$$

30. What size of engine would you recommend to operate a 155V DC electric generator having a maximum output of 96 amperes and an operating efficiency of 79%.

Solution:

$$P = \frac{(155\text{V})(96\text{amperes})(1.34)}{(1000)(0.79)} = 25.24\text{hp}$$

31. Determine the effective capacity of a twelve-bottom plow with a rated width of 74cm each operating at 7.25kph if field efficiency is 87%.

Solution:

$$\text{EFC} = \frac{\text{SWEff}}{10} = \frac{(0.74\text{m})(12)(7.25\text{kph})(0.87)}{10} = 5.6 \frac{\text{ha}}{\text{hr}}$$

32. Compute the firing interval for a V-12 four stroke-cycle engine having a crank shaft speed of 3200rpm.

Solution:

$$\text{FI} = \frac{(360)(2)}{12} = 60$$

33. Determine the compression ratio of four cylinder 150mmx150mm engine turning at 1600rpm with an individual clearance of 155cm³

Solution:

$$PD = \frac{\pi D^2 L}{4} = \frac{\pi (15\text{cm})^2 (12\text{cm})}{4} = 10,602.88 \text{ cm}^3$$

$$CV = 155\text{cm}^3 \times 4 = 620\text{cm}^3$$

$$TV = 11,222.88\text{cm}^3$$

$$CR = \frac{11,222.88\text{cm}^3}{10,602.88\text{cm}^3} = 18.10:1$$

34. A tractor has a 2-cylinder engine with 6.35x9.25in cylinder and develops 55.9hp at 1250rpm. What will be its total piston displacement per minute per horsepower?

Solution:

$$PD = \frac{\pi D^2 L}{4} = \frac{\pi (6.35\text{in})^2 (9.25\text{in})}{4} = \frac{2(92.94\text{in}^3)(1250\text{rpm})(2)}{55.9\text{hp}} = 13,101\text{in}^3$$

35. Determine the rated width of each of the twelve-bottom plow operating at 10kph with field efficiency of 87%. Assume that the effective capacity is 5.714ha/hr

Solution:

$$EFC = \frac{SWEff}{10}; W = \frac{\frac{5.714\text{ha}}{\text{hr}}}{12 \times 0.87} = 54.73\text{cm}$$

36. Determine the depth of cut of a plow with 3-14bottom with a given rate of pulling of 4.75pmh and draft of 8psi of furrow section. Assume that the DHP required is 25hp.

Solution:

$$DHP = FV; F = \frac{(25\text{hp}) \left(76.2\text{kg} \cdot \frac{\text{m}}{\text{s}} \right)}{\frac{2.12\text{m}}{\text{s}}} = 898.60\text{kg}$$

Note: 4.75mph = 2.12m/s

$$\text{Depth of cut} = \frac{898.60\text{kg}}{8\text{psi} \left(\frac{3 \times 14\text{in}}{2.2\text{lb}} \right)} = 5.88\text{in}$$

37. A four cylinder, four stroke cycle engine with 3x4in cylinders develops 19hp at 1650rpm. Assuming a mechanical efficiency of 85%, compute the Hp.

Solution:

$$HP = \frac{19\text{hp}}{0.85} = 22.35\text{hp}$$

38. How long will it take to pump water weighing 23kg up to 352m hill if the power required to do the work is 2.33kW?

Solution:

$$P = QH$$

$$T = \frac{(23\text{kg})(352\text{m})(9.8)}{2,330\text{kg} \cdot \frac{\text{m}}{\text{s}}} = 34.05 \text{ seconds}$$

Note:

$$1W = 1J/s = 1N \cdot m/s = 9.8\text{kg} \cdot m/s$$

39. Compute the crankshaft rear axle speed ration required to give travel speed of 4.5mph for tractor having an engine speed of 2,000rpm and rear wheels are 60in diameter.

Solution:

$$V(\text{tractor}) = 4.5\text{mph}$$

$$V(\text{engine}) = \frac{\pi(1.524\text{m})(2000\text{rpm})(60\text{min})(1\text{km})(1\text{mile})}{(1000\text{m})(1\text{hr})(1.609\text{km})} = 357\text{mph}$$

$$SR = \frac{57\text{mph}}{4.5\text{mph}} = 79.35:1$$

40. What is the dimension of the bore of a tractor given that the stroke measures 5.75in and the bore-stroke ratio equal to 2.35?

Solution:

$$\frac{D}{L} = 2.35, D = 2.35L = 2.35(5.75\text{in}) = 13.5\text{in}$$

41. 5 pieces of work benches are needed for the Agricultural Engineering Review 2011. The engineers found out that they needed 10 pieces of 2"x3"x12' and 6 pieces of 2"x3"x6' for each work benches. How many board feet of lumber would they need for a bench?

Solution:

$$\text{BF} = \frac{(10)(2"x3"x12')}{12} + \frac{0(2"x3"x6')}{12} = 72 \text{ board feet}$$

42. Considering problem #41. How much money do they need to produce the wood materials for the benches? Assume that a board foot of lumber cost 30 pesos.

Solution:

$$X = 72 \times 3 \times 5 = \text{P}10,800$$

43. A sugarcane fertilizer applicator is to apply 16 bags of fertilizer per hectare. The fertilizer applicator is to be calibrated by driving the applicator a distance of 200 meters. What is the weight of fertilizer that should be collected from the delivery tube to get the correct application rate? (Assume 1 bag = 50kg, row spacing = 1.8m).

Given:

$$\text{No. of bags} = 16 \text{ bags } D = 200\text{m}$$

$$\text{Row spacing} = 1.8\text{m } 1\text{bag} = 50\text{kg}$$

Solution:

$$\frac{(16\text{bags})(1\text{ha})(50\text{kg})}{1\text{ha}(10,000\text{m}^2)(1\text{bag})} = 0.08\text{kg/m}^2$$

$$w_{\text{fertilizer}} = \left(\frac{0.08\text{kg}}{\text{m}^2} \right) (200\text{m})(1.8\text{m}) = 28.8\text{kg}$$

44. The loaded radius of the pneumatic tire of the ground wheel driven grain drill is 16 inches. The estimated slippage is 18% and the required speed of the grain drill is 7kph. How many revolutions should the wheel be turned to achieve this?

Solution:

$$V = 2\pi DN; N = \frac{V}{2\pi D} = \frac{\left(\frac{7,000\text{m}}{\text{hr}} \right) \left(\frac{1\text{hr}}{60\text{mins}} \right)}{2\pi(16\text{in})(0.0254\text{m})} = \frac{45.70\text{rpm}}{0.82} = 55.72\text{rpm}$$

45. In the expanded yellow corn production, the recommended plant population for a population for a hybrid yellow corn is 67,000 plants/ha to be planted in furrows 75cm. the ground wheel of the planter has an effective radius of 45cm and equipped with sprocket with 40 teeth. The ground wheel sprocket in turn drives a vertical seed plate with a sprocket of 20 teeth. How many cells should be needed in the seed plate with to achieve planting pattern (Assume 88% germination rate)

Given: radius = 45cm

Solution:

$$\text{Seed plate \# of cells} = \frac{2\pi rSR}{FS} = \frac{2\pi(0.45)(2)}{0.75} = 7.5 \approx 8 \text{ cells}$$

$$SR = \frac{T_1}{T_2} = 2$$

46. A newly purchased grain drill is to be calibrated to determine the approximate seeding rate. The drill has a 10 furrow openers spaced 25cm apart and is equipped with pneumatic tires. When the grain hopper is filled at half its capacity and the tires are inflated at the recommended pressure, the loaded radius is measured at 38cm. after turning the wheels at 60rpm for a period of 10 minutes, 13.4kg of palay were collected from the seed tubes. What is the seeding rate of the drill, in cavan per hectare assuming 1 cavan = 44kg.

Solution:

$$13.4\text{kg} \left(1 \frac{\text{cav}}{44\text{kg}}\right) = 0.304\text{cav}$$

$$X = \frac{\left(\frac{10,000\text{m}^2}{1\text{ha}}\right)}{((10\text{mins})(60\text{rpm})(2\pi(0.38\text{m})(0.25\text{m})(10)))} = \left(\frac{2.80}{\text{ha}}\right)(0.304\text{cavan}) = \frac{0.85\text{cavan}}{\text{ha}}$$

47. In a 200ha corn project, the fields have to be tilled in 30days. What is the width of the offset disk harrow should be selected if the tractor to be used operates at 3.125mph. Assume 75% Field Efficiency and 8hrs working day.

Solution:

Given:

A = 200ha, t = 10days, 8hrs/day S = 3.124mph, FE = 75%

Solution:

$$\text{EFC} = \frac{\text{SWEff}}{10} \\ \frac{200\text{ha}}{(30\text{days})\left(\frac{8\text{hrs}}{\text{day}}\right)} = \frac{(3.125\text{mph} \times 1.609\text{mile})(0.75)W}{10}$$

$$W = 2.21\text{m}$$

48. A paddy seeder was imported from Timbuktu with the following specs. Ground wheel diameter is 47.24in; fixed row spacing is 250mm; hopper capacity is 20kg; speed = 3mph. Engr. Drunken is to calibrate the seeder for paddy sowing. What ground wheel rpm must be set to simulate the designed working speed?

Given:

GWD = 47.24 inches RS = 250mm Capacity = 20kg S = 3mph

Solution:

$$V = \pi DN,$$

$$N = \frac{V}{\pi D} = \frac{\left(\frac{3\text{mile}}{\text{hr}}\right)\left(\frac{1\text{hr}}{60\text{mi}}\right)\left(\frac{1.609\text{km}}{1\text{mile}}\right)\left(\frac{1000\text{m}}{1\text{km}}\right)}{\pi(47.24\text{in})\left(\frac{0.0254\text{m}}{1\text{in}}\right)} = 21.34\text{rpm}$$

49. Remote control hydraulic cylinders with internal diameters of 2.5 inches and 3.5 inches with 1000psi maximum pressure are available for the tractors. What is the maximum force that can be developed by the 3.5 inch cylinders?

Solution:

$$X = \left(\frac{1000\text{lb}}{\text{in}^2}\right)\left(\frac{\pi(3.5)^2}{4}\right) = 9621\text{lbs}$$

50. The application rate of a certain chemical is recommended at 10kg/ha. The calibration test of a 15 liter knapsack sprayer is as follows: Nozzle discharge = 0.75lit/min; Swath = 1.5m; S = 20m/min. Determine the amount of chemical to be mixed with water per loading.

Solution:

Let x be the amount of chemical to be mixed with water per load.

$$\frac{10\text{kg}}{\text{ha}} \approx \frac{10\text{lit}}{\text{ha}}$$

Time per load:

$$T = \frac{\frac{15\text{lit}}{\text{load}}}{\frac{0.75\text{lit}}{\text{min}}} = 20 \frac{\text{min}}{\text{load}}$$

Area per load:

$$A = (1.5\text{m}) \left(\frac{20\text{m}}{\text{min}} \right) \left(\frac{20\text{min}}{\text{load}} \right) = \frac{600\text{m}^2}{\text{load}}$$

$$X = \left(\frac{600\text{m}^2}{\text{load}} \right) \left(\frac{1\text{ha}}{10,000\text{m}^2} \right) \left(\frac{10\text{lit}}{\text{ha}} \right) = 0.6\text{lit/load}$$

51. A pony brake test of a motor give the following: force = 1.5lb; length of arm = 2ft; rpm = 1720; electrical meter reading current = 7.8 amperes; voltage = 220; Power = 1,100Watts. What is the power factor?

Solution:

$$\text{PF} = \frac{1,100\text{Watts}}{220\text{V}(7.8\text{amp})} = 0.64$$

52. A tractor travels at a speed of 4mph. it pulls a drawbar load of 2000lb. what is the drawbar power by the tractor?

Solution:

$$\text{DHP} = \frac{\left(\frac{2000\text{lb}}{2.2\text{lb}} \right) (4\text{mph} \times 1609\text{m})}{3600\text{sec} \left(76.2\text{kg} - \frac{\text{m}}{\text{s}} \right)} = 21\text{hp}$$

53. The engine crankshaft produces 30hp at 1700rpm. What is the torque exerted?

Solution:

$$P = 2\pi TN$$

$$T = \frac{P}{2\pi N} = \frac{30 \times 33,000 \text{ ft} - \frac{\text{lb}}{\text{min}}}{2\pi(1700\text{rpm})} = 92.7 \text{ ft} - \text{lb}$$

54. Determine the theoretical power available from a stream using the following data. Average width = 6m, average depth = 16 inches, Speed = 10ft/min, H= 1.5m

Solution:

$$\text{Theoretical Power} = QH$$

$$P = \rho AVH = \frac{\left(\frac{62.4\text{lb}}{\text{ft}^3} \right) (6\text{m} \times 3.28\text{ft}) \left(\frac{10\text{ft}}{\text{min}} \right) \left(\frac{16\text{in}}{12\text{in}} \right) (1.5\text{m} \times 3.28\text{ft})}{33,000\text{ft} - \frac{\text{lb}}{\text{min}}} = 2.44\text{hp}$$

55. A diesel engine was tested for continuous 3hrs operation. It consumed a total of 5 liters and developed a constant power of 7.5kW. If the fuel contains 45.46MJ/kg, what is the approximate thermal efficiency?

Solution:

$$\left(\frac{5\text{liters}}{3\text{hrs}} \right) \left(\frac{45.46\text{MJ}}{\text{kg}} \right) \left(\frac{1\text{hr}}{3600\text{sec}} \right) = 21\text{kW}$$

$$\eta_{th} = \frac{7.5kW}{21kW} = 36\%$$

56. What is the piston displacement volume in liters of a 4-stroke 4 cylinder 15x18cm engine rotating at 2000rpm?

Solution:

$$PD = \frac{\pi D^2 L}{4} = \frac{\pi (15cm)^2 (18cm) (4)}{4} = 12.72 \text{ liters}$$

57. Tractor A can cover 1-ha field in 2hrs, tractor B in 3hrs and C in 6hrs. How long will it take cover 1ha field if the three tractor worked together?

Solution:

Let X be the time it takes for the three tractors to cover 1ha field.

$$X = \frac{1}{A} + \frac{1}{B} + \frac{1}{C} = \frac{1}{2hrs} + \frac{1}{3hrs} + \frac{1}{6hrs} = 1hr$$

58. At a wind speed of 10kph, a horizontal wind rotor rotates at 100rpm. If the wind rotor diameter is 3.18m, what is the tip speed ratio?

Solution:

$$V = \pi DN = \frac{\pi (3.18m) (100) (60min)}{1000m} = 60kph$$

$$SR = \frac{60kph}{10kph} = 6:1$$

59. An engine uses about 0.24kg of fuel/kW-hr. if the heating value of the fuel is 50MJ/kg, what is the thermal efficiency?

Solution:

$$\left(\frac{0.24kg}{kW-hr} \right) \left(\frac{1kW-hr}{3.6MJ} \right) = \frac{0.067kg}{MJ} \approx \frac{15MJ}{kg}$$

$$\eta_{theoretical} = \frac{15}{50} = 30\%$$

60. A tractor PTO has been tested for its power output. It was made to run a dynamometer with a brake arm of 42 inches at a speed of 1540rpm. If the total of load is 58lb, what is the approximate power develop?

Solution:

$$P = 2\pi FN\alpha = \frac{2\pi (58lb) (1540rpm) (42in)}{\left(33,000ft - \frac{lb}{min} \right) (12)} = 44.4kW \text{ or } 60hp$$

61. Determine the time in hours to plow a 3-ha filed using a 15cm animal-drawn plow at a speed of 4kph and efficiency of 75%.

Solution:

$$EFC = \frac{SWEff}{10}$$

$$\left(\frac{3ha}{T} \right) = \frac{(4kph)(0.15)(0.75)}{10}$$

$$T = 66.67hrs$$

62. Determine the number of days (8hrs/day) to plow a 10-ha filed using a 3 bottom moldboard plow drawn by a 4-wheel tractor at a speed of 5kph and efficiency of 76%. Width of cut of one bottom is 33cm.

Solution:

$$\left(\frac{10\text{ha}}{T}\right) = \frac{(5\text{kph})(0.33)(0.76)}{10}$$

$$T = 26.58\text{hrs}$$

$$X = 26.58\text{hrs} \left(\frac{\text{day}}{8\text{hrs}}\right) = 3.32\text{days}$$

63. If the speed ratio of the ground wheel over seed plate of a two-row corn planter is 5.0, determine the plant population per hectare. Row spacing is 75cm. ground wheel diameter is 60cm. average number of seeds per hill is 2. Number of cells of seed plate is 25. Average emergence of seeds is 85%.

Solution:

$$HS = 2\pi r \frac{SR}{25} = \frac{2\pi(0.3)(5)}{25} = 0.377\text{m}$$

$$\text{Area per hill} = 0.377\text{m} \times 0.75\text{m} = 0.283\text{m}^2$$

$$\text{Plant Population} = \frac{\left(\frac{10,000\text{m}^2}{\text{ha}}\right) \left(\frac{2\text{seeds}}{\text{hill}}\right) (0.85)}{\frac{0.283\text{m}^2}{\text{hill}}} = \frac{60,071\text{plants}}{\text{ha}}$$

64. Considering that there are 3M farmers, 12M-ha of agricultural lands and assuming five farmers per farm. The human power contribution in the farm is about?

Solution:

$$\left(\frac{5\text{farmers}}{\text{farm}}\right) (3,000,000 \text{ farms}) = 15\text{M} \frac{\text{farmers}}{12\text{M} - \text{ha}} = \left(\frac{1.25\text{farmers}}{\text{ha}}\right) \left(\frac{0.1\text{hp}}{\text{farmer}}\right) = 0.125\text{hp/ha}$$

65. A 4-ha field was prepared using a 1.2m floating power tiller. It took 20hrs to complete the operation at a speed of 2.5kph. Determine the field efficiency.

Solution:

$$\text{Field Efficiency} = \frac{\frac{4\text{ha}}{20\text{hrs}}}{\left(\frac{2.5\text{kph} \times 1.2\text{m}}{10}\right)} = 67\%$$

66. Determine the tractor power required to work the soil using the four 14-inch moldboard plow bottoms at a depth of 10 inches and 6kph. F.E = 75%, soil specific draft = 8psi, increase in draft due to speed = 138%.

Solution:

$$D_a = D_s \times D_{\text{increased}} = \frac{(8\text{psi})(1.38)(4 \times 14\text{in} \times 10\text{in})}{2.2\text{lb}} = 2,810\text{kg}$$

$$\text{HP}_{\text{draft}} = D_a \times \frac{S}{274} = \frac{(2,810\text{kg})(6\text{kph})}{274} = 61.54\text{hp}$$

$$\text{HP}_{\text{tractor}} = \frac{61.54\text{hp}}{0.8} = 76.9\text{hp}$$

= use 80hp

67. Determine the number of 13cm-animal drawn plows needed to complete plowing a 3-ha field in one day (8hrs/day). F.E = 78% and speed of travel is 3kph.

Solution:

$$\text{No. of implements} = \frac{\text{Area}}{\text{time} \times \text{EFC}} = \frac{3\text{ha}}{8\text{hrs} \times \frac{0.03042\text{ha}}{\text{hr}}} = 12.3 \text{ plows, use 13 plows}$$

68. Determine the size of the diesel engine to be used for a 4 x 4 centrifugal pump. Pump discharge is 200gpm. Total head is 24ft. Pump Eff. = 55%, V-belt Eff. = 95%, Air-cooled diesel engine delivery rating is 70%.

Solution:

$$HP = \frac{QH}{3960 \text{Eff's.}} = \frac{(200\text{gpm})(24\text{ft})}{3960(0.55)(0.95)(0.70)} = 3.3\text{hp}$$

Note: this formula works only with the same units as the problem given. (ft. & gpm)

Other solution: $200\text{gpm} = \frac{26.74\text{ft}^3}{\text{min}}$

$$HP = \frac{\gamma QH}{\text{Eff's.}} = \frac{\left(\frac{62.4\text{lb}}{\text{ft}^3}\right)\left(\frac{26.74\text{ft}^3}{\text{min}}\right)(24\text{ft})}{(0.70)(0.55)(0.95)\left(\frac{33,000\text{ft-lb}}{\text{min}}\right)} = 3.3\text{hp}$$

69. A 4m combine travelling at 5kph can empty its 1.5ton grain tank in 2 minutes when unloading on-the-go, it has an 86% field efficiency. Field yield is 3.5tons/ha. What would be the efficiency if the combine stopped to load?

Solution:

$$\text{EFC} = \frac{\text{SWEff}}{10} = \frac{(5\text{kph})(4\text{m})(0.86)}{10} = \frac{1.72\text{ha}}{\text{hr}}$$

Number of loading in an hour:

$$X = \left(\frac{3.5\text{tons}}{\text{ha}}\right)\left(\frac{1.72\text{ha}}{\text{hr}}\right)\left(\frac{\text{loading}}{1.5\text{tons}}\right) = 4.01 \text{ or } 4 \text{ loading per hour.}$$

Efficiency if the combine stopped to load:

$$\text{Eff}_{\text{new}} = \frac{T_{\text{total}}}{T_0 + T_1} = \frac{0.86\text{hr}}{1\text{hr} + 0.13\text{hr}} = 76\%$$

$$T_0 = 1\text{hr}; T_1 = 4 \text{ loadings} \times 2\text{mins} = 0.13\text{hr}; T_{\text{total}} = (1\text{hr})(0.86) = 0.86\text{hr}$$

70. The calibration of the 9x7 (9 rows, 7 inches RS) grain drill is shown below. Row spacing to be adjusted to 14 inches. Ground wheel diameter is 1.22m. What is the seeding rate ¼ open seeding adjustment?

Solution:

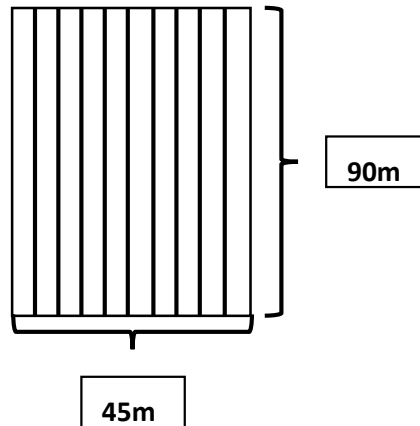
Seeding adjustment	Discharge (gm) (9row/10rev)
¼ open	140
½ open	460
¾ open	740
Full open	1100

$$\text{Area}_{10\text{rev}} = \text{SW} = (\pi(10\text{rev})(1.22\text{m}))(9\text{rows})(14\text{inx}0.0254\text{m})) = 122.66\text{m}^2$$

Seeding rate ¼ open:

$$X = 0.14\text{kg} \left(\frac{\frac{10,000\text{m}^2}{\text{ha}}}{122.66\text{m}^2} \right) = \frac{11.41\text{kg}}{\text{ha}}$$

71. A field measuring 45x90m is to be prepared using a 1m upland power tiller at a speed of 3kph. Assume a 10% overlap between passing's and 9 seconds turnaround time at the end of each pass. Determine actual field capacity when tilling is done along the longer field direction.
Solution:



$$\text{Number of rows} = \frac{45\text{m}}{(1\text{m})(0.9)} = 50 \text{ rows}$$

Total Area = 0.405ha

Total time = T1 + T2 = 1.625hr

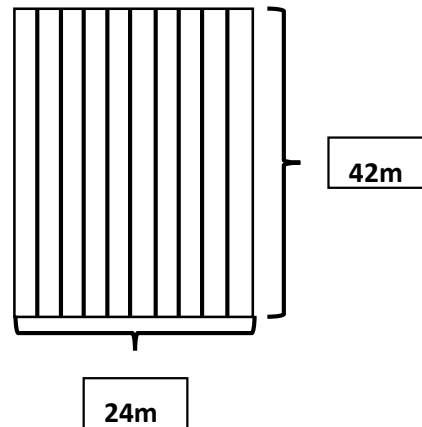
T1 (time to cover 50 rows) = 50rows $\left(\frac{9\text{sec}}{\text{row}} \right) = 450\text{sec}$ or 0.125hr

T2 = $\frac{90\text{m} \times 50\text{rows}}{\left(\frac{3000\text{m}}{\text{hr}} \right)} = 1.5\text{hr}$

C_{actual} = $\frac{A_{\text{total}}}{T_{\text{total}}} = \frac{0.405\text{ha}}{1.625\text{hr}} = \frac{0.25\text{ha}}{\text{hr}}$

72. A 1.2m wide floating tiller is used for rotary tilling/pudding a 24m x 42m wetland paddy field. The average speed is 3kph and turning at the headlands takes 10 seconds. What is the field efficiency if the field is worked along its length?

Solution:



$$\text{Total Area} = 0.1008\text{ha}; \text{ Number of rows} = \frac{24\text{m}}{1.2\text{m}} = 20 \text{ rows}$$

$$T1 = 20\text{rows} \left(\frac{10\text{sec}}{\text{row}} \right) = 200\text{sec or } 0.056\text{hr}$$

$$T2 = \frac{42\text{m} \times 20\text{rows}}{\frac{3,000\text{m}}{\text{hr}}} = 0.28\text{hr}$$

$$\text{Total time} = 0.336\text{hr}$$

$$C_{\text{actual}} = \frac{0.1008\text{ha}}{0.336\text{hr}} = \frac{0.3\text{ha}}{\text{hr}}$$

$$C_{\text{actual}} = \frac{\text{SWEff}}{10} = \frac{(3\text{kph})(1.2\text{m})}{10} = \frac{0.36\text{ha}}{\text{hr}}$$

$$\text{Efficiency} = \frac{\frac{0.3\text{ha}}{\text{hr}}}{\frac{0.36\text{ha}}{\text{hr}}} = 83.33\%$$

73. A 4-wheel tractor with 3x14 inches moldboard plow is to operate on clay-loam soil ($SD = 0.56\text{kg/cm}^2$) at a depth of 25.4cm. The maximum draft and percent increase in draft due to speed are given below. Determine the highest gear setting that can sustain the draft requirement for the operation.

Solution:

Gear setting	Speed (kph)	Maximum draft (kg)	Increase in draft due to speed
1L	3.2	9,000	14
2L	4.8	3,100	28
3L	6.4	2,200	42
4L	8.0	1,300	56

$$D_{\text{specific}} = \left(\frac{0.56\text{kg}}{\text{cm}^2} \right) (25.4\text{cm}) (2.54\text{cm}) (3) (14\text{inches}) = 1517.42\text{kg}$$

$$D_{\text{actual}} = 1517.42\text{kg} (1.42) = 2,154.74\text{kg use } 2,200\text{kg} \\ = 3\text{L}$$

74. Determine the horsepower required to perform the operation in problem #73.

Solution:

$$D_{\text{actual}} = 1517.42\text{kg} (1.42) = 2,154\text{kg}$$

$$D_{\text{draft}} = \frac{(2,154\text{kg})(6.4)}{274} = 50.4\text{hp use } 50\text{hp}$$

75. The fertility of a particular field is such that maximum corn yield is obtained with a population of 54,000 plant per hectare. A checkrow corn planter is used which discharges 2 seeds/hill. Assuming emergence is 85%, RS = HS, find the spacing.

Solution:

$$\text{If } RS = HS \rightarrow RS^2 = \text{Area}$$

$$A = \frac{\left(\frac{10,000\text{m}^2}{\text{ha}}\right) \left(\frac{2\text{seeds}}{\text{hill}}\right) (0.85)}{\frac{54,000\text{plants}}{\text{ha}}} = 0.31\text{m}^2$$

$$RS = \sqrt{A} = \sqrt{0.31} = 0.56\text{m}$$

76. The loaded radius of the pneumatic tire of the ground wheel driven grain drill is 16 inches. The estimated slippage is 18% and the required ground speed of the grain is 7kph. How many revolutions should the wheel be turned to achieve this?

Solution:

$$V = \pi DN$$

$$N = \frac{\frac{116.67\text{m}}{\text{min}}}{2.553\text{m}} = \frac{45.70\text{rpm}}{0.82} = 56\text{rpm}$$

77. A 4 cylinder 4-stroke cycle engine with 2.5"x4" cylinder develops 12hp at 1850rpm. What is the BMEP?

Solution:

$$\text{BMEP} = \frac{(2)(12\text{hp}) \left(33,000\text{ft} \cdot \frac{\text{lb}}{\text{min}}\right)}{\left(\frac{4\text{in}}{12\text{in}}\right) \left(\frac{\pi(2.5)^2}{4}\right) \left(\frac{1850}{\text{min}}\right) (4)} = \frac{65.41\text{lb}}{\text{in}^2}$$

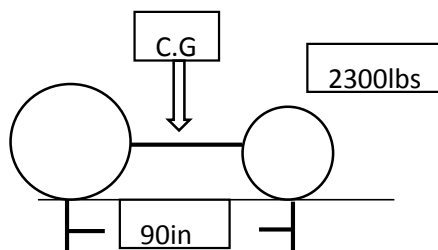
78. A tractor with a 9in pulley is belted to pony brake having a 24in pulley. If the tractor engine is 950rpm, the brake arm is 54in, and the net load is 60lb. what is the horsepower developed?

Solution:

$$\frac{N_1}{N_2} = \frac{D_1}{D_2} \rightarrow N_1 = \frac{D_1 N_2}{D_2} = \frac{(950\text{rpm})(9\text{in})}{24\text{in}} = 356.25\text{rpm}$$

$$P = 2\pi F N \alpha = \frac{2\pi(60\text{lbs})(356.25\text{rpm})(54\text{in} \cdot 12\text{in})}{33,000\text{ft} \cdot \frac{\text{lb}}{\text{min}}} = 18.3\text{hp}$$

79. Calculate the location of the center of gravity of a 7800lb-tractor with respect to the rear axle if the wheel base is 90in. The static weight on the front wheels is 2300lb.



Solution:

$$C.G. = \frac{(2300\text{lb})(90\text{in})}{7800\text{lb}} = 26.5 \text{ inches}$$

80. What is the compression ratio of a 2-stroke cycle 3"x4" engine if the clearance volume is 5in³?
The speed of operation is 1000rpm.

Solution:

$$PD = \frac{\pi D^2 L}{4} = \frac{\pi (3\text{in})^2 (4\text{in})}{4} = 28.27\text{in}^3$$

$$TV = 28.27\text{in}^3 + 5\text{in}^3 = 33.27\text{in}^3$$

$$CR = \frac{33.27\text{in}^3}{5\text{in}^3} = 6.7:1$$

81. Calculate the theoretical power available from a stream using the following data: average width = 12ft; average depth = 2ft; speed = 15ft/min; height of drop = 4ft.

Solution:

$$P = \rho AVH = \frac{\left(\frac{62.4\text{lb}}{\text{ft}^3}\right) (12\text{ft})(2\text{ft}) \left(\frac{15\text{ft}}{\text{min}}\right) (4\text{ft})}{33,000\text{ft} - \frac{\text{lb}}{\text{min}}} = 2.7\text{hp}$$

82. How much power can a horse deliver if it is continuously pulling 120lb load for several hours and walking at a rate of 2.5mph?

Solution:

$$DHP = \frac{(120\text{lb})(2.5\text{mph})(1609\text{m})(3.28\text{ft})}{60\text{min}} = 0.8\text{hp}$$

83. Determine the drawbar power required to pull a plow with a 3-14in bottom at a rate of 3.5mph if the unit draft is 8psi of furrow section and depth of cut is 7in.

Solution:

$$DHP = \frac{\left(\frac{8\text{lb}}{\text{in}^2}\right) (3)(14\text{in})(7\text{in})(3.5\text{mph})(1609\text{m})(3.28\text{ft})}{(60\text{min})(33,000\text{ft} - \text{lbmin})} = 21.94\text{hp}$$

84. A diesel engine burns a 0.25kg of fuel/kW-hr. What is the approximate thermal efficiency?

Solution:

$$X = \frac{1}{\left(\frac{0.25\text{kg}}{\text{kW} - \text{hr}}\right) \left(\frac{45.5\text{MJ}}{\text{kg}}\right) \left(\frac{\text{kW} - \text{hr}}{3.6\text{MJ}}\right)} = 32\%$$

85. An ordinary gasoline engine uses about 0.4kg of fuel/kW-hr. What is the approximate thermal efficiency?

Solution:

$$X = \frac{1}{\left(\frac{0.4\text{kg}}{\text{kW} - \text{hr}}\right) \left(\frac{34,330\text{MJ}}{\text{m}^3}\right) \left(\frac{1\text{lit}}{0.73\text{kg}}\right) \left(\frac{1\text{m}^3}{1000\text{lit}}\right) \left(\frac{1\text{kW} - \text{hr}}{3.6\text{MJ}}\right)} = 19\%$$

86. Juan will plow the field of 120m wide and 426.4 feet long using a carabao and moldboard plow with a 10-inch width of cut. If his speed of plowing is 2mph, how long will he finish plowing if he plows along the length of the field with no headland left unplowed?

Solution:

$$\frac{1.56\text{ha}}{t} = \frac{(3.218\text{kph})(0.254\text{m})}{10} = 19.08\text{hrs}$$

Other solution: # of rows = $120/0.254 = 472$ rows

$$t = \frac{472 \text{ rows} \times 130 \text{ m}}{3218 \text{ m/hr}} = 19.07 \text{ hrs}$$

87. In problem # 86. How long will it take for Juan to finish plowing if he plows along the width of the field with 2m headland unplowed?

Solution:

$$\frac{1.534 \text{ ha}}{t} = \frac{(3.218 \text{ kph})(0.254 \text{ m})}{10} = 18.77 \text{ hrs}$$

Other solution: # of rows = 512

$$t = \frac{512 \text{ rows} \times 118 \text{ m}}{3218 \text{ m/hr}} = 18.77 \text{ hrs}$$

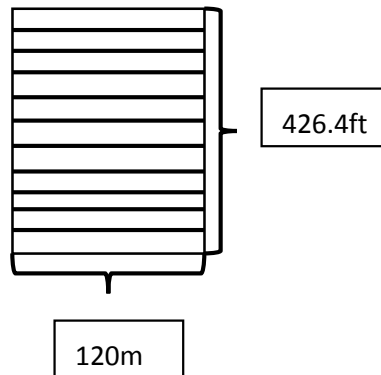
88. What is the plowing capacity in problem #87?

Solution:

$$C = \frac{1.534 \text{ ha}}{18.77 \text{ hrs}} = 0.082 \text{ ha/hr.}$$

89. In problem #86, if his plowing speed is 2kph, how long will he finish plowing if he plows along the width of the field with no headlands left unplowed?

Solution:



$$\frac{1.56 \text{ ha}}{t} = \frac{(2 \text{ kph})(0.254 \text{ m})}{10} = 30.71 \text{ hrs}$$

Other solution: # of rows = 511.81 rows

$$t = \frac{511.81 \text{ rows} \times 120 \text{ m}}{2000 \text{ m/hr}} = 30.71 \text{ hrs}$$

90. In accordance with conditions in problem # 89, how long will he finish if he plows along the length with no headland left unplowed?

Solution:

$$\frac{1.56\text{ha}}{t} = \frac{(2\text{kph})(0.254\text{m})}{10} = 30.71\text{hrs}$$

Other solution: # of rows = 511.81rows

$$t = \frac{472\text{rows} \times 130\text{m}}{2000\text{m/hr}} = 30.71\text{hrs}$$

91. What is the plowing capacity in problem #89?

Solution:

$$C = \frac{1.560\text{ha}}{30.70\text{hrs}} = \frac{0.051\text{ha}}{\text{hr}}$$

92. If the width of cut or his plow is 30cm, by how much will his plowing capacity (ha/hr) increase in accordance with the conditions in problem #88.

Solution:

$$C = \frac{1.534\text{ha}}{15.89\text{hrs}} = \frac{0.0965\text{ha}}{\text{hr}} = \frac{0.0965\text{ha}}{\text{hr}} - \frac{0.082\text{ha}}{\text{hr}} = \frac{0.015\text{ha}}{\text{hr}}$$

Note: S = 3218m/hr. 2m headlands. Plow direction along its width. # of rows = 130/0.30m = 433.33rows

$$t = \frac{433.33\text{rows} \times 118\text{m}}{3218\text{m/hr}} = 15.89\text{hrs}$$

93. A farmer uses a hand tractor with 20cm moldboard plow. How many kilometers must he walk to plow one hectare?

Solution:

$$X = \frac{10,000\text{m}^2}{0.2\text{m}} = 50,000\text{m} \text{ or } 50\text{km}$$

94. In problem #93, if he overlaps each pass by 10%, how many kilometers more will he have to walk to plow one hectare?

Solution:

$$X = \frac{10,000\text{m}^2}{(0.2\text{m})(0.9)} = 55.6\text{km}$$

95. If his walking speed is 1.55mph, how many will it take him to plow in each case in problem #93 and #94? What will be his plowing capacity (ha/hr.) in each case?

Solution:

Case 1 (problem #93):

$$\frac{10,000\text{m}^2}{t} = \frac{(1.55\text{mph} \times 1.609\text{km})(0.2\text{m})}{10} = t_1 = 200\text{hrs}; C = \frac{1\text{ha}}{200\text{hrs}} = \frac{0.005\text{ha}}{\text{hr}}$$

Case 2 (problem #94):

$$\frac{10,000\text{m}^2}{t} = \frac{(1.55\text{mph} \times 1.609\text{km})(0.2\text{m})(0.9)}{10}; t_2 = 222.76\text{hrs}; C = \frac{1\text{ha}}{222.76\text{hrs}} = \frac{0.0045\text{ha}}{\text{hr}}$$

Note: I don't know if the given speed in the problem (1.55mph) is correct. If it is, then it won't match the answer (Capacity) provided in the book. But I'm sure that my answer is correct. The capacity provided in the book are 0.05ha/hr. and 0.045ha/hr. If you change the speed to 15.5mph. It will give the same answers as provided.

96. An engine at 2000rpm is to drive a pump to run at 1000rpm. The engine driven pulley is 15cm in diameter and there is 10% belt slippage. What should be the diameter of the pulley on the pump?

Solution:

$$\frac{N_1}{N_2} = \frac{D_1}{D_2} \rightarrow D_2 = \frac{D_1 N_2}{N_1} = \frac{(15\text{cm})(2000\text{rpm})(0.90)}{1000\text{rpm}} = 27\text{cm}$$

97. How many hectares can be plowed in one hour by a tractor operating at 5kph and is pulling 3 moldboard plows with total width of cut of 160cm at a depth of 14cm?

Solution:

$$X = \frac{(5\text{kph})(16\text{m})}{10} = \frac{0.8\text{ha}}{\text{hr}}$$

98. What is the total draft in kilogram pull if there is a draft of 0.45kg/cm²? Refer to problem #97.

Solution:

$$D = \left(\frac{0.45\text{kg}}{\text{cm}^2}\right)(160)(14) = 1008\text{kg}$$

99. How many hectares can be plowed in 3hrs if 8% of the time is spent on headland turning? Refer to problem #97.

Solution:

$$A = \left(\frac{0.8\text{ha}}{\text{hr}}\right)(3\text{hrs})(0.92) = 2.208\text{ha}$$

100. A 2ha farm is to be planted by a 4-row rear mounted corn planter traveling 5kph with a row spacing of 1m. The field is twice as long as its width. If 3 minutes is spent for each headland turn,

- a. How many turns will the planter make to plant the whole area?

Solution:

$$X = \left(\frac{100}{4}\right) = 25\text{turns} - 1 = 24\text{turns}$$

- b. How many hours are needed to plant the whole field if the plant rows run along the length?

Solution:

$$t_1 = (24\text{turns})(200\text{m})\left(\frac{1\text{hr}}{5000\text{m}}\right) = 0.96\text{hr or } 1\text{hr}$$

$$t_2 = (3\text{minutes})(24\text{turns})\left(\frac{1\text{hr}}{60\text{min}}\right) = 1.2\text{hr}$$

$$T_{\text{total}} = 2.2\text{hrs}$$

- c. How many percent of the total time in (b) was spent in headland turning?

Solution:

$$X = \frac{1.2\text{hr}}{2.2\text{hrs}} = 54.54\%$$

101. A 4-cylinder four-stroke cycle engine with 4 in cylinders develops 19hp at 1650rpm. Compute the Indicated horsepower if the mechanical efficiency is 85%.

Solution:

$$\text{IHP} = \frac{19\text{hp}}{0.85} = 22.4\text{hp}$$

102. What is the BMEP for the problem #101?

Solution:

$$\begin{aligned} \text{BMEP} &= \frac{22.4\text{hp}}{(4\text{in})\left(\frac{\pi(3\text{in})^2}{4}\right)(1650\text{rpm})(4)} = \frac{(2)(22.4\text{hp})\left(\frac{33,000\text{ft} - \text{lb}}{\text{min}}\right)(12\text{in}1\text{ft})}{\frac{186,610.60\text{in}^3}{\text{min}}} \\ &= \left(\frac{95.07\text{lb}}{\text{in}^2}\right)(0.85) = \frac{80.80\text{lb}}{\text{in}^2} \end{aligned}$$

103. A 100mm x 100mm engine runs at 2,400rpm. What is the maximum piston speed inside the cylinder?

Solution:

$$V = 2\pi DN = \frac{2\pi(0.10\text{m})(2400\text{rpm})}{60\text{sec}} = \frac{25\text{m}}{\text{s}}$$

104. An engine uses about 0.6lb of fuel/hp.-hr. If the heating value of the fuel is 20,000Btu/lb., what is the thermal efficiency?

Solution:

$$\left(\frac{1\text{hp} - \text{hr}}{0.8\text{lb}}\right)\left(\frac{2544.4\text{Btu}}{1\text{hp} - \text{hr}}\right)\left(\frac{1\text{lb}}{20,000\text{Btu}}\right) = 15.9\%$$

105. How many hectares will a disk plow finish in 10hrs if the effective cut is 12"/disk (5disk) at a speed of 5kph and an average of 30% time loss.

Solution:

$$\frac{X}{10} = \frac{(5)(12\text{in})(0.0254\text{m})(5\text{disk})(0.70)}{10} = 5.334\text{ha}$$

106. A 4m combine traveling at 5kph can empty its 1.64ton grain tank in 60 seconds. When unloading on the go, it has 85% Field Efficiency. Field yield is 5tons/ha. Field speed is 5kph. The combine material efficiency is 95%. Determine the material harvested per hour?

Solution:

$$X = \frac{(5\text{kph})(4\text{m})(0.85)(0.95)(3\text{ton})}{10} = 4.845\text{kg/hr}$$

107. A field measuring 45x90m is to be prepared using 1m upland power tiller at a speed of 3kph. Assume a 10% overlap between passing and 9 seconds turn-around time at the end of each pass. Determine the actual field capacity when tilling is done along the longer field direction.

Solution:

$$\text{Number of rows} = \frac{45\text{m}}{1\text{m} \times 0.9} = 50\text{rows}$$

$$t_1 = 50\text{rows} \times 9\text{seconds} = 450\text{seconds or } 0.125\text{hr}$$

$$t_2 = \frac{50\text{rows} \times 90\text{m}}{10} = 1.5\text{hr}$$

$$CA = \frac{0.405\text{ha}}{1,625\text{hr}} = 0.25\text{ha/hr}$$

108. A tractor engine running at 1,200rpm transmits an input torque of 272N.m to the tractor gear train with the following speed reduction: a.) transmission 3:1, b.) differential 4:1, c.) final drive 5:1. The tractor is required with the drive tires indicating a static load, radius of 61cm. assuming an overall power transmission efficiency of 96% and an average wheel slippage of 15%. Calculate the following:

- a. Tractor forward speed.

Solution:

From engine to final drive:

$$= (1200\text{rpm}) \left(\frac{1}{3}\right) \left(\frac{1}{4}\right) \left(\frac{1}{5}\right) = 20\text{rpm (Rear - axle)}$$

$$V = \pi DN = \frac{\pi(1.22\text{m}) \left(\frac{20\text{rev}}{\text{min}}\right) (60)(0.85)}{1000\text{m}} = 3.11\text{kph}$$

- b. Crankshaft-rear axle speed ratio.

Solution:

$$\frac{\text{Crankshaf}}{\text{trear - axle}} = \frac{1200}{20} = 60:1$$

- c. Output torque available at the drive wheels, Nm

Solution:

$$T_2 = \frac{N_1 T_1}{N_2} = \frac{(1200\text{rpm})(272\text{N.m})}{20\text{rpm}} = 15,667.2\text{Nm}$$

109. A gas engine works on a 4-stroke cycle and has the following dimensions: cylinder diameter is 10 inches, stroke bore ratio of 1.8, clearance volume is 27cm³, engine speed is 240rpm, mean effective pressure is 100psi, and the mechanical efficiency is 75%.

- a. Calculate the Indicated horsepower.

Solution: since stroke-bore ratio is 1.8, it indicates that Length is 18in

$$\text{IHP} = \frac{\left(\frac{100\text{lb}}{\text{in}^2}\right) (18\text{in}) \left(\frac{\pi(18)^2}{4}\right) (240\text{rpm})}{(2)(12\text{in}) \left(33,000\text{ft} - \frac{\text{lb}}{\text{in}^2}\right)} = 42.83\text{hp}$$

- b. The brake horsepower is?

Solution:

$$\text{BHP} = 42.83\text{hp} \times 0.75 = 32.12\text{hp}$$

- c. Compression ratio.

Solution:

$$\text{CR} = \frac{\left(\frac{\pi(10)^2}{4}\right) (18\text{in}) + 27\text{in}^3}{27\text{in}^3} = 53.3:1$$

110. An engine crankshaft provides 30hp at 1700rpm. What is the torque exerted?

Solution:

$$P = 2\pi FN\alpha = 2\pi TN$$

$$T = \frac{(30\text{hp}) \left(\frac{33,000\text{ft} - \text{lb}}{\text{min}} \right)}{2\pi(1700\text{rpm})} = 92.68\text{ft} - \text{lb}$$

111. Using the results of the calibration test of the 8x20cm drum seeder, determine the seeding rate (kg/ha) if Q_{medium} is used. Row spacing is 40cm, Ground wheel diameter is 50cm, Discharge of one row per 10 revolutions of the ground wheel. $Q_{\text{low}}=16\text{gms}$, $Q_{\text{medium}}=22\text{gms}$, $Q_{\text{high}}=38\text{gms}$.

Solution:

$$Q = 0.022\text{kg} \left(\frac{A_T}{A_{10}} \right) = 0.022\text{kg} \left(\frac{10,000 \frac{\text{m}^2}{\text{ha}}}{6.28\text{m}^2} \right) = 35\text{kg/ha}$$

$$A_T = \frac{10,000\text{m}^2}{\text{ha}} =$$

$$A_{10} = RS \times S_{10} = (0.4\text{m}) (\pi(0.5\text{m}) (10\text{revs})) = 6.28\text{m}^2$$

112. Determine the number of days (8hrs/day) to plow a 10ha field using a 3 bottom moldboard plow drawn by a 4-wheel tractor at a speed of 5kph and field efficiency of 76%. Width of cut of one bottom is 33cm.

Solution:

$$\frac{10\text{ha}}{t} = \frac{(5\text{kph})(3 \times 0.33\text{m})(0.76)}{10} = \frac{26.58\text{hrs}}{\frac{8\text{hrs}}{\text{day}}} = 3.3 \text{ days}$$

113. Determine the size of air-cooled diesel engine to be used for a 1.5ha lowland rice. The field is to be irrigated daily for 8hrs. source of water is a 3-inch diameter drilled well with the following data: static water level is 9ft, specific yield of well is 15gpm/ft., discharge head is 6ft, and the friction and velocity head loss is 2ft. Pump efficiency is 45%, transmission efficiency is 95% and power unit continuous delivery rating is 70%.

Solution:

Operating (hrs./day)	Area that can be irrigated (ha)		
	30gpm	60gpm	90gpm
8	0.5	1.0	1.5
12	0.75	1.5	2.25
16	1.0	2.0	3.0
20	1.25	2.5	3.75
24	1.5	3.0	4.5

$$HP = \frac{QHT}{3960\text{Eff}'s_T} = \frac{(90\text{gpm})(23\text{ft})}{3960(0.29925)} = 1.74\text{hp use } 2\text{hp}$$

For area of 1.5ha and 8hrs a day, discharge is 90gpm

$$\frac{90\text{gpm}}{15\frac{\text{gpm}}{\text{ft}}} = 6\text{ft}$$

$$H_T = H_{dd} + H_f + H_s + H_d = 23\text{ft}; \text{Eff}'s_T = (\text{Eff}_p)(\text{Eff}_t)(\text{Eff}_{pm}) = 0.29925$$

114. Using the results of the calibration test of the 8x20cm drum seeder, determine the seeding rate adjustment to use if it is desired to plant at the rate of 75kg/ha at row spacing of 40cm. ground wheel diameter is 50. Discharge of seeder per 10 revolutions of ground wheel. Low = 128grams, Medium = 176grams, High = 304grams.

Solution:

$$Q = m \left(\frac{A_t}{A_{10}} \right)$$

$$A_{10} = RS \times S_{10} = (0.4)(\pi)(0.5)(10) = 6.28m^2$$

$$A_t = \frac{10,000m^2}{ha}$$

$$Q_{low} = m \left(\frac{A_t}{A_{10}} \right) = (0.128kg) \left(\frac{\frac{10,000m^2}{ha}}{5.28} \right) = \frac{208.82kg}{ha}$$

$$Q_{medium} = m \left(\frac{A_t}{A_{10}} \right) = (0.176kg) \left(\frac{\frac{10,000m^2}{ha}}{5.28} \right) = \frac{280kg}{ha}$$

$$Q_{low} = m \left(\frac{A_t}{A_{10}} \right) = (0.304kg) \left(\frac{\frac{10,000m^2}{ha}}{5.28} \right) = \frac{483.8kg}{ha}$$

115. Using the specifications of the transmission system of the metering device (horizontal seed plate) of the two-row planter, determine the expected hill spacing. Row spacing is 75cm. ground wheel diameter is 60cm, number of cells of seed plate is 20, T1 = 6t, T2 = 18t, T3 = 10t and T4 = 25t.

Solution:

$$\text{Hill spacing} = \frac{2\pi r SR}{SP} = \frac{2\pi(0.3m) \left(\frac{7.5rev}{SP} \right)}{20} = 0.71m$$

$$N_2 T_2 = N_3 T_3; N_3 = \frac{\left(\frac{1rev}{SP} \right) (25t)}{10t} = \frac{2.5rev}{SP}$$

$$N_3 = N_2, \text{ thus } N_1 = 7.5rpm = SR$$

116. From problem #73, determine the drawbar horsepower to perform the operation.

Solution:

$$\text{Draft} = \left(\frac{0.56kg}{cm^2} \right) (3 \times 14in) (2.54cm) (2.54cm) = 1517.4kg$$

$$HP_{draft} = \frac{(\text{Draft}_{adjusted})(\text{Speed})}{274} = \frac{(1517.4kg)(1.42)(6.4kph)}{274} = 50.32hp$$

$$HP_{tractor} = \frac{50.32hp}{0.8} = 62hp$$

117. A certain lift pump has a cylinder diameter of 100mm and pump stroke of 145mm. results of calibration test of this pump gave water discharge of 17kg/min at 17 pump stroke/min. Determine its volumetric efficiency. Density of water is 1g/cc.

Solution:

$$\text{Volumetric Efficiency} = \frac{V_{\text{actual}}}{V_{\text{theoretical}}} = \frac{1000\text{cm}^3}{1138.82\text{cm}^3} = 88\%$$

$$V_{\text{theoretical}} = \frac{\pi D^2 L}{4} = \frac{\pi (10\text{cm})^2 (14.5\text{cm})}{4} = 1138.82\text{cm}^3$$

$$V_{\text{actual}} = \left(\frac{17\text{kg}}{\text{min}}\right) \left(\frac{\text{min}}{17\text{strokes}}\right) = \frac{1\text{kg}}{\frac{0.001\text{kg}}{\text{cm}^3}} = 1000\text{cm}^3$$

118. A gasoline engine has a thermal efficiency of 23%. Test shows it uses 1.2 liter of premium gasoline per hour. What is the approximate power it generates?

Solution:

$$\left(\frac{1.2\text{liter}}{\text{hr}}\right) \left(\frac{34330\text{MJ}}{\text{m}^3}\right) \left(\frac{1\text{m}^3}{1000\text{liter}}\right) = \left(\frac{41.196\text{MJ}}{\text{hr}}\right) \left(\frac{1\text{hr}}{3600\text{sec}}\right) = 11.44\text{kW}$$

$$P_{\text{approximate}} = 11.44\text{kW} \times 0.23 = 2.632\text{kW}$$

119. A 15.6liter of engine produces a brake mean effective pressure (BMEP) of 1077KPa. Calculate the brake torque.

Solution:

$$T = \frac{(15.6\text{liters}) \left(\frac{1077 \times 10^3 \text{N}}{\text{m}^2}\right) \left(\frac{\text{m}^3}{1000\text{liters}}\right)}{4\pi} = 1337\text{N} - \text{m}$$

120. An imported paddy seeder has the following specifications: GWD = 47.24in, RS = 0.82ft, Hopper capacity = 44lbs, Designed speed = 4mph. If you were the engineer and you are to calibrate the seeder. What ground wheel rpm should be set to simulate the designed working speed?

Solution:

$$V = \pi D N$$

$$N = \frac{V}{\pi D} = \frac{(4\text{mph})(1609\text{m})}{\pi (60\text{min})(47.24\text{in})(0.0254\text{m})} = 28.45\text{rpm}$$

121. What is the volumetric efficiency of a lift pump having the following specifications and test data:

Brand name: Eagle Peachier Pump

Pump Lever: 400mm

Piston rod: 100mm

Piston rod-stroke: 150mm

Cylinder diameter: 80mm

Suction pump length: 650mm

Water temperature: 250 C

Density (water): 997.1kg/m³

Trial	Weight/minute (kg/min)	No. of strokes/minute
1	26	40
2	27	42
3	27	45
4	25	43
5	24	41

Solution:

$$\text{Eff}_{\text{volumetric}} = \frac{V_{\text{actual}}}{V_{\text{theoretical}}} = \frac{613.15\text{cm}^3}{753.982\text{cm}^3} = 81.32\%$$

$$V_{\text{theoretical}} = \frac{\pi D^2 L}{4} = \frac{\pi(8\text{cm})^2(15\text{cm})}{4} = 753.982\text{cm}^3$$

$$V_{\text{actual}} = \left(\frac{25.8\text{kg}}{\text{min}}\right)\left(\frac{\text{min}}{42.2}\right) = \frac{0.611\text{kg}}{\frac{997.1\text{kg}}{\text{m}^3}} = 613.15\text{cm}^3$$

122. What should be the size of the air-cooled diesel engine (delivery rating of 70%) available in the market needed to power a centrifugal pump use for irrigating a field planted with lowland rice if the pump is operating at 20lps and the source of water is a well with the following data:

Static level of water = 2.75m

$$\text{Specific yield} = \frac{\frac{7.57 \times 10^{-4} \text{m}^3}{\text{sec}}}{0.3\text{mDD}}$$

Discharge head = 0.3m

Friction velocity head = 0.6m

Power transmission = flat belt and pulley (90%)

Normal efficiencies of centrifugal pumps at 150gpm and above = 55%.

Solution:

$$Q = \left(\frac{20\text{liter}}{\text{sec}}\right)\left(\frac{1\text{gal}}{3.785\text{liter}}\right)\left(\frac{60\text{sec}}{\text{min}}\right) = 317\text{gpm}$$

$$H_{\text{DD}} = \left(\frac{\frac{7.57 \times 10^{-4} \text{m}^3}{\text{sec}}}{0.3\text{m}}\right)\left(\frac{60\text{sec}}{\text{min}}\right)\left(\frac{1000\text{liters}}{1\text{m}^3}\right)\left(\frac{1\text{gal}}{3.785\text{liter}}\right) = \frac{40\text{gpm}}{3.28\text{ft}} = \frac{12.1938\text{gpm}}{\text{ft}}$$

$$= \frac{317\text{gpm}}{\frac{12.1938\text{gpm}}{\text{ft}}} = 25.997\text{ft}$$

$$H_T = (25.997\text{ft}) + (9.02\text{ft}) + (0.984\text{ft}) + (1.968\text{ft}) = 37.971\text{ft}$$

$$\text{Eff}_T = 0.3465$$

$$\text{HP} = \frac{\text{QTDH}}{39860\text{Eff}_T} = \frac{(317\text{gpm})(37.971\text{ft})}{3960(0.3465)}$$

123. After the calibration test of a rice planter, data showed that 2.0% of the rice grains were totally damage by the metering device and after germination test, only 92% seed viability was achieved. The seeds weighed 60grams per 1000grain, what weeding rate in (kg/ha) should you calibrate the seeder to attain a crop stand of 7.0×10^5 plants per hectare?

Solution:

$$X = \frac{\left(\frac{(700,000\text{plants})}{\text{ha}}\right)\left(\frac{60\text{grams}}{1000\text{grams}}\right)}{(0.98)(0.92)} = \frac{46.58\text{kg}}{\text{ha}}$$

124. If the mechanical work is done at a rate of 1ft-lb/sec, what is its equivalent in watts?

Solution:

$$\left(\frac{1\text{ft} - \text{lb}}{\text{second}}\right)\left(\frac{60\text{seconds}}{1\text{min}}\right) = \frac{\left(\frac{60\text{ft} - \text{lb}}{\text{min}}\right)}{\frac{33,000\text{ft} - \text{lb}}{\text{min} - \text{hp}}} = \frac{0.0018\text{hp}}{1.43\text{hp}} = 0.00136\text{kW}$$

125. A water stream had the following data:

Average depth = 2ft

Average width = 12ft

Velocity per minute = 15ft/min

Head = 4ft

What is the theoretical power available from the stream?

Solution:

$$P_{th} = \rho AVH = \frac{\left(\frac{62.4\text{lb}}{\text{ft}^3}\right)(12\text{ft})(2\text{ft})\left(\frac{15\text{ft}}{\text{min}}\right)(4\text{ft})}{\frac{33,000\text{ft} - \text{lb}}{\text{min}}} = 2.72\text{hp}$$

126. A ball and a roller bearing of the same size, are used to carry the same load of 300kgf. The rated capacity of the bearing are 2500kgf for the ball bearing and 3000kgf for the roller bearing. What is the rated life of each of the bearing, in number of revolutions?

Solution:

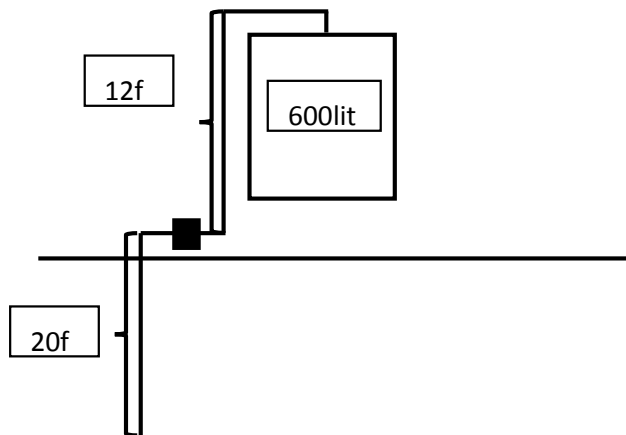
For ball bearing:

$$L_{BB} = \left(\frac{C}{F}\right)^n = \left(\frac{2500\text{kgf}}{350\text{kgf}}\right)^3 = 364.4 \times 10^6 \text{revolutions}$$

For roller bearing:

$$L_{RB} = \left(\frac{C}{F}\right)^n = \left(\frac{3,000\text{kgf}}{350\text{kgf}}\right)^{3.33} = 1279.60 \times 10^6 \text{revolutions}$$

127. A lift pump having a diameter of 4 inches and a stroke of 6 inches is used to lift water from a 20ft well and deliver it to a cylindrical tank at a height of 12ft. If the volumetric efficiency of the pump at 10 lifting strokes per minute is 90%, what is the pump capacity? Also compute for the power required to operate the pump manually if its mechanical efficiency is 80%? How long will it take the pump to fully fill the 600liters tank if its operating efficiency is 70%?



Solution:

$$C_{\text{pump}} = \left(\frac{\pi D^2 L}{4}\right)(\text{Eff}_{\text{Vol}})(S_{\text{Pump}}) = \left(\frac{\pi(10.16\text{cm})^2(15.24\text{cm})}{4}\right)(0.9)\left(\frac{10\text{stroke}}{\text{min}}\right) = \frac{11,120\text{cm}^3}{\text{min}}$$

$$= \left(\frac{11,120\text{cm}^3}{\text{min}}\right)\left(\frac{1\text{liter}}{1000\text{cm}^3}\right) = \frac{11.12\text{liter}}{\text{min}}$$

$$P_{\text{required}} = \left(\frac{11.12\text{liter}}{\text{min}}\right)\left(\frac{0.76\text{kg}}{\text{lit}}\right)(32\text{ft})\left(\frac{2.2\text{lb}}{1\text{kg}}\right) = \frac{\left(\frac{594\text{ft} - \text{lb}}{\text{min}}\right)}{\frac{33,000\text{ft} - \text{lb}}{\text{min}}} = 0.02\text{hp}$$

Time to fill the tank:

$$T = \frac{600\text{liters}}{\frac{11.12\text{lit}}{\text{min}}} = 53.96\text{min} \times 0.70 = 37.80\text{mins}$$

128. If the humidity ratios of the drying air and the ambient air are 0.002lb moisture/lb. dry air and 0.001lb moisture/lb. of dry air, what is the drying time of 2 tons of paddy if its moisture is to be reduced from 24% to 14%? Consider an airflow rate of 100m³ of air per minute cubic meter of grain and density of paddy of 55kg/m³.

Solution:

Difference in Humidity Ratio:

$$X = \frac{0.001\text{lbm}}{\text{lb.-DA}}$$

Weight of moisture after drying:

$$W = 2000\text{kg} - 2000\text{kg} \left(\frac{1-0.24}{1-0.14} \right) = 232.56\text{kg}$$

$$\text{Volume} = \frac{2,000\text{kg}}{\frac{559\text{kg}}{\text{m}^3}} = 3.64\text{m}^3$$

$$\text{Air - flow} = 3.64\text{m}^3 \left(\frac{100\text{m}^3\text{air}}{\text{min} - \text{m}^3\text{grain}} \right) = \frac{364\text{m}^3}{\text{min}} \quad (\text{air - flow in a volume of grain})$$

$$\text{Drying time} = (232.5\text{kg}) \left(\frac{2.2\text{lb}}{\text{kg}} \right) = \left(\frac{511.632\text{lb}}{0.001\text{lb} \frac{\text{moisture}}{\text{lb dry air}}} \right) \left(\frac{364\text{m}^3}{\text{min}} \right) \left(\frac{1.2\text{kg}}{\text{m}^3} \right) \left(\frac{2.2\text{lb}}{\text{kg}} \right)$$

$$\frac{532.42\text{min}}{60\text{min}} = 8.87\text{hrs}$$

129. An engine flat pulley is used to drive a vertical feed mixer input shaft at a speed of 350rpm. The engine pulley diameter is 4 inches and runs at a speed of 2000rpm. If the shaft center distance is 8ft, what is the length of the belt needed for the drive?

Solution:

$$L = 1.57(D_1 + D_2) + \sqrt{C^2 + D_1^2} + \sqrt{C^2 + D_2^2}$$

$$L = 1.57(4\text{in} + 32\text{in}) + \sqrt{(96\text{in})^2 + (4)^2} + \sqrt{(96)^2 + (32)^2}$$

$$L = 56.52\text{in} + 96.08\text{in} + 101.93\text{in} = 253.80 \text{ inches or } 21.15\text{ft}$$

130. A horizontal mixer is designed to incorporate soil and charcoal for composting laboratory experiment. The motor used drives series of belt and pulleys as follows: D1 = 4in; D2 = 12in; D3 = 2in; D4 = 12in; D5 = 2in and D6 = 12in. Determine the speed of the shafts in every transmission drive if the motor runs at a speed of 1720rpm.

Solution:

$$N_1D_1 = N_2D_2 \rightarrow N_2 = 573.33\text{rpm}$$

$$N_2D_3 = N_3D_4 \rightarrow N_3 = 95.56\text{rpm}$$

$$N_3D_5 = N_4D_6 \rightarrow N_4 = 15.93\text{rpm}$$

131. Compute the apparent air velocity in a 1ton capacity flatbed dryer having a dimension of 2.4mx2.4m square bin. The specific airflow rate of the dryer is 50m³ /min-m³. Assume a density of grain of 500kg/m³.

Solution:

$$V = \frac{1000\text{kg}}{\frac{500\text{kg}}{\text{m}^3}} = (2\text{m}^3) \left(\frac{0.50\text{m}^3}{\text{min} - \text{m}^3} \right) = \frac{100\text{m}^3}{2.4\text{m}^2} = \frac{17.36\text{m}}{\text{min}}$$

132. What is the annual rate of interest if ₦265 is earned in four months on an investment of ₦15,000?

Solution:

$$I = Pni$$

$$i = \frac{I}{pn} = \frac{₦265}{(₦15,000)(412)} = 5.3\%$$

133. A loan of ₦2,000 is made for a period of 13 months, from January 1 to January 31 the following year, at a simple interest of 20%. What is the future amount is due at the end of the loan period?

Solution:

$$F = P(1 + ni) = ₦2,000 \left(1 + \left(\frac{13}{12} \right) (0.20) \right) = ₦2,433.33$$

134. If you barrow money from your friend with a simple interest of 12%, find the present worth of ₦20,000, which is due at the end of nine months.

Solution:

$$F = P(1 + ni)$$

$$P = \frac{₦20,000}{1 + \left(\frac{9}{12} \right) (0.12)} = ₦18,348.82$$

135. Determine the exact simple interest on ₦5,000 for the period from January 15 to November 18, 1992. If the interest rate is 22%.

Solution:

$$I = Pni = (₦5,000) \left(\frac{318\text{days}}{366\text{days}} \right) (0.22) = ₦955.74$$

136. A man wishes his son to receive ₦200,000 ten years from now. What amount should he invest if it will earn interest of 10% compounded annually during the first 5 years and 12% compounded quarterly during the next 5 years?

Solution:

$$P_2 = ₦200,000(1 + 0.03)^{-20} = ₦110,735.15$$

$$P_1 = ₦110,735.15(1 + 0.1)^{-5} = P_0 = ₦68,758$$

137. By the condition of a will the sum of ₦25,000 is left tot held in trust by her guardian until it amounts ₦45,000. When will the girl receive the money if the fund is invested at 8% compounded quarterly?

Solution:

$$F = P(1 + i)^n$$

$$₦45,000 = ₦25,000(1 + 0.02)^n$$

$$\ln(1.8) = 4n\ln(1.02)$$

$$n = \frac{\ln(1.8)}{4\ln(1.02)} = 7.42\text{years}$$

138. A man barrowed ₦5,000 from a bank and agreed to pay the loan at the end of 9 months. The bank discounted the loan and gave him ₦4,000 in cash. (a.) what was the rate of discount? (b.) what was the rate of interest? (c.) what was the rate of interest for one year?

Solution:

$$a. D = \frac{P_{1000}}{P_{5000}} = 20\%$$

$$b. i = \frac{d}{1-d} = \frac{0.20}{1-0.20} = 25\%$$

$$c. i = \frac{I}{Pn} = \frac{P_{1,000}}{(P_{4,000})\left(\frac{9}{12}\right)} = 33.33\%$$

139. An item presently costs ₱1,000. If inflation is at the rate of 8% per year. What will be the cost of the item in two years?

Solution:

$$F = P(1 + f)^n = ₱1,000(1 + 0.08)^2 = ₱1,166.40$$

140. An economy is experiencing inflation at an annual rate of 8%. If this continues, what will ₱1,000 be worth two years from now in terms of today's pesos?

Solution:

$$F = \frac{p}{(1 + f)^n} = \frac{₱1,000}{(1 + 0.08)^2} = ₱857.34$$

141. What is the present worth of ₱500 deposited at the end of every three months for 6 years if the interest rate is 12% compounded semiannually?

Solution:

$$P = A \left(\frac{(1 + i)^n - 1}{i(1 + i)^n} \right)$$

$$P = ₱500 \left(\frac{(1 + 0.02956)^{24} - 1}{0.02956(1 + 0.02956)^{24}} \right) = ₱8,508$$

$$(1 + i)^4 = \left(1 + \left(\frac{0.12}{2} \right) \right)^2$$

$$i = 2.956\%$$

142. Sandy, a manufacturing engineer, just received a year-end bonus of \$10,000 that will be invested immediately. With the expectation of earning 8% per year, sandy hopes to take the entire amount out in exactly 20 years to pay for a family vacation when the oldest daughter is due to graduate from college. Find the amount of funds that will be available in 20 years.

Solution:

$$F = P(1 + i)^n = ($10,000)(1 + 0.08)^{20} = $46,610$$

143. How much money should you be willing to pay now for a guaranteed \$600 per year for 9 years starting next year, at a rate of return of 16% per year?

Solution:

$$P = \frac{A((1 + i)^n - 1)}{i(1 + i)^n} = \frac{\$600((1 + 0.16)^9 - 1)}{0.16(1 + 0.16)^9} = \$2,763.90$$

144. A certain interest rate compounded semiannually ₱5,000 will amount to ₱20,000 after 10 years. What is the amount of the end of 15 years?

Solution:

$$F_{10} = P_{10}(1 + i)^n$$

$$₱20,000 = ₱5,000(1 + \frac{i}{2})^{20}$$

$$i = 14.35\%$$

$$F_{15} = P_{15}(1 + \frac{i}{2})^{30} = ₱5,000 \left(1 + \left(\frac{0.1435}{2} \right) \right)^{30} = ₱39,973.74$$

145. A college student borrows \$1500 during his senior year. The loan is to be repaid in 20 equal quarterly installments. The interest rate is 5% per year with the first payment to be made 3 years after the date of the loan. What will be the approximate amount of the quarterly payment?

Solution:

$$F_3 = P_3(1+i)^n = (\$1,500)(1+0.05)^3 = \$1,736.44 = P_2$$

$$A = P_2 \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right) = \left(\frac{0.0123(1+0.0123)^{20}}{(1+0.0123)^{20} - 1} \right) = \$98.44$$

$$i_e = \left(1 + \frac{r}{m}\right)^m - 1$$

$$0.05 = \left(1 + \frac{r}{4}\right)^4 - 1; \rightarrow \frac{r}{4} = \text{quarterly} = 0.0123$$

146. Assume an effective interest rate of 15% per year compounded annually. An investment requires \$1500 at the end of each year for the next 5 years plus a final investment of \$3000 in 5 years. What is the equivalent lump sum investment now?

Solution:

$$P = A \left(\frac{P}{A}, 0.15, 5 \right) + F \left(\frac{P}{F}, 0.15, 5 \right) = (\$1500) \left(\frac{(1+0.15)^5 - 1}{(0.15)(1+0.15)^5} \right) + \frac{\$3,000}{(1+0.15)^5}$$

$$= \$5028.23 + \$1491.53$$

$$= \$6519.76$$

147. An investor is to receive an annuity of \$1000 per year for 15 years, with the first payment to be made on March 1, 2011. The investor offers to sell the annuity on March 1, 2008. With interest at 3.5% compounded annually, what is a fair price?

Solution:

March 1, 2011:

$$P = \$1,000 \left(\frac{(1+0.035)^{15} - 1}{(0.035)(1+0.035)^{15}} \right) = \$11,517.41$$

$$P = (\$11,517.41)(1+0.035)^{-2} = \$10,751.63$$

148. Jones Corporation borrowed ₱9,000 from Brown Corporation on Jan. 1, 1978 and ₱12,000 on Jan. 1, 1980. Jones Corporation made a partial payment of ₱7,000 on Jan. 1, 1981. It was agreed that the balance of the loan would be amortized by two payments one of Jan. 1, 1982 and the other on Jan. 1, 1983, the second being 50% larger than the first. If the interest rate is 12%. What is the amount of each payment?

Solution:

$$₱9,000(1+0.12)^5 + ₱12,000(1+0.12)^3 = ₱7,000(1+0.12)^2 + X_1(1+0.12) + 1.5X_1$$

$$X_1 = \frac{₱23,939.41}{2.62} = ₱9,137.18$$

$$X_2 = ₱9,137.18(1.5) = ₱13,705.77$$

149. A woman borrowed ₱3,000 to be paid after 112 years with interest at 12% compounded semiannually and ₱5,000 to be paid after 3 years at 12% compounded monthly. What single

payment must she pay after 312 years at an interest rate of 16% compounded quarterly to settle the two obligations?

Solution:

For ₱3,000:

$$F_1 = P(1 + \frac{i_1}{2})^{2n_1} = ₱3,000(1 + \frac{0.12}{2})^3 = ₱3,573.05$$

For ₱5,000:

$$F_2 = P(1 + \frac{i_2}{2})^{2n_2} = ₱3,000(1 + \frac{0.12}{2})^{36} = ₱7,153.84$$

For single payment after 3 ½ years:

$$\begin{aligned} F_3 &= F_1(1 + \frac{i_3}{2})^{4(n_2 - n_1)} + F_2(1 + \frac{i_3}{4})^{4(n_2 - n_1)} \\ F_3 &= (₱3,573.05) \left(1 + \frac{0.16}{4}\right)^{4(3\frac{1}{2} - 1\frac{1}{2})} + (₱7,153.84) \left(1 + \frac{0.16}{4}\right)^{4(3\frac{1}{2} - 3)} \\ &= ₱4,889.96 + ₱7,737.59 \\ &= ₱12,627.55 \end{aligned}$$

150. Mr. J. de la Cruz borrowed money from a bank. He received from the bank P1, 342 and promises to repay P1, 500 at the end of 9 months. Determine the simple interest rate and the corresponding discount rate or often referred to as the "Banker's discount."

Solution:

$$d = \frac{\text{discount}}{\text{principal}} = \frac{158}{1500} = 10.53\%$$

$$i = \frac{d}{1 - d} = \frac{0.1053}{1 - 0.1053} = 11.77\%$$

151. A man deposits P50, 000 in a bank account at 6% compounded monthly for 5 years. If the inflation rate of 6.5% per year continues for this period, will this effectively protect the purchasing power of the original principal?

Solution:

$$i_e = (1 + \frac{r}{m})^m = (1 + \frac{0.06}{12})^{12} = 6.1678\%$$

$$F = P(\frac{1+i}{1+f})^n = (₱50,000)(\frac{1+0.061678}{1+0.065})^5 = ₱49,225$$

152. What is the future worth of P600 deposited at the end of every month for 4 years if the interest rate is 12% compounded quarterly?

Solution:

$$(1+i)^{12} - 1 = \left(1 + \frac{r}{4}\right)^4 - 1 = (1+i)^{12} - 1 = \left(1 + \frac{0.12}{4}\right)^4 - 1$$

$$i = 0.99\%$$

$$F = A \left(\frac{(1+i)^n - 1}{i} \right) = \frac{(₱600)((1+0.0099)^{48} - 1)}{0.0099} = ₱36,641.32$$

153. Mr. Reyes borrows P600, 000 at 12% compounded annually, agreeing to repay the loan in 15 equal annual payments. How much of the original principal is still unpaid after he has made the 8th payment?

Solution:

$$P = A \left(\frac{1 - (1 + i)^n}{i} \right)$$

$$₱600,000 = A \left(\frac{1 - (1 + 0.12)^{-15}}{0.12} \right)$$

$$A = ₱88,094.54$$

$$P = A \left(\frac{1 - (1 + i)^n}{i} \right) = (₱88,094.54) \left(\frac{1 - (1 + 0.12)^{-7}}{0.12} \right) = ₱402,042.033$$

154. A ₱10,000 loan is to be repaid at the rate of ₱200 per month, with an annual effective interest rate of 19.56% change against the unpaid balance. What principal remains to be paid after the third payment?

Solution:

$$0.1956 = \left(1 + \frac{r}{12}\right)^{12} - 1; \frac{r}{12} = 0.01499 \text{ or } 1.499\%$$

$$F = (₱200) \left(\frac{(1 + 0.01499)^3 - 1}{0.01499} \right) = ₱609.04$$

$$F_t = (₱10,000)(1 + 0.01499)^3 = ₱10,456.47$$

$$X = ₱10,456.47 - ₱609.04 = ₱9847.43$$

155. Find the nominal rate which if converted quarterly could be used instead of 12% compounded monthly. What is the corresponding effective rate?

Solution:

$$\left(1 + \frac{0.12}{12}\right)^{12} - 1 = \left(1 + \frac{r}{4}\right)^4 - 1$$

$$\frac{r}{4} = 0.0303 \text{ or } 3.03\% \text{ quarterly, nominal}$$

$$r = 12.12\% \text{ annual, quarterly}$$

156. Find the amount at the end of 2 years and 7 months if ₱1,000 invested at 8% compounded quarterly using simple interest for anytime less than a year interest period.

Solution:

$$F_1 = (₱1,000)(1 + 0.02)^8 = ₱1,171.66$$

$$F_2 = (₱1,171.66) \left(1 + \left(\frac{7}{12}\right)(0.08)\right) = ₱1,226.34$$

157. A ₱2,000 loan was originally made at 8% simple interest for 4 years. At the end of this period, the loan was extended for 3 years. Without the interest being paid. But the new interest rate was made 10% compounded semiannually. How much should be the borrower pay at the end of 7 years?

Solution:

$$F_4 = P(1 + ni) = ₱2,000(1 + (4)(0.08)) = ₱2,640$$

$$F_7 = ₱2,640(1 + 0.05)^6 = ₱3,537.86$$

158. A man bought a lot worth ₱1,000,000 if paid in cash. On the installment basis, he paid a down payment of ₱200,000, ₱300,000 at the end of the year. ₱400,000 the end of three years and a final payment at the end of five years. What was the final payment if interest was 20%?

Solution:

$$₱800,000 = ₱300,000(1 + 0.2)^{-1} + ₱400,000(1 + 0.2)^{-3} + Q(1 + 0.2)^{-5}$$

$$Q = ₱792,531.77$$

159. A man invested ₱10,000 at an interest rate of 10% compounded annually. What will be the final amount of his investment, in term of today's pesos, after five years? If inflation remains the same at the rate of 8% per year?

Solution:

$$F = \frac{P(1+i)^n}{(1+f)^n} = \frac{₱10,000(1+0.1)^5}{(1+0.08)^5} = ₱10,960.86$$

160. What are the present worth and the accumulated amount of a 10-year annuity of ₱10,000 at the end of each year, with interest at 15% compounded annually?

Solution:

$$F = A \left(\frac{(1+i)^n - 1}{i} \right) = ₱10,000 \left(\frac{(1+0.15)^{10} - 1}{0.15} \right) = ₱203,037.18$$

161. Assume effective interest rate of 15% per year compounded. An investment requires \$1,500 at the end of each year for the next 5 years plus a final invest met of \$3,000 in 5 years. What is the equivalent lump sum investment now?

Solution:

$$P = A \left(\frac{(1+i)^5 - 1}{i(1+i)^5} \right) + \$3,000(1+i)^{-5}$$

$$P = \$1,500 \left(\frac{(1+0.15)^5 - 1}{i(1+0.15)^5} \right) + ₱3,000(1+0.15)^{-5} = \$6519.76$$

162. The fixed cot incurred by manufacturing plant are \$200,000 per year. Variable cost are 60% of the annual sales. If annual sales are \$300,000, the annual profit/loss is?

Solution:

$$X = \$300,000 - (\$200,000 + 0.6(\$300,000)) = -\$80,000 \text{ loss}$$

163. I will invest \$500 per quarter for my retirement at 7.3% compounding quarterly for 32 years. I have a choice of making that payment of \$500 at the beginning or the end of the year. In which account will I have more money and by how much? Which account will earn the most interest and by how much?

Solution:

Regular annuity:

$$F = A \left(\frac{(1+i)^{nt} - 1}{i} \right) = \$500 \left(\frac{(1+0.0734)^{4(32)} - 1}{\frac{0.073}{4}} \right) = \$249,981.2$$

$$X_1 = \$249,981.2 - \$500(128) = \$185,981.20$$

Regular annuity:

$$F = A \left(\frac{(1+i)^{n+1} - 1}{i} \right) - \$500 = \left(\frac{(1+0.0734)^{128+1} - 1}{0.0734} \right) - \$500 = \$254,543.36$$

$$X_2 = 190,543.36$$

Answer: the end of each year

164. You are interested in buying a farm worth of ₱80,000. Your available cash is only ₱25,000 to pay for the down payment. The remaining balance will be amortized over a period of 18 months at an interest of 24% compounded monthly. What is the amount of monthly payment?

Solution:

$$₱80,000 - ₱25,000 = ₱55,000$$

$$\left(1 + \frac{0.24}{18}\right)^{18} - 1 = \left(1 + \frac{r}{12}\right)^{12} - 1$$

$$\frac{r}{12} = 2\%$$

$$A = P \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right) = ₱55,000 \left(\frac{(0.02)(1+0.02)^{18}}{(1+0.02)^{18} - 1} \right) = ₱3,668.62$$

165. Mang Andoy secured a loan from a bank in the amount of ₱250,000 at an interest rate of 15% compounded quarterly. The loan will be paid back in 36 equal monthly installments. What is the closest to Mang Andoy's monthly payment?

Solution:

$$(1+i)^n - 1 = \left(1 + \frac{r}{n}\right)^n - 1$$

$$\left(1 + \frac{0.15}{36}\right)^{36} - 1 = \left(1 + \frac{r}{12}\right)^{12} - 1$$

$$\frac{r}{12} = 0.01255 \text{ or } 1.255\%$$

$$A = P \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right) = ₱250,000 \left(\frac{(0.01255)(1+0.01255)^{36}}{(1+0.01255)^{36} - 1} \right) = ₱8,673.68$$

166. If you invest ₱20,000 now in a business venture that promises to return ₱39,479 in 6 years, how many years would it take to get a return of at least ₱30,000?

Solution:

$$F = P(1+i)^n$$

$$₱39,479 = ₱20,000(1+i)^6$$

$$i = 12\%$$

$$₱30,000 = ₱20,000(1+0.12)^t$$

$$\ln(1.5) = t \ln(1.12)$$

$$t = 4 \text{ years}$$

167. The monthly interest equivalent to an effective semi-annual rate of 4% is?

Solution:

$$(1+0.04)^2 = (1+i)^{12}$$

$$i = 0.656\%$$

168. What is the annual effective rate for a rate advertised as 18% computed monthly?

Solution:

$$i = \left(1 + \frac{0.18}{12}\right)^{12} - 1 = 19.56\%$$

169. A new machine cost ₱76,000. How much monthly payments be to finance it at a nominal rate 18% per year for 30 months? Consider that the rate is the equivalent of monthly compounding.

Solution:

$$(1+0.1830)^{30} = \left(1 + \frac{r}{12}\right)^{12}$$

$$\frac{r}{12} = 1.5\%$$

$$A = P \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right) = (₱76,000) \left(\frac{(0.015)(1+0.015)^{30}}{(1+0.015)^{30} - 1} \right) = ₱3,165$$

170. The value of the interest factor needed to find a series of equal revenues that must be received every year for 12 years to realize at return of 25% from an initial investment of ₱25M is?

Solution:

$$X = \left(\frac{i(1+i)^n}{(1+i)^n - 1} \right) = \frac{(0.25)(1+.25)^{12}}{(1+0.25)^{12} - 1} = 0.268$$

171. Engr. Carl Vincent Ramos is due to pay ₱50,000 at the end of the year and ₱150,000 five years from today. However, he was able to pay ₱10,000 today and he plan to pay the balance at the end of the three years if money is worth 10% per annum. His payment is closest to?

Solution:

$$F = ₱150,000(1.1)^{-2} - ₱60,000(1.1)^2 = ₱51,366.94$$

172. A savings loan offers a 5.25% per annum compounded daily. What is the effective annual rate?

Solution:

$$\left(1 + \frac{0.0525}{12}\right)^{12} = (1 + i)$$

$$i = 5.38\%$$

Or

$$\left(1 + \frac{0.0525}{365}\right)^{365} = (1 + i)$$

173. A new Agricultural Engineer gets loan of ₱10,000 today at an interest rate of 15% and the first payment of ₱3,000 is made four years later. The amount that is still due on the loan after the first payment is most nearly

Solution:

$$X = ₱10,000(1.15)^4 - ₱3,000 = ₱14,490 \text{ or } ₱14,500$$

174. With an interest rate of 8% compounded semi-annually, the value of ₱75,000 investment after 5 years is most nearly:

Solution:

$$F = P(1 + i)^n = ₱75,000\left(1 + \frac{0.08}{2}\right)^{10} = ₱111,018.32$$

175. If you invest ₱50,000 today at 9% interest, how much time is required for the investment to be worth ₱125,000?

Solution:

$$₱125,000 = ₱50,000(1 + 0.09)^t$$

$$t = 10.6 \text{ or } 11 \text{ years}$$

176. If the annual cost of lining now for a farmer's household consisting of 5 members is ₱180,000 per year, how much will it cost ten years from now if inflation rate is 4.5%?

Solution:

$$X = ₱180,000(1 + 0.045)^{10} = ₱279,534.50$$

177. The simple interest that results when ₱1,000 accumulates to ₱1250 in 3 years is?

Solution:

$$F = P(1 + ni)$$

$$i = 8.33\%$$

178. The interest earned if ₱1,000 is invested for 1 year at 6% compounded quarterly is?

Solution:

$$F = P(1 + i)^n = \text{P}1,000\left(1 + \frac{0.06}{4}\right)^4 = \text{P}61.36$$

179. A loan of ₱10,000 is to be amortized through 5 equal semi-annual payments, which the first payment to be made after 3 years. What is the semi-annual payment if interest rate was 20% compounded semi-annually?

Solution:

$$i = (1 + 0.20)^2 - 1 = 21\%$$

$$F = \text{P}10,000(1 + 0.21)^5 = \text{P}25,937.42$$

Semi-annual:

$$i = (1 + 0.22) = 10\%$$

$$A = \text{P}25,937.42 \left(\frac{1.2}{(1.10)^5 - 1} \right) = \text{P}4,248.48$$

180. In buying a machine worth ₱30,000, the farmer pays ₱10,000 cash and agrees to pay in balance, including interest at 5.5% compounded semiannually, by series of 8 equal semi-annual payments. The first due at the end of 4 years. Find the semi-annual payment.

Solution:

$$i = \left(1 + \frac{0.055}{2}\right)^2 - 1 = 0.05576$$

$$F = \text{P}30,000(1 + 0.05576)^{7.5} - \text{P}10,000(1 + 0.05576)^{7.5} = \text{P}30,044.78$$

$$A = \text{P}30,044.78 \left(\frac{0.0275}{(1 + 0.0275)^8 - 1} \right) = \text{P}3,408.82$$

181. Suppose a farm laborer deposits ₱500 in a savings account at the end of each year for the next 12 years. If the bank pays an interest of 8% per year, compounded annually, how much money will be accumulated by the end of the 12 year period?

Solution:

$$F = A \left(\frac{(1 + i)^n - 1}{i} \right) = \text{P}500 \left(\frac{(1 + 0.08)^{12} - 1}{0.08} \right) = \text{P}9,489$$

182. A farmer lends ₱10,000 for 4 years at 12% simple interest. At the end of the 4 years, he invests the entire amount, which he then has, for 10 years at 8% per year compounded annually. How much money will he have at the end of the 14 year period?

Solution:

$$F = P(1 + ni) = \text{P}10,000(1 + (4)(0.12)) = \text{P}14,800$$

$$F = \text{P}14,800(1 + 0.08)^{10} = \text{P}31,952$$

183. How many years will be required for an investment of ₱3,000 to increase to ₱4,081.47 at an interest of 8% per year compounded annually?

Solution:

$$\text{P}4,081.47 = \text{P}3,000(1.08)^n$$

$$\ln(1.36049) = n \ln(1.08)$$

$$n = 4 \text{ years}$$

184. On the first day of the year, a farmer deposits ₱10,000 in a bank at 8% per year, compounded annually. He withdraws ₱800 at the end of the 1st year, ₱900 at the end of the 2nd year, and the remaining balance at the end of the 3rd year. How much does he withdraw at the end of the 3rd year?

Solution:

$$P = (\text{P}800)(1 + 0.08)^{-1} + \text{P}900(1 + 0.08)^{-2} = \text{P}1,512.34$$

$$₱10,000 - ₱1,512.34 = ₱8,487.66$$

$$F_{3rd} = ₱8,487.66(1 + 0.08)^3 = ₱10,692$$

185. A diesel engine operating at 2500rpm will be used to drive an irrigation pump operating at 1000rpm. The pulley diameter for the pump is 4 inches. What is the recommended size of pulley for the engine?

Solution:

$$N_1D_1 = N_2D_2$$

$$D_2 = 1.6\text{in}$$

186. Determine the length of the flat-belt pulley drive having the following data: diameter of first pulley is 300mm, diameter of the 2nd pulley is 400mm and the center distance of the pulley shaft is 10m.

Solution:

$$L = 2C + 1.57(D_1 + D_2) + \frac{(D_1 - D_2)^2}{4C}$$

$$= 2(10\text{m}) + 1.57(0.3\text{m} + 0.4\text{m}) + \frac{(0.3 - 0.4)^2}{4(10\text{m})} = 21.099\text{m}$$

187. The cylinder of a corn Sheller will be driven by a 10hp diesel engine using v-belt drive. The engine has speed of 1800rpm while the corn Sheller requires an engine speed of 650rpm. The effective diameter of the driver pulley is 12 inches. What should be the diameter of the driven pulley?

Solution:

$$N_1D_1 = N_2D_2;$$

$$D_2 = 33.23\text{in}$$

188. What is the required length of the V-belt if the center C is 60 inches? Refer to problem #187.

Solution:

$$L = 2C + 1.57(D_1 + D_2) + \frac{(D_1 - D_2)^2}{4C}$$

$$L = 2(60\text{in}) + 1.57(12\text{in} + 33.23\text{in}) + \frac{(12 - 33.23)^2}{4(60)} = 192.50 \text{ inches}$$

189. If a belt has a capacity rating of 4.5hp per belt, approximately how many belts are recommended for the drive? Refer to problem #187.

Solution:

$$X = \frac{10\text{hp}}{4.5\text{hp}} = 2.2\text{hp use } 3\text{hp}$$

190. What is the recommended shaft diameter? Stress = 6000psi. Refer to problem #187.

Solution:

$$T = \frac{P}{2\pi N} = \frac{(10\text{hp}) \left(33,000\text{ft} - \frac{\text{lb}}{\text{min}} \right) (12\text{in})}{2\pi(1800\text{rpm})} = 350\text{lb} - \text{in}$$

$$D = \sqrt[3]{\frac{16T}{\pi\sigma}} = \sqrt[3]{\frac{16(350\text{lb} - \text{in})}{\pi\left(\frac{6000\text{lb}}{\text{in}^2}\right)}} = 0.67\text{in}$$

191. Compute the torque acting on a pulley rotating at 1800 rpm and delivering 10-kW power. If the pulley used in transmitting power that runs at 1800 rpm has a diameter of 8 in., what is the

force transmitted by the belt? Also, compute the actual load applied to the pulley shaft. If 6200 single-row deep-groove ball bearing will be used to drive the pulley, what would be the rating fatigue life of the bearing? What is the rated life, in hours, of the bearing?

Required:

Torque Acting on the Pulley, Belt Force, Actual Load at the Pulley Shaft,

Rated Fatigue Life of the Bearing, Rated Life of the Bearing

Solution:

$$T = \frac{974000P}{N} = \frac{974000(10\text{kW})}{1800\text{rpm}} = 5411.11\text{kgf} - \text{mm}$$

$$F_b = \frac{5411\text{kgf} - \text{mm}}{(4\text{in})(25.4\text{mm})} = 53.26\text{kgf}$$

$$F_a = f_b F_b = (2.25)(53.26\text{kgf}) = 119.83\text{kgf}$$

$$L_{\text{fatigue}} = \left(\frac{C}{P}\right)^3 = \left(\frac{400\text{kgf}}{119.83\text{kgf}}\right)^3 = 37.20 \times 10^6 \text{rev}$$

$$L_{\text{rated}} = \left(\frac{10^6}{60n}\right) \left(\frac{C}{F}\right)^3 = (10660(1800\text{rpm})) \left(\frac{400\text{kgf}}{119.83\text{kgf}}\right)^3 = 344.40\text{hrs}$$

192. Recommend a shaft size required for the main transmission shaft of a rice mill's prime mover with the following specifications: Engine, Isuzu 8 PA; Hp rated, 225Hp; and Engine Speed, 3600 rpm. Use 6,000-psi design stress for the shaft. Recommend a square-key size for the calculated shaft. Assume a ½" square size of the key.

Solution:

$$T = \frac{P}{2\pi N} = \frac{(225\text{hp}) \left(33,000\text{ft} \cdot \frac{\text{lb}}{\text{min}}\right) (12\text{in})}{2\pi(3600\text{rpm})} = 3939\text{lb} - \text{in}$$

$$D = \sqrt[3]{\frac{16T}{\pi\sigma}} = \sqrt[3]{\frac{16(3939\text{lb} - \text{in})}{\pi(6,000\text{psi})}} = 1.50\text{in use } 2\text{in}$$

$$F_s = \frac{3939\text{lb} - \text{in}}{1\text{in}} = 3939\text{lb}$$

$$\text{Length}_{\text{key}} = \frac{F}{\sigma_{\text{allow}} W} = \frac{3939\text{lb}}{(25,000\text{psi})(0.5\text{in})} = 0.315\text{in}$$

193. A shaft delivers 10lbs at 10-cm crank radius. What is the torque delivered by the shaft? If the said shaft rotates at 80 rpm, what is its power delivered?

Solution:

$$T = (10\text{lb})(10\text{cm}) = 100\text{lb} - \text{cm or } 45.45\text{kgf} - \text{cm or } 3.28\text{ft} - \text{lb}$$

$$P = 2\pi TN = \frac{2\pi(3.28\text{ft} - \text{lb})(80\text{rpm})}{33,000\text{ft} - \frac{\text{lb}}{\text{min}}} = 0.05\text{hp}$$

194. In a 200-hectare corn project, the field has to be tilled within 30 days. What size of a heavy-duty offset-disk harrow should be selected if the tractor to be used will be operated at 5-kph speed? Assume a 75% field efficiency and an 8-hour working day.

Solution:

$$\frac{200\text{ha}}{(30\text{days}) \left(\frac{8\text{hrs}}{\text{day}}\right)} = \frac{(5\text{kph})(0.75)W}{10};$$

$$W = 2.22\text{m}$$

195. In a field demonstration of a rice reaper, with 1.2-m cutter bar, performance showed that the average travel speed of the machine is 3kph. The average width of cutter bar actually utilized is 1.0 m. Time lost in crossing the dike is 5 min/ha and 10 min/ha for repair. Turning and other idle time is 10% of the effective operating time while removing field obstruction and clogging is 15 min/ha. What is the theoretical and effective field capacity of the machine? What is the time lost in turning and other idle time per hectare of the reaper?

Solution:

$$TFC = \frac{SW}{10} = \frac{(3kph)(1.2m)}{10} = \frac{0.36ha}{hr}$$

$$EFC = \frac{SWEff}{10} = \frac{(3kph)(1m)}{10} = \frac{0.30ha}{hr}$$

$$t_{lost} = \frac{10.30ha}{hr} = \left(\frac{3.33hr}{ha} \right) \left(\frac{60min}{1hr} \right) (0.1) = \frac{19.98min}{ha} \text{ or } \frac{20min}{ha}$$

196. One-thousand gallons of water is to be pumped from a river per minute (1000gpm) to a farm 150 feet above the river. What is the required horsepower to drive the pump, considering a 70% pump efficiency and 20% friction losses and static head?

Given:

Pumping rate - 1000gpm

Pumping head - 150ft

Pump efficiency - 70%

Friction and other loss - 20% of the static head

Solution:

$$P = \frac{\gamma QTDH}{Eff} \left(Q = \frac{m^3}{sec} \right) \text{ or } \frac{QTDH}{3960Eff} (Q = gpm)$$

$$P = \frac{\left(\frac{62.4lb}{ft^3} \right) \left(\frac{133.68ft^3}{min} \right) (150ft \times 1.20)}{(0.70) \left(33,000ft - \frac{lb}{min} \right)} = 64.90hp$$

197. If a diesel engine will be used to drive the pump in Problem #196, what is the recommended horsepower for the engine? Engine efficiency is 80%.

Solution:

$$P = \frac{64.90hp}{0.8} = 81.125hp$$

198. If a gasoline engine is used in problem #196 what is the recommended horsepower for the engine? Engine efficiency is 70%

Solution:

$$P = \frac{64.90hp}{0.7} = 92.70hp$$

199. A farmer wishes to spray his vegetable farm at a rate of 300 gallons per hectare. A 16-nozzles, 18-inches apart, sprayer will be used. The forward speed of the tractor is at 5kph. Compute the field capacity of the sprayer in hectare per hour. What is the required pump capacity for the sprayer? Also, what is the capacity per nozzle?

Given:

Spraying rate - 300 gal per ha

Number of nozzles - 16

Nozzle spacing - 18in.

Tractor speed - 5kph

Solution:

$$FC = \frac{SWEff}{10} = \frac{(5kph)(18inx0.0254mx16)}{10} = \frac{3.66ha}{hr}$$

$$Q_p = \left(\frac{3.66ha}{hr}\right)\left(\frac{300gal}{ha}\right)\left(\frac{1hr}{60min}\right) = 18.3gpm$$

$$Q_n = \frac{18.3gpm}{16} = 1.14gpm$$

200. The specific fuel consumption of a tractor is 0.3lbs/BHP-hr. If the heating value is 19,500BTU/lb., what is the Brake Thermal Efficiency?

Solution:

$$BTE = \left(\frac{19,500BTU}{lb}\right)\left(\frac{1hp-hr}{2545BTU}\right) = \frac{\left(\frac{7.662hp-hr}{lb}\right)}{\frac{3.33BHP-hr}{lb}} = \frac{\left(\frac{3.33BHP-hr}{lb}\right)}{\left(\frac{7.662hp-hr}{lb}\right)} = 43.46\%$$

201. A corn plantation requires a population of 37,000 plants/ha. The average emergence is 90% and the row spacing is 1.2 meters. If the seeds are to be drilled. What should be the seed spacing?

Solution:

Number of hills:

$$\left(\frac{\frac{37,000plants}{ha}}{0.9}\right) = 41,111.11Plants/ha$$

Seeds spacing:

$$X = \left(\frac{41,111plants}{ha}\right)\left(\frac{1ha}{10,000m^2}\right) = \frac{4.11plants}{m^2} \text{ or } \frac{0.24324m^2}{plant} = \frac{\frac{0.24324m^2}{plant}}{1.2m} = 0.2m$$

202. A tractor drive wheel under load rotates at 1200 revolution passing a newly cultivated field. At no land, the tractor drive wheel is able to rotate at 1,125 revolutions. What is the percentage slippage of the tractor drive wheel?

Solution:

$$\% \text{ slippage} = \frac{1200 - 1125}{1125} = 6.67\%$$

203. What is the application rate of a granular broad cast spreader required to apply 1ton of manure at a distance of 1000m. The machine swath is 2 meters.

Solution:

$$Q = \left(\frac{1,000kg}{(1000m)(2m)}\right)\left(\frac{10,000m^2}{1ha}\right) = \frac{5000kg}{ha}$$

204. Assume that you are the Agricultural Engineer of a 500-hectare corn farm with soil type that gives 10 psi draft when plowing at a speed of 4 km/hr. Compute the drawbar horsepower required to pull a three-disk plow with effective cut of 12 inches per disk at a plowing depth of 6 inches.

Solution:

$$DHP = \frac{\left(\frac{10lb}{in^2}\right)(12in)(6in)(3disk)\left(\frac{4km}{hr}\right)\left(\frac{1kg}{2.2lb}\right)}{274} = 14.33hp$$

205. Referring to Problem #204, how many hectares does a tractor pulling a three-disk plow would be able to plow in one day (16 hours operation) assuming a total time loss of 30%?

Solution:

$$EFC = \frac{SWEff}{10} = \frac{(4kph)(3)(12 \times 0.0254m)(0.7)}{10} = \left(\frac{0.256ha}{hr}\right) \left(\frac{16hrs}{day}\right) = 4.096ha/day$$

206. Referring to Problem #204, how many days are required to plow the 500-hectare farm using ten units of the same kind of tractor and of plow?

Solution:

$$X = \frac{500ha}{\left(\frac{4.096ha}{day}\right)(10)} = 12.2days$$

207. Still referring to Problem 204, what is the amount of fuel consumed in liters per 16 hours day of one tractor unit if the specific fuel consumption is 1lb/dhp-hr. and the specific gravity of the fuel is 0.8?

Solution:

$$FCR = \frac{\left(\frac{1lb}{dhp-hr}\right)(14.33hp)}{(0.8)\left(\frac{62.4lb}{ft^3}\right)} = \left(\frac{0.287ft^3}{hr}\right) \left(\frac{1m}{3.28ft}\right)^3 \left(\frac{100cm}{1m}\right)^3 \left(\frac{1lit}{1000cm^3}\right) = \frac{8.13lit}{hr}$$

$$V_{fuel} = \left(\frac{8.13lit}{hr}\right) \left(\frac{16hrs}{day}\right)(12.2days) = 1586liters$$

208. What is the cutting width of the single-disk harrow set at 18-degree working angle containing 20 disks spaced at 30 cm? Disk diameter is 40 cm.

Solution:

$$W = 0.95(20disks)(30cm) + (0.3)(40) = 5.82m$$

209. If the harrow in Problem #208 is a tandem-disk harrow, what is the cutting width of the machine?

Solution:

$$W = 0.95NS + 1.2D = 0.95(20disks)(30cm) + 1.2(40) = 6.18m$$

210. A 30-disk double-offset-disk harrow is travelling at a speed of 10kph. If the disk diameter is 40 cm spaced at 30cm, what is the cutting width of the harrow? What is its theoretical field capacity?

Solution:

$$W = 0.95NS + 0.6D = 0.95(30disks)(30cm) + 0.6(40) = 8.79m$$

$$TFC = \frac{SW}{10} = \frac{(10kph)(8.79m)}{10} = \frac{8.79ha}{hr}$$

211. Compute the number of plants required for a 2-hectare corn farm with the following seed spacing: Plant spacing, 0.25 m and row spacing, 0.45 m.

Solution:

$$N_P = \frac{A_T}{RS \times PS} = \frac{20,000m^2}{(0.25m)(0.45m)} = 177,778 \text{ plants Or hills}$$

212. The axle of a power tiller was tested on a Pony-brake dynamometer. The force delivered at 1-m brake arm was 300 kg at a speed of 25 rpm. Determine the torque and the shaft power of the machine. Also, compute the specific fuel consumption of the engine driving the power tiller if it consumes diesel at a rate of 0.9 liters per hour.

Solution:

$$T = (300kg)(1m) = 300kgf - m$$

$$P = \frac{FN}{1340} = \frac{(300\text{kg})(25\text{rpm})}{1340} = 5.597\text{kW or } 7.50\text{hp}$$

$$\text{SFC} = \left(\frac{0.9\text{liter}}{\text{hr}}\right)\left(\frac{0.84\text{kg}}{\text{lit}}\right) = \frac{0.756\text{kg}}{\text{hr}} = \left(\frac{0.0734\text{kg}}{\text{kW} - \text{hr}}\right) = \frac{0.135\text{kg}}{\text{kW} - \text{hr}} \text{ or } \frac{135\text{g}}{\text{kW} - \text{hr}}$$

Note: this problem is from Engr. Belonio's reviewer. The formula $\frac{FN}{1340}$ is supposed to give a power directly in kW not hp. This is according to his formula. The given answer in the book is 5.597hp. So, it might be good to change it to 5.597kW. This formula can only be used in a power tiller. If you try to use the formula $P = 2\pi TN$, it will not give the same answer as the formula above.

213. At 18-degrees working angle, a tandem- disk harrow is used to pulverize a newly plowed farm at a rate speed of 10 km/hour. Compute the effective width of the implement, the theoretical field capacity, and the effective field capacity of the harrow. Assume a field efficiency of 90%. The harrow has 24 pieces of 30-cm disk spaced at 20 cm.

Given:

Harrow = tandem
 Work angle = 18deg
 Speed = 10kph
 Field Eff. = 90%
 No. Disks = 24 units
 Disk Diameter = 30 cm
 Spacing = 20 cm

Required: Effective Width, Theoretical Field Capacity, and Effective Field Capacity.

Solution:

$$W_e = 0.95NS + 1.2D = 0.95(24)(20\text{cm}) + 1.2(30\text{cm}) = 492\text{cm}$$

$$\text{TFC} = \frac{SW}{10} = \frac{(10\text{kph})(4.92\text{m})}{10} = \frac{4.92\text{ha}}{\text{hr}}$$

$$\text{EFC} = \frac{\text{SWEff}}{10} = \frac{(10\text{kph})(4.92\text{m})(0.9)}{10} = \frac{4.428\text{ha}}{\text{hr}}$$

214. How many grams of cotton seeds are required per row for a hectare of farm, assuming a seeder is to be used in planting the seeds at 0.20-m row spacing and 0.50-m plant spacing? The seeding rate is 10 kg/ha.

Given:

Crop - cotton
 Seeding rate - 10 kg/ha
 Row spacing - 0.5 m
 Plant spacing - 0.20

Solution:

$$X = \left(\frac{10\text{kg}}{\text{ha}}\right)(0.20)(0.50\text{m})\left(\frac{1\text{ha}}{1000\text{m}^2}\right) = 0.001\text{kg or } 1\text{g}$$

215. Walking in the rice field operating the sprayer with a spray swath of 1 m, a person covers 20 m/min. If the nozzle discharge is 0.4 liter/min, what is the application rate per hectare?

Given:

Swath - 1 m

Walking speed - 20 m/min

Nozzle discharge - 0.4 liter/min

Solution:

$$Q = (1\text{m}) \left(\frac{20\text{m}}{\text{min}} \right) = \left(\frac{20\text{m}^2}{\text{min}} \right) = \frac{\left(\frac{0.4\text{liter}}{\text{min}} \right)}{\frac{20\text{m}^2}{\text{min}}} = \frac{0.02\text{lit}}{\text{m}^2} \times \frac{10,000\text{m}^2}{1\text{ha}} = \frac{200\text{lit}}{\text{ha}}$$

216. A 120-hectare farm is to be planted with corn. The row and hill spacing are 50 cm and 40 cm, respectively. How many hills are to be planted in the farm? If a 12-row drill planter is to be used at 10 kph implement speed, what is the theoretical and effective field capacities of the machine? Assume a field efficiency of 80%. If the field requires 3 seeds per hill and has a percentage emergence of 90%, how many kilos of seeds are needed for the entire area? Assume 0.1 g per seed.

Given:

Area - 120 hectares

Row spacing - 50 cm

Hill spacing - 40 cm

Machine - 12- row seeder/planter

Implement Speed - 10 kph

Field efficiency - 80%

% Emergence - 90%

No. of Seeds/hill - 3 seed

Wt. per seed - 0.1 g / seed

Solution:

$$\# \text{ of hills} = \frac{1,200,000\text{m}^2}{(0.5\text{m})(0.4\text{m})} = 6,000,000 \text{ hills}$$

$$\text{TFC} = \frac{\text{SW}}{10} = \frac{(10\text{kph})(12)(0.5\text{m})}{10} = \frac{6\text{ha}}{\text{hr}}$$

$$\text{EFC} = \frac{\text{SWEff}}{10} = \frac{(10\text{kph})(12)(0.5\text{m})(0.8)}{10} = \frac{4.8\text{ha}}{\text{hr}}$$

$$X = (6,000,000\text{hills}) \left(\frac{3\text{seed}}{\text{shil}} \right) \left(\frac{0.1\text{g}}{\text{seed}} \right) = \frac{1800\text{kg}}{0.9} = 2000\text{kg}$$

217. If a boom sprayer has 6 nozzles and each nozzle discharges 0.6 liter/ min, what is the application rate if the spray swath and walking speed is 25 m/min? Swath is 3m.

Solution:

$$A_{1\text{min}} = \left(\frac{25\text{m}}{\text{min}} \right) (3\text{m}) = \frac{75\text{m}^2}{\text{min}}$$

$$Q_{1\text{min}} = \left(\frac{0.6\text{lit}}{\text{min}} \right) (6) = \frac{3.6\text{liters}}{\text{min}}$$

$$Q_T = \left(\frac{3.6\text{lit}}{\text{min}} \right) \left(\frac{1\text{min}}{75\text{m}^2} \right) \left(\frac{10,000\text{m}^2}{1\text{ha}} \right) = \frac{480\text{lit}}{\text{ha}}$$

218. A pump was tested in measuring the flow rate using a 90-degree triangular weir. Tests have shown that the head of water into the weir averages at 13 cm. What is the average rate of flow of the pump?

Given:

Machine - pump

Test rig - 90 degree triangular weir

Ave head reading - 15 cm

Solution:

$$Q = 0.0138H^{\frac{5}{2}} = 0.0138(13)^{\frac{5}{2}} = \frac{8.41\text{lit}}{\text{sec}}$$

219. Compute the friction head from a 6-meter-depth suction pipe of a centrifugal pump which discharges 0.012 m³/sec of water. The pipe nominal diameter is 4 inches.

Given:

Length of pipe - 6 meters

Discharge - 0.012 m³/sec

Pipe diameter - 4 inches

Solution:

$$H_f = \frac{kLQ^2}{C^2D^2} = \frac{(10)(6m)\left(\frac{0.012\text{m}^3}{\text{sec}}\right)^2}{(1)^2(4 \times 0.0254)^2} = 0.837\text{m}$$

220. A right-angled V-notch was used measure the discharge of a centrifugal pump. If the depth of water at V-notch is 200mm, calculate the discharge over the notch in liters per minute, assume coefficient of discharge of 0.62.

Solution:

$$Q = \frac{8}{12} C_d \sqrt{2g} \tan \frac{\theta}{2} H^{\frac{5}{2}} = \frac{8}{12} (0.62) \sqrt{2(9.81)} \tan \frac{90}{2} (0.2\text{m})^{\frac{5}{2}}$$

221. A pump was installed with a suction head of 6 meter and a discharge head of 10 meters. The discharge rate of the pump is 60 liters per minute. What is the water horsepower of the pump?

Given:

Suction head - 6 m

Discharge head - 10 meters

Discharge rate - 60lpm

Solution:

$$\text{WHP} = \frac{\left(\frac{62.4\text{lb}}{\text{ft}^3}\right) \left(\frac{60\text{liter}}{\text{min}}\right) (16\text{m}) (3.28\text{ft}) \left(\frac{1\text{ft}}{30.48\text{cm}}\right)^3 \left(\frac{1000\text{cm}^3}{1\text{lit}}\right)}{33,000\text{ft} - \text{lb}} = 0.21\text{hp}$$

222. A pump, which discharges 4 liters per second at a head of 6 meters, is driven by an electric motor. The input current of the motor is 1.5 amp while the input voltage is 220 volt. What is the overall efficiency of the pump if the motor power factor is 0.98?

Give:

Pump discharge - 4 lps

Head - 6 meters

Input current - 1.5 amp

Input voltage - 220 volt

Power factor - 0.98

Solution:

$$\text{WHP} = \frac{\left(\frac{62.4\text{lb}}{\text{ft}^3}\right)\left(\frac{4\text{liter}}{\text{min}}\right)\left(\frac{60\text{sec}}{\text{min}}\right)\left(\frac{1\text{ft}}{30.48\text{cm}}\right)^3\left(\frac{1\text{lit}}{1000\text{cm}^3}\right)}{\frac{33,000\text{ft} \cdot \text{lb}}{\text{min}}} = 0.235\text{kW}$$

Or

$$\text{WHP} = \frac{(4\text{lbs})(6\text{m})}{102} = 0.235\text{kW}$$

$$P = VI = (220\text{V})(1.5\text{amp})(0.98) = 323.4\text{W or } 0.3234\text{kW}$$

$$\text{Eff} = \frac{\text{WHP}}{P} = \frac{0.235\text{kW}}{0.3234\text{kW}} = 73\%$$

223. At 1200 rpm, a rotary pump is used to deliver milk at a head of 6 meters. The rate of flow of the pump is 0.01 m³/sec. What is the fluid and brake horsepower of the pump shaft if it has an efficiency of 75%. What is the motor horsepower if the transmission and motor efficiencies are 91% and 99%, respectively?

Given:

Flow rate - 0.1 m³/s

Total head - 6 m

Specific gravity of milk - 1.1

Pump efficiency - 75%

Transmission efficiency - 91%

Motor Efficiency - 99%

Solution:

$$\text{FHP or WHP} = \frac{\left(\frac{1000\text{kg}}{\text{m}^3}\right)\left(\frac{0.01\text{m}^3}{\text{sec}}\right)(6\text{m})(1.1)}{\frac{76.2\text{kg} \cdot \text{m}}{\text{sec}}} = 0.87\text{hp}$$

$$\text{BHP} = \frac{0.87\text{hp}}{0.75} = 1.15\text{hp}$$

$$P_M = \frac{1.15\text{hp}}{(0.91)(0.99)} = 1.28\text{hp use } 112\text{hp}$$

224. Refer to problem # 223. What will be the flow rate, working head, and power requirement of the pump system above if its speed is increased to 2100 rpm?

Given:

Speed1 - 1200 rpm

Flow rate - 0.1 m³/s

Head1 - 6 m

Horsepower1 - 0.87hp

Speed2 - 2100 rpm

Solution:

$$\frac{Q_2}{Q_1} = \frac{N_2}{N_1}; Q_2 = \left(\frac{N_2}{N_1}\right) Q_1 = \left(\frac{2100\text{rpm}}{1200\text{rpm}}\right)\left(\frac{0.01\text{m}^3}{\text{sec}}\right) = \frac{0.0175\text{m}^3}{\text{sec}}$$

$$\frac{H_2}{H_1} = \frac{N_2}{N_1}; H_2 = \left(\frac{N_2}{N_1}\right) H_1 = \left(\frac{2100\text{rpm}}{1200\text{rpm}}\right)^2(6\text{m}) = 18.375\text{m}$$

$$\frac{HP_2}{HP_1} = \frac{N_2}{N_1}; HP_2 = \left(\frac{N_2}{N_1}\right) HP_1 = \left(\frac{2100\text{rpm}}{1200\text{rpm}}\right) 3(0.87\text{hp}) = 4.66\text{hp}$$

225. Estimate the pull that can be developed from a 900-kg water buffalo pulling a load at an angle of 30 deg. The horizontal distance between the front and the rear foot is 1.4 m. The center of gravity of the animal is located 0.7 m from the foot. The horizontal distance from the yoke to the rear foot is 0.3 m. The height of the neck yoke was measured 1.2 m from the neck load point to the ground. Assume a coefficient of friction of 0.95 between the roof and the ground surface.

Given:

$$W = 900 \text{ kg}$$

$$h = 1.2 \text{ m}$$

$$L = 1.4 \text{ m}$$

$$\alpha = 30 \text{ deg}$$

$$L_1 = 0.7 \text{ m}$$

$$\mu = 0.95$$

$$L_2 = 0.3 \text{ m}$$

Required: Pull

Solution:

$$P = \frac{WL_1\mu_f}{(L + h\mu_f)\cos\theta + L_2\mu_f\sin\theta}$$

$$P = \frac{900 \text{ kg} (0.7 \text{ m}) (0.95)}{(1.4 \text{ m} + 1.2 \text{ m} (0.95)) \cos 30 + 0.3 \text{ m} (0.95) \sin 30} = 255.76 \text{ kg}$$

226. If a cow can provide a draft of 60 kg, how much draft can be provided by 2 cows together?

Given:

$$\text{No. of cows} = 2$$

$$\text{Draft} = 60 \text{ kg}$$

Required: Total Draft

Solution:

$$D_t = (\# \text{ of animals}) \left(\frac{\text{draft}}{\text{animal}} \right) (\text{factor}) = (2) \left(\frac{60\text{kg}}{\text{animal}} \right) \left(\frac{1.9}{2} \right) = 114\text{kg}$$

227. Calculate the BHP of a 4-cylinder, 2-stroke cycle internal combustion engine with 4" diameter and 5" stroke piston. The mean effective pressure is 102 psi and the number of explosion per minute is 600. The mechanical efficiency of the engine is 80 percent.

Given:

$$\text{No. of cylinder} = 4$$

$$\text{Mean Eff. Pressure} = 102 \text{ psi}$$

$$\text{No. of stroke} = 2$$

$$\text{No. of exp. / min.} = 600$$

$$\text{Bore (diameter)} = 4 \text{ in.}$$

$$\text{Mech. Eff.} = 80\%$$

$$\text{Stroke} = 5 \text{ in.}$$

Required: BHP

Solution:

$$\frac{\text{No. of explosion}}{\text{min}} = N \left(\frac{\text{rev}}{\text{min}} \right) (\# \text{ of cylinder})$$

$$600 = N(4); N = 150\text{rpm}$$

$$\text{BHP} = \text{IHP} \times \text{Eff}_{\text{Mechanical}} = (9.71 \text{hp})(0.8) = 7.80 \text{hp}$$

$$\text{IHP} = \frac{\left(\frac{102 \text{lb}}{\text{in}^2}\right) \left(\frac{5 \text{in}}{12 \text{in}}\right) \left(\frac{\pi(4)^2}{4}\right) (150 \text{rpm}) \times 4}{\frac{33,000 \text{ft} - \text{lb}}{\text{min}}} = 9.71 \text{hp}$$

228. A person is working for 4 hours in a farm harvesting. Estimate the power that can be generated by the person in doing the job.

Solution:

$$P = 0.35 - 0.092 \log \left(4 \text{ hours} \left(60 \frac{\text{min}}{\text{hr}} \right) \right) = 0.13 \text{ hp}$$

229. Two persons are doing the threshing operation by “Hampasan” method. What is the total human power generated by the two persons in one day if they work an average of 6 hours?

Given:

Work - threshing

No. of persons - 2

Time - 6 hours

Required: Power generated by the two persons

Solution:

$$P_g = 0.35 - 0.092 \log \left(6 \text{ hours} \left(\frac{60 \text{min}}{\text{hr}} \right) \right) = 0.11482 \text{hp/person}$$

$$P_T = \frac{0.11482 \text{hp}}{\text{person}} (2 \text{ person}) = 0.23 \text{hp}$$

230. The engine fuel tank was completely filled with gasoline before testing. After 4 hours of continuous test, 3.7 liters of fuel was used to reload the tank completely. Test has shown that the engine shaft power was 10 hp. What is the specific fuel consumption of the engine?

Given:

Fuel - gasoline

Operating time - 4 hours

Fuel consumed - 3.7 liters

Shaft power - 10 horsepower

Required: Specific fuel consumption

Solution:

$$\text{SFC} = \frac{(3.7 \text{ liters}) \left(\frac{0.743 \text{kg}}{\text{liter gasoline}} \right) \left(1000 \frac{\text{g}}{1 \text{kg}} \right)}{(4 \text{ hours})(10 \text{ hp}) \left(0.746 \frac{\text{kW}}{\text{hp}} \right)} = \frac{92.13 \text{g}}{\text{kW} - \text{hr}}$$

231. A spark ignition engine was tested with BHP output of 9.6 horsepower. If the ambient condition during the test is 30°C ambient air temperature, what is the corrected BHP of the engine? The total atmospheric air condition is 1011 mbar.

Given:

BHP of engine tested - 9.6hp

Ambient air temp - 30°C

Total atmospheric air condition - 1011mb

Required: Corrected BHP

Solution:

$$\text{BHP}_{\text{corrected}} = \text{BHP} \times \text{CF} = (9.6\text{hp})(1.02) = 9.80\text{hp}$$

$$\text{BHP} = \frac{1013}{P_{\text{atm}}} \sqrt{\frac{T + 273}{293}} = \frac{1013}{1011} \sqrt{\frac{30 + 273}{293}} = 1.02$$

232. A four-stroke naturally-aspirated diesel engine was tested on a pony brake dynamometer and gave a BHP output of 14.2 horsepower. During the test, the ambient air temperature was 27°C while the total atmospheric pressure was 1020mb. What is the corrected brake horsepower of the engine?

Given:

Engine - 4 stroke naturally aspirated

BHP output - 14.2 horsepower

Ambient air temperature - 27°C

Total atm. pressure - 1020mb

Required: Corrected BHP

Solution:

$$\text{BHP}_{\text{corrected}} = \text{BHP} \times \text{CF} = (14.2\text{hp})(1.007) = 14.30\text{hp}$$

$$\text{BHP} = \left(\frac{1013}{P_{\text{atm}}}\right)^{0.65} \sqrt{\frac{T + 273}{293}} = \left(\frac{1013}{1020}\right)^{0.65} \sqrt{\frac{27 + 273}{293}} = 1.007$$

233. Compute the power that can be generated by a hydro-turbine with water input of 40lps with a head of 20 m. The turbine efficiency is 70%.

Given:

Q - 40lps or 0.04 m³/s

H - 20 m

Eff_t - 0.7

Required: Power

Solution:

$$P = \rho g Q H = \left(\frac{9810\text{kg}}{\text{m}^3\text{sec}^2}\right) (0.7)(20\text{m}) \left(\frac{0.04\text{m}^3}{\text{sec}}\right) = 5493.6\text{Watts or } 5.5\text{kW}$$

234. A hydro site offers a head of 50 m and a flow of 1.2 m³/s. With an alternator speed requirement of 1500 rpm in a direct-drive transmission, compute the specific speed of the turbine for the system. Assume a turbine efficiency of 70%.

Given:

Q - 1.2 m³/sec

H - 50 m

η_t - 70%

N_a - 1500

G - 1

Required: Turbine specific speed

Solution:

$$P = \rho g Q H = \left(\frac{9810\text{kg}}{\text{m}^3\text{sec}^2}\right) \left(1.2 \frac{\text{m}^3}{\text{s}}\right) (50\text{m})(0.70) = 412\text{kW}$$

$$N_s = \frac{1500 \text{ rpm } \sqrt{412 \text{ kW}}}{(50 \text{ m})^{1.25}} = 230$$

235. A wood charcoal stove consumes 0.5 kg of fuel per hour. If 95% is volatile matter, what is the design power of the stove? Assume a 6000 kcal/kg heating value for wood charcoal.

Given:

Fuel consumption rate - 0.5 kg/hr.

Percentage volatile matter - 95 %

Required: Design power

Solution:

$$P_d = 0.7(P_c + P_{mv}) = 0.7 \left(\left(\frac{0.5 \text{ kg}}{\text{hr}} \right) \left(\frac{6000 \text{ kcal}}{\text{kg}} \right) + (0.95) \left(\frac{0.5 \text{ kg}}{\text{hr}} \right) \left(\frac{6000 \text{ kcal}}{\text{kg}} \right) \right) = \frac{4095 \text{ kcal}}{\text{hr}}$$

236. Compute the average power output of a stove with a fuel consumption rate of 5 kg of rice hull in 4 hours operation. The net calorific value of rice hull is 3000 kcal/kg.

Given:

Total fuel used - 5 kg

Net calorific value - 3000 kcal/kg

Total burning time - 4 hours

Required: Average power output

Solution:

$$P_{\text{average}} = \frac{(\text{FCR})(\text{HV})}{T} = \frac{(5 \text{ kg}) \left(\frac{3,000 \text{ kcal}}{\text{kg}} \right)}{4 \text{ hrs}} = \frac{3750 \text{ kcal}}{\text{hr}}$$

237. What is the probable height of the flame in a step-grate stove with a power output of 1000 BTU/hr.?

Given:

Power output - 1000 BTU/hr.

Type of stove - with grate

°C - 75 mm/kW

Required: Flame height

Solution:

$$FH = CP^{\frac{2}{5}} = \left(\frac{75 \text{ mm}}{\text{kW}} \right) \left(\left(\frac{1000 \text{ BTU}}{\text{hr}} \right) \left(\frac{1 \text{ kW} - \text{hr}}{3412.2 \text{ BTU}} \right) \right)^{\frac{2}{5}} = 45.90 \text{ mm}$$

238. Two kilogram of rice is to be prepared for cooking. Estimate the time required to cook it.

Given:

Weight of rice - 2 kg

Required: Cooking Time

Solution:

$$t_c = 550(M_{\text{food}})^{0.38} = 550(2 \text{ kg})^{0.38} = 716 \text{ seconds or } 11.92 \text{ minutes}$$

239. Two liters of water will be boiled in the stove. The desired time to boil the water is 20 minutes. If the initial temperature of water is 28°C and the boiling temperature is 100°C, what is the maximum power required for the stove? Assume a thermal efficiency of 15% for the stove.

Given:

Volume of water - 2 liters

Boiling time - 20 minutes

Initial temperature of water - 28°C

Final temperature of water - 100°C

Thermal efficiency - 15%

Required: Maximum power

Solution:

$$P_{\text{maximum}} = \frac{M_f C_p (T_2 - T_1)}{T_c \text{Eff}_{\text{thermal}}} = \frac{(2\text{lit}) \left(\frac{1\text{kg}}{1\text{lit}} \right) \left(\frac{1\text{kcal}}{\text{kg} \cdot ^\circ\text{C}} \right) (100^\circ\text{C} - 28^\circ\text{C})}{(20\text{min})(0.15)} = \left(\frac{48\text{kcal}}{\text{min}} \right) \left(\frac{60\text{min}}{1\text{hr}} \right) \\ = \frac{2880\text{kcal}}{\text{hr}}$$

240. A 2-ton capacity flat-bed dryer dries paddy from 20 to 14 percent in 10 hours. What is the drying capacity of the dryer? Also compute the moisture reduction per hour of the dryer.

Given:

Capacity of the bin - 2 tons

Drying time - 10 hours

Required: Drying capacity

Solution:

$$C_{\text{dryer}} = \frac{2\text{tons}}{10\text{hrs}} = \frac{0.2\text{ton}}{\text{hr}} \\ \text{MR} = \frac{20\% - 14\%}{10\text{hours}} = \frac{0.6\text{percent}}{\text{hour}}$$

241. Five hundred hectares of farm is to be plowed using 5-bottom moldboard plow. The cutting width of each plow bottom is 20 cm with a cutting depth of 15 cm. If three (3) tractors are pulling a plow at an average speed of 15kph, how many days does the whole area be finished? Assume a plowing efficiency of 85% and a plowing operation per day of 6 hours. Also, compute the drawbar hp. of the plow if the specific draft of soil is equal to 12 psi.

Given:

Area - 500 has

No. of Plow - 5

Width of Cut - 20 cm

Depth of Cut - 15 cm

No of Tractors - 3 units

Speed of Tractor - 15kph

Plowing Eff. - 85%

Operation - 6 hr. /day

Soil Draft - 12 psi

Required: Days to finish plowing, Plow drawbar horsepower

Solutions:

$$\text{TFC} = \frac{\text{SW}}{10} = \frac{(15\text{kph})(0.2\text{m})(5)}{10} = \left(\frac{1.5\text{ha}}{\text{hr}} \right) \\ \text{EFC} = \frac{\text{SWEff}}{10} = \frac{(15\text{kph})(0.2\text{m})(5)(0.85)}{10} = \frac{1.275\text{ha}}{\text{hr}} \text{ (per tractor)}$$

Total capacity for 3 tractors:

$$C = \frac{1.275\text{ha}}{\text{hr} - \text{tractor}} (3 \text{ tractors}) = \left(\frac{3.825\text{ha}}{\text{hr}} \right) \left(\frac{6\text{hrs}}{\text{day}} \right) = \frac{22.95\text{ha}}{\text{day}}$$

Days to finish 500ha:

$$X = \frac{500\text{ha}}{\left(\frac{22.95\text{ha}}{\text{day}}\right)} = 21.78\text{days}$$

$$\text{DHP} = \frac{\left(\frac{12\text{lb}}{\text{in}^2}\right)(20\text{cm} \times 15\text{cm} \times 5) \left(\frac{1\text{in}}{2.54\text{cm}}\right)^2 \left(\frac{15000\text{m}}{\text{hr}}\right) \left(\frac{1\text{hr}}{60\text{min}}\right) \left(\frac{3.28\text{ft}}{1\text{m}}\right)}{\frac{33,000\text{ft} - \text{lb}}{\text{min}}} = 69\text{hp}$$

242. A tractor, has the following engine specifications: number of cylinders, 6; cylinder bore, 0.15 m; piston stroke, 0.20 m; engine speed, 2200 rpm; mean effective pressure, 180 psi; and number of stroke, 4. What is the indicated horsepower and brake horsepower of the engine, assuming an engine efficiency of 80%? What is the estimated PTO power, wheel axle power, and drawbar power of the tractor if its transmission efficiencies are 90%, 92%, and 81%, respectively?

Given:

No. of cylinders - 6
 Engine speed - 2200 rpm
 Cylinder bore - 0.15 m
 Mean Eff. Pressure - 180 psi
 Piston stroke - 0.20 m
 Engine # of stroke - 4
 Engine efficiency - 80%
 Transmission efficiency - 90% to 92%
 Wheel axle, and 81% db.

Required: Indicated horsepower and brake horsepower of the tractor engine, PTO power, wheel axle power, and drawbar power

Solution:

$$\text{IHP} = \frac{\left(\frac{180\text{lb}}{\text{in}^2}\right)(0.2\text{m}) \left(\frac{3.28\text{ft}}{1\text{m}}\right) \left(\frac{\pi \left(\frac{0.15\text{m}}{0.0254}\right)^2}{4}\right) (2200\text{rpm})(6)}{\frac{33,000\text{ft} - \text{lb}}{\text{min}}(2)} = 647\text{hp}$$

$$\text{BHP} = (\text{IHP})(\text{Eff}_{\text{engine}}) = (647\text{hp})(0.80) = 517.6\text{hp}$$

$$P_{\text{PTO}} = (\text{BHP})(\text{Eff}_{\text{PTO}}) = (517.6\text{hp})(0.90) = 465.84\text{hp}$$

$$P_{\text{wheel axle}} = (\text{BHP})(\text{Eff}_{\text{wheel axle}}) = (517.6\text{hp})(0.9) = 476\text{hp}$$

$$\text{DHP} = (\text{BHP})(\text{Eff}_{\text{drawbar}}) = (517.6\text{hp})(0.81) = 419\text{hp}$$

243. During transport, a 3,500-kg tractor passes through an earth road at a speed of 15kph. The rated brake power of the tractor is 125 horsepower while the transmission efficiency from the engine to the wheel axle is 80%. If the 2-m rear diameter wheel of the tractor, at 100 meter distance, can only rotate 11 revolutions during operation, determine the following: (1) percentage slip of the tractor wheel; (2) drawbar horsepower of the tractor; and (3) tractive efficiency of the tractor wheel.

Given:

Tractor weight - 3500 kg
 Running velocity - 15kph
 Brake horsepower - 125hp
 Transmission efficiency - 80% engine to wheel axle

Wheel diameter - 2 m

Coefficient of rolling resist - 0.07

Required: Slip, drawbar horsepower, Tractive efficiency, and torque

Solutions:

$$\% \text{slip} = 100 \left(\frac{A_n - A_l}{A_l} \right) = 100 \left(\frac{100\text{m} - \pi(2\text{m})(11\text{rev})}{\pi(2\text{m})(11\text{rev})} \right) = 44.70\%$$

$$\text{WAP (wheel axle power)} = (125\text{hp})(0.8) = 100\text{hp}$$

$$P_s(\text{slip}) = (100\text{hp})(0.447) = 44.70\text{hp}$$

$$\text{Resistance} = (3500\text{kg})(0.07) = 245\text{kgf}$$

$$P_R(\text{resistance}) = \frac{(245\text{kgf}) \left(\frac{15000\text{m}}{\text{hr}} \right) \left(\frac{1\text{hr}}{3600\text{sec}} \right)}{\frac{76.2\text{kg} \cdot \text{m}}{\text{sec} \cdot \text{hp}}} = 13.40\text{hp}$$

$$P_{\text{loss}} = P_s(\text{slip}) + P_R(\text{resistance}) = 44.70\text{hp} + 13.40\text{hp} = 58.1\text{hp}$$

$$\text{DHP} = 100\text{hp} - 58.1\text{hp} = 41.9\text{hp}$$

$$\text{Eff}_{\text{tractive}} = \frac{\text{DHP}}{\text{WAP}} = \frac{41.9\text{hp}}{100\text{hp}} = 41.9\%$$

$$N_{\text{wheel}} = \frac{\left(\frac{15,000\text{m}}{\text{hr}} \right) \left(\frac{1\text{hr}}{60\text{min}} \right)}{\pi(2\text{m})(0.4470)} = 89\text{rpm}$$

$$T = \frac{1000(100\text{hp})(0.746\text{kW})}{2\pi(89\text{rpm})} = 133.4\text{N} \cdot \text{m}$$

244. A 15 wheel axle hp tractor is running at a speed of 10kph during harrowing operation on a deep mud soil. The tractors weighs 1500 kg. The wheel diameter of the tractor is 2 meters and is able to rotate at 14 revolutions in 100 meter distance. What is speed of the tractor wheel in revolution per minute? What are the power losses of the tractor by wheel slip and rolling resistances?

Given:

Tractor WAP - 15hp

Tractor speed - 10kph

Tractor weight - 500 kg

Soil condition - deep mud soil

Wheel rotation - 28 revolutions

Distance traveled - 200 meters

Wheel diameter - 2 meters

Required: Wheel slip and speed of tractor wheel in rpm and power losses due to wheel slip, and power loss due to running resistance.

Solution:

$$\% \text{slip} = 100 \left(\frac{A_n - A_l}{A_l} \right) = 100 \left(\frac{100 - \pi(2)(14)}{\pi(2)(14)} \right) = 13.70\%$$

$$N_{\text{wheel}} = \frac{\left(\frac{10,000\text{m}}{\text{hr}} \right) \left(\frac{1\text{hr}}{60\text{min}} \right)}{\pi(2\text{m})(0.137)} = 193.6\text{rpm}$$

$$P_{\text{slip}} = \text{WAP} \times \% \text{slip} = (15\text{hp})(0.137) = 2.055\text{hp}$$

$$\text{Resistance} = 1500\text{kg} \times 0.25 \text{ for mud soil} = 375\text{kg}$$

$$P_{\text{resistance}} = \frac{(375\text{kg}) \left(\frac{10,000\text{m}}{\text{hr}} \right) \left(\frac{1\text{hr}}{3600\text{sec}} \right)}{\frac{76.2\text{kg} \cdot \text{m}}{\text{sec}}} = 13.70\text{hp}$$

245. What is the wheel axle power of a four-wheel tractor whose engine power is equal to 150hp? The mechanical efficiency is 85%.

Solution:

$$P_{\text{wheel axle}} = (150\text{hp})(0.85) = 127.5\text{hp}$$

246. What is the wheel axle torque of a tractor above if its wheel rotates at 40rpm? Refer to problem #245.

Solution:

$$T = \frac{1000(P_{\text{wheel axle}})}{2\pi N} = \frac{1000(127.5\text{hp})(0.746\text{kW})}{2\pi(40\text{rpm})} = 378.45\text{N} \cdot \text{m}$$

247. If the tractor wheel above has a diameter of 2 meters, what is the force available at the wheel of the tractor? Refer to problem #246

Solution:

$$F_{\text{available}} = \frac{T}{r} = \frac{378.45\text{N} \cdot \text{m}}{1\text{m}} = 378.45\text{N}$$

248. The wheel of tractor above rotates by two-half revolution at a distance of 20 meters. What is the percentage slip of the tractor? Refer to problems # above.

Solution:

$$\% \text{ slip} = 100 \left(\frac{20 - \pi(2)(2.5)}{\pi(2)(2.5)} \right) = 27.32\%$$

249. If a tractor weighs 2.5 tons and runs in a deep mud soil. What is the rolling resistance of the tractor? Refer to problems above.

Solution:

$$R = (2.5\text{tons})(1000\text{kg})(0.25) = 625\text{kgf}$$

250. If the tractor in Problem #249 runs at a velocity of 15kph, what is the power loss?

Solution:

$$P_{\text{resistance}} = \frac{(625\text{kg}) \left(\frac{15,000\text{m}}{\text{hr}} \right) \left(\frac{1\text{hr}}{3600\text{sec}} \right)}{\frac{76.2\text{kg} \cdot \text{m}}{\text{sec} \cdot \text{hp}}} = 34.18\text{hp}$$

$$P_L = P_s + P_r = (127.5\text{hp})(0.2732) + 34.18\text{hp} = 69\text{hp}$$

251. A 2.4 meter sugar planter is used to transplant sugarcane to a 150-hectare farm. The plant spacing is 60 cm while the row spacing is 120 cm. compute the number of hills the machine can plant for the entire area. What is the rate of planting in hills per hectare if the machine travels at a speed of 6kph? Estimate the number of tractors and of planter units required to finish the entire area in 10 days with 8 hours/day operation. Assume a planting efficiency of 80%.

Given:

Area - 150 ha

Plant spacing - 60 cm

Row spacing - 120 cm

Speed - 6kph

Operating time required - 10 day (8hr/day)

Field efficiency - 80 %

Required: No. of hills to be planted, Rate of planting in hills/ha, No. of tractors/planter unit

Solution:

$$\text{No. of hills} = \frac{150 \text{ ha} \times 10,000}{1.2\text{m} \times 0.6\text{m}} = 2,083,333.3 \text{ hills}$$

$$\text{Rate of planting} = \frac{\text{SWEff}}{10} = \frac{(6\text{kph})(1.2\text{m})(0.8)}{10} = \frac{0.576\text{ha}}{\text{hr}}$$

$$= \frac{2,083,333.3 \text{ hills}}{\frac{0.576\text{ha}}{\text{hr}}} = \frac{3616898.09 \text{ hill} - \text{hr}}{\text{ha}} = 45211.23 \text{ hills/ha}$$

$$\text{TFC} = \frac{(6\text{kph})(2.4\text{m})(0.8)}{10} = \frac{1.152\text{ha}}{\text{hr} - \text{tractor}}$$

$$\text{No. of Tractors} = \frac{150\text{ha}}{\frac{1.152\text{ha}}{\text{hr} - \text{tractor}}} = \frac{130.2\text{hr} - \text{tractor}}{80\text{hrs}} = 1.63 \text{ tractor use 2 tractor}$$

252. Two pulleys of a tractor-generator drive system are running in opposite direction. The pulleys were measured to have 6-meters center distance from each other. If the tractor pulley is 8 inches and the generator pulley is 6 inches, what would be the total length of the belt in the drive system?

Given:

Center distance - 6 m

Tractor pulley - 8 inches

Generator pulley - 6 inches

Required: Length of belt

Solution:

$$L = 2C + 1.57(D + d) + \frac{(D + d)^2}{4C}$$

$$= 2(236.22 \text{ in.}) + 1.57(8 \text{ in.} + 6 \text{ in.}) + \frac{(8 \text{ in.} + 6 \text{ in.})^2}{(4 \text{ in.})(236.22 \text{ in.})} = 494.63 \text{ in.}$$

253. An axial-flow biomass shredder is equipped with a 5-inch diameter double-groove V-belt at its cylinder axle. The engine used to drive the machine is rated at 2400 rpm at an optimum engine throttle opening. If the engine pulley is 4 inches, what is the speed of the shredder axle? What is the speed ratio of the engine and the shredder pulleys?

Given:

Shredder pulley - 5 inches

Engine pulley - 4 inches

Engine speed - 2400 rpm

Required: Shredder speed and speed ratio

Solution:

$$N_1 D_1 = N_2 D_2$$

$$N_1 = \frac{N_2 D_2}{D_1} = \left(\frac{2400 \text{ rpm} \times 4 \text{ in.}}{5 \text{ in.}} \right) = 1920 \text{ rpm}$$

$$\text{SR} = \frac{2400 \text{ rpm}}{1920 \text{ rpm}} = 1.25:1$$

254. Design a water pump system required to supply water to 10,000 heads of broiler in a poultry farm. The source of water is 20ft below the ground and is required to be elevated to 30-ft high into a water tank. The specific water requirement of the broiler is 0.025 m³/day-head. Assume a 20% friction head load from the suction and discharge head loss. Pump operating time is 20hrs per day. The Pump, Transmission and Engine efficiencies are 60%, 95% and 80 % respectively.

Given:

No. of broilers - 10,000 heads

H_s - 20ft

H_d - 30ft

SWR - 0.025 m³/day-chicken

Friction Head - 20% of the H_s & H_d

Operating time - 20 hr. per day

Required:

Pump flow rate, total head, water hp, brake hp, and motor hp

Solution:

$$\text{Pump flow rate} = \left(\frac{0.025 \text{m}^3}{\text{day} - \text{head}} \right) (10,000 \text{heads}) \left(\frac{1 \text{day}}{20 \text{hrs}} \right) = \frac{12.50 \text{m}^3}{\text{hr}}$$

$$\text{total head} = (20 \text{ft} + 30 \text{ft})(1.2) = 60 \text{ft} \left(\frac{1 \text{m}}{3.28 \text{ft}} \right) = 18.30 \text{m}$$

$$\text{WHP} = \frac{\left(\frac{12.50 \text{m}^3}{\text{hr}} \right) \left(\frac{1000 \text{kg}}{\text{m}^3} \right) (18.30 \text{m}) \left(\frac{1 \text{hr}}{3600 \text{sec}} \right)}{\left(\frac{76.2 \text{kg} - \text{m}}{\text{sec}} \right)} = 0.84 \text{hp}$$

$$\text{BHP} = \frac{0.84 \text{hp}}{0.60} = 1.4 \text{hp}$$

$$\text{MHP} = \frac{1.4 \text{hp}}{(0.95)(0.80)} = 1.84 \text{hp}$$

255. Design a water pump system required to supply water to a 5-hectare rice farm. The source of water is from a river 10 feet below, and discharges water 2ft above the ground surface. Consider 25% friction head from the suction and from the discharge head of the piping system. Assume a crop requirement for rice of 10 mm/day and losses due to percolation and seepage of 2 mm/day. Conveyance losses is 10% of the irrigation requirement. Assume a 60% pump efficiency, an 80% transmission efficiency, and an 80% engine efficiency. Operating time per day is 8 hrs.

Given:

Area of the farm - 5 hectare rice farm

H_s - 10ft

H_d - 2ft

H_f - 25% of H_s & H_d

Crop Requirement - 10 mm/day

Seepage & percolation - 2 mm/day

Conveyance losses - 10% of IR

Required:

Pump capacity, total head loss, water hp, brake hp, motor hp

Solution:

$$\text{irrigation requirement} = (50,000\text{m}^2) \left(\frac{12\text{mm}}{\text{day}} \right) \left(\frac{1\text{m}}{1000\text{mm}} \right) \left(\frac{1\text{day}}{8\text{hrs}} \right) = \frac{600\text{m}^3}{\text{day}}$$

$$\text{Diversion requirement} = \left(\frac{600\text{m}^3}{\text{day}} \right) (1.1) = \frac{660\text{m}^3}{\text{day}}$$

$$C_p = \left(\frac{660\text{m}^3}{\text{day}} \right) \left(\frac{1\text{day}}{8\text{hrs}} \right) = \frac{82.5\text{m}^3}{\text{hr}}$$

$$H_T = (10\text{ft} + 2\text{ft})(1.25) \left(\frac{1\text{m}}{3.28\text{ft}} \right) = 4.57\text{m}$$

$$\text{WHP} = \frac{\left(\frac{1000\text{kg}}{\text{m}^3} \right) \left(\frac{82.5\text{m}^3}{\text{hr}} \right) (4.57\text{m}) \left(\frac{1\text{hr}}{3600\text{sec}} \right)}{\frac{76.2\text{kg} \cdot \text{m}}{\text{sec}}} = 1.37\text{hp}$$

$$\text{BHP} = \frac{1.37\text{hp}}{0.6} = 2.28\text{hp}$$

$$\text{MHP} = \frac{2.28\text{hp}}{(0.8)(0.8)} = 3.56\text{hp use 5hp engine}$$

256. What is the wind power of a wind pump having 5 meter diameter rotor at a wind speed of 4 m/sec?

Given:

Rotor diameter - 5 m

Wind velocity - 4 m/s

Required:

Wind Power

Solution:

$$P = \frac{\rho_a A V^3}{2} = \frac{\left(\frac{1.225\text{kg}}{\text{m}^3} \right) \left(\frac{\pi(5\text{m})^2}{4} \right) \left(\frac{4\text{m}}{\text{sec}} \right)^3}{2} = 785.40\text{watts}$$

257. What is the power of the rotor if the 3-meter wind pump operates at 3mps wind speed? Assume a power coefficient for the wind pump rotor of 0.4.

Given:

Rotor diameter - 3 m

Average wind speed - 3 m/s

Power coefficient - 0.4

Required:

Rotor Power

Solution:

$$P = \frac{\rho_a A V^3}{2} = \frac{\left(\frac{1.225\text{kg}}{\text{m}^3} \right) \left(\frac{\pi(3\text{m})^2}{4} \right) \left(\frac{3\text{m}}{\text{sec}} \right)^3 (0.4)}{2} = 47.70\text{Watts}$$

258. What is the rotor speed of a wind pump having a rotor diameter of 4 m at a design speed of 3 m/s? The rotor will contain 12 blades with expected TSR of 2.

Given:

Rotor diameter - 4 m

Design velocity - 3 m/s

TSR - 2

Required:

Rotor speed

Solution:

$$TSR = \frac{\pi DN}{V}$$

$$N = \frac{V(TSR)}{\pi D} = \frac{\left(\frac{3m}{s}\right)(2)\left(\frac{60sec}{1sec}\right)}{\pi(4m)} = 28.65rpm$$

259. What is the hydraulic power required by a wind pump to deliver water at a rate of 15 m³/day with a head of 10 meters? Assume a 10 hours water pumping per day.

Given:

Discharge - 15 m³/day

Head - 10 m

Pumping time - 10 hours per day

Required: Hydraulic Power

Solution:

$$P = \frac{\left(\frac{1000kg}{m^3}\right)\left(\frac{15m^3}{day}\right)(10m)\left(\frac{1day}{10hrs}\right)\left(\frac{1hr}{3600s}\right)}{\frac{76.2kg-m}{sec}} = (0.0567hp)(0.746kW) = 40.8Watts$$

260. Compute the amount of power that can be developed by a 600-kg draft animal pulling an 80-kg load at a speed of 0.9 m/sec.

Given:

Weight of animal = 600 kg

Load = 80 kg

Speed = 0.9m.sec

Required: Hp.

Solution:

$$HP = \frac{(80kg)\left(\frac{0.9m}{s}\right)}{\frac{76.2kg-m}{s}} = 0.945hp$$

261. If the amount of heat used in removing the moisture from the grains in a continuous-flow dryer is 3000 kJ/hr., what is the heat utilization efficiency of the dryer if the burner uses 6000 kJ/hr. of fuel?

Given:

Heat utilized = 3000 kJ/hr.

Heat supplied = 6000 kJ/hr.

Required:

Heat utilization efficiency

Solution:

$$HUE = \frac{\left(\frac{3000kJ}{hr}\right)}{\frac{6000kJ}{hr}} = 50\%$$

262. A 16-hp rotary tiller is tilling at a depth of 10 cm. Its rotating tiller has a width of 0.5 m. What is the specific power output of the machine if it is traveling at a speed of 0.5 m/sec? Assume a power transmission efficiency for the tiller of 0.80.

Given:

Power Input = 16hp

Depth of Cut = 10 cm

Width of Cut = 0.5 m

Speed of Tiller = 0.5 m/sec

Power Trans Eff. = 0.80

Required: Specific Power Output

Solution:

$$SPO = (16hp)(746Watts)(0.8) = \frac{9.548.8Watts}{(10cm)(50cm)\left(\frac{50cm}{s}\right)} = \frac{382Watt - sec}{cm^3}$$

263. A 4 x 6 rubber roller is used as husker for a rice mill. The faster roller is directly driven by an electric motor at a speed of 1720 rpm. What would be the required speed for the faster and slower rollers?

Given:

Size of Roller - 4 x 6

Speed of motor - 1720 rpm

Required:

Speed faster and slower rollers

Solution:

Speed of faster roller = 1720 rpm

Speed of slower roller = 1720 rpm(0.75) = 1290 rpm

264. Compute the drawbar pull of plow on a power tiller whose depth and width of cut are equal to 10 cm x 8 cm. The number of plow bottom is 2. Soil draft is equal to 5 psi. If the plow is drawn at a speed of 10 km/hr., what is the drawbar horsepower of the plow?

Given:

Width of Cut = 8 cm

Depth of Cut = 10 cm

No. of Plow = 2

Draft = 5 psi

Speed = 10kph

Required:

Drawbar pull

Solution:

$$DHP = \frac{\left(\frac{5lb}{in^2}\right)\left(\frac{10cm}{2.54cm} \times \frac{8cm}{2.54cm}\right)\left(\frac{1kg}{2.2lb}\right)\left(\frac{10,000m}{hr}\right)\left(\frac{1hr}{3600s}\right)(2)}{\frac{76.2kg - m}{s}} = 2.05hp$$

265. A sprayer containing 100 liters of mixture operates at a nozzle discharge of 0.2 liters per minute per nozzle. If the sprayer boom consist of 6 nozzles, at what time does the sprayer mixture be changed?

Given:

V_m – 100 liters

D_n - 0.2 lpm

N_n – 6 nozzles

Required: Time to change mixture

Solution:

$$\text{Time to change mixture} = \frac{100\text{liter}}{\left(\frac{0.2\text{lit}}{\text{min} - \text{nozzle}}\right) (6\text{nozzles})} = 83.33 \text{ min} \left(\frac{1\text{hr}}{60\text{min}}\right) = 1.389\text{hr}$$

266. A rice husk gas stove has a reactor diameter of 0.15 m. The amount of rice husk that can be loaded in the reactor is 1.20 kg. If the specific gasification rate of the stove is 110 kg/m²-hr, how long does the gasifier will operate producing combustible gases? Assume a gasifier efficiency of 20%.

Given:

$$D_r - 0.15 \text{ m}; W_f - 1.20 \text{ kg}$$

$$\text{SGR} - 110 \text{ kg/m}^2\text{-hr}$$

Required: Operating time

Solution:

$$\text{Operating time} = \frac{\left(\frac{110\text{kg}}{\text{m}^2 - \text{hr}}\right) \left(\frac{\pi(0.15\text{m})^2}{4}\right) (0.20)}{(1.20\text{kg})} = 0.32\text{hr or 20minutes}$$

267. Determine the diameter of a gasifier required to provide power for a gas burner that consumes 10 kg of rice husk per hour. The specific gasification rate of the gasifier is 110 kg/m² - hr. Also determine the amount of air needed for gasification in m³/hr. Assume a gasifier efficiency of 54% and equivalence ratio of 0.3.

Given:

$$\text{FCR} - 10 \text{ kg/hr.}$$

$$\text{SGR} - 110 \text{ kg/m}^2 - \text{hr.}$$

$$\text{Eff}_g - 54\%$$

$$\text{Stoichiometric Air} - 4.7 \text{ kg air/kg fuel}$$

Required: Diameter of reactor and Air required for combustion

Solution:

$$A_{\text{reactor}} = \frac{\text{FCR}}{\text{SGR} \times \text{Eff}_g} = \frac{\frac{10\text{kg}}{\text{hr}}}{\left(\frac{110\text{kg}}{\text{m}^2 - \text{hr.}}\right) (0.54)} = 0.17\text{m}^2$$

$$D_{\text{reactor}} = \sqrt{\left(\frac{4A_r}{\pi}\right)} = \sqrt{\left(\frac{4(0.17\text{m}^2)}{\pi}\right)} = 0.46 \text{ m}$$

$$\text{Air}_{\text{required}} = \text{SA} \times \delta_a \times \text{FCR} \times \varepsilon = \frac{\left(\frac{4.7\text{kg} - \text{air}}{\text{kg} - \text{fuel}}\right) \left(\frac{10\text{kg}}{\text{hr}}\right) (0.3)}{\left(\frac{1.25\text{kg}}{\text{m}^3}\right)} = \frac{11.28\text{m}^3}{\text{hr}}$$

268. A gasifier reactor has a diameter of 0.78 m and a length of 2.4 m. If the reactor will be loaded with rice husk at 80% load, how long does the gasifier would operate to totally consume the rice husk fuel? How much air is needed for the gasification of fuel? Assume a specific gasification rate of 110 kg/m² -hr., equivalence ratio of 0.3, rice husk density of 100 kg/m³, stoichiometric air requirement of 4.7 kg air per kg rice husk, and gasifier efficiency of 20%.

Given:

$$D_r - 0.78 \text{ m}$$

$$L_r - 2.4 \text{ m}$$

%Load – 80%

SGR - 110 kg/m² –hr.

ε- 0.3

δ_{rh}- 100 kg/m³

SAR - 4.7 kg air per kg rice husk

Eff_g – 20%

Required: Time to operate the gasifier and the air required for gasification

Solution:

$$V_{\text{fuel}} = \frac{\pi D^2 L}{4} = \frac{\pi (0.78\text{m})^2 (2.4\text{m})}{4} = (1.15\text{m}^3)(0.8) = 0.92\text{m}^3$$

$$W_{\text{fuel}} = (0.92\text{m}^3) \left(\frac{100\text{kg}}{\text{m}^3} \right) = 92\text{kg}$$

$$\text{Time} = (W_{\text{fuel}}) \frac{(\text{Eff}_g)}{(\text{SGR})(A_r)} = \frac{92\text{kg} \times 0.20}{\left(\frac{110\text{kg}}{\text{m}^2\text{-hr}} \right) \left(\frac{\pi (0.78)^2}{4} \right)} = 0.35\text{hr}$$

$$\text{Air}_{\text{required}} = (92\text{kg}) \left(\frac{4.7\text{kg air}}{\text{kg rice husk}} \right) \left(\frac{0.3}{0.385\text{hr}} \right) = \frac{370.63\text{kg}}{\text{hr}} \text{ or } \frac{296.50\text{m}^3}{\text{hr}}$$

269. An 8 x 10 rubber-roll-huller head was purchased for installation in an existing rice milling plant. The proposed drive for the huller is an electric motor with operating speed of 1720 rpm. If the motor will use a 4-inche pulley, what will be the pulley for the huller?

Given:

Rubber huller size - 8 x 10

Motor speed - 1720 rpm

Motor pulley - 4 inches

Required: Huller pulley

Solution:

$$\text{Huller speed} = \frac{V \left(14 \frac{\text{m}}{\text{s}} \right) \left(3.28 \frac{\text{ft}}{\text{m}} \right) \left(60 \frac{\text{sec}}{\text{min}} \right)}{\pi D \left(\frac{8\text{in}}{12} \right) \pi} = 1315.51\text{rpm}$$

$$\text{Huller pulley} = \frac{(1720 \text{ rpm})(4 \text{ in})}{1315.8 \text{ rpm}} = 5.23\text{in}$$

270. A disk huller having a diameter of 600 mm will be installed as return huller for a rice milling plant. As an Agricultural Engineer, determine the diameter of the pulley required for the huller. The motor for the huller will have a pulley diameter of 4 inches and will be expected to run at 1740 rpm.

Given:

Diameter of huller - 600 mm

Motor pulley - 4 in.

Motor speed - 1740 rpm

Required: Huller pulley diameter

Solution:

$$\text{Huller rpm} = \frac{V}{\pi D} = \frac{\left(\frac{14\text{m}}{\text{s}} \right) \left(\frac{3.28\text{ft}}{\text{m}} \right) \left(\frac{60\text{s}}{\text{min}} \right)}{\pi (0.6\text{m} \times 3.28\text{ft})} = 445.63\text{rpm}$$

$$\text{Huller Pulley D} = \frac{(4\text{in})(1740 \text{ rpm})}{445.63 \text{ rpm}} = 15.60\text{in}$$

271. A rice milling plant is to be designed to run at a throughput rate of 5 tons per hour. The design milling recovery is 69% while the hulling efficiency is 95%. How many compartments are needed for the paddy separator of the rice mill? Assume 50 kg br/hr-comp.

Given:

Throughput capacity - 5 tph

Milling recovery - 69%

Hulling efficiency - 95%

Required: No. of compartments

Solution:

$$C_{\text{paddy}} = \frac{5\text{tph}}{0.69} = 7.24\text{tph of paddy}$$

$$C_{\text{br}} = (7.24\text{tph})(0.95) = 6.878\text{tph brown rice}$$

$$\text{No. of compartments} = (6.9 \text{ tph}) \left(\frac{1000\text{kg}}{\text{ton}} \right) \left(\frac{1\text{comp}}{50 \text{ kg} - \text{hr}} \right) = 138 \text{ compartments}$$

272. Wheel slip test of a tractor had shown that a wheel rotates by two-and-a-half revolutions at a distance of 20 meters. If the tractor wheel has a diameter of 2 meters, what is the percentage slip of the tractor wheel?

Given:

Wheel diameter – 2 meters

Wheel rotation – 2-1/2 rev,

Distance travel – 20 meters

Required: Percentage wheel slip

Solution:

$$\text{No load} = \pi D N_r = \pi(2\text{m})(2.5\text{rev}) = 15.7\text{m}$$

$$\text{Under load} = 20\text{m}$$

$$\% \text{slippage} = \frac{20\text{m} - 15.7\text{m}}{15.7\text{m}} = 27.40\%$$

273. A four-wheel tractor weighs 2.5 tons and runs in a deep mud soil during rotary tilling operation. What is the rolling resistance of the tractor wheel?

Given:

Tractor weight – 2.5 tons

Coefficient of rolling resistance – 0.25

Required: Rolling resistance of tractor wheel

Solution:

$$\text{RR} = (0.25)(2500\text{kg}) = 625\text{kg}$$

274. Compute the traveling speed of a tractor at 2nd gear with engine speed set at 2400 rpm. The diameter of wheel of the tractor is 2 meters.

Given:

Gear position - 2nd

Engine speed – 2400 rpm

Wheel diameter - 2 m

Required: Tractor speed

Solution:

$$V = \frac{(0.033)(2400\text{rpm})(2\text{m})}{2.49} = 63.61\text{kph}$$

275. A furnace is required to supply heat to a water boiler of a kiln dryer. The boiler tank contains 2 cubic meters of water and needs to be heated from 27°C to 100°C within 1 hour. The boiler is required to maintain the water temperature at 100°C and allow the water to evaporate at 100 kg per hour. What is the power input of the furnace in raising the temperature of the water and in maintaining its operation? Assume a boiler efficiency of 70% and a furnace efficiency of 40%, and a heat of vaporization of water of 540 kCal/kg.

Given:

$$V - 2\text{m}^3$$

$$T_1 - 27^\circ\text{C};$$

$$T_2 - 100^\circ\text{C};$$

$$M - 100 \text{ kg/hr.}$$

$$\text{Eff}_b - 70\%$$

$$\text{Eff}_f - 40\%$$

Required:

Power input in raising the temperature of water and maintaining its operation

Solution:

$$Q_h = mcp(T_s - T_1) = (2\text{m}^3) \left(\frac{1000\text{kg}}{\text{m}^3} \right) \left(\frac{1\text{kcal}}{\text{kg} - ^\circ\text{C}} \right) (100 - 27) = \frac{146\text{kCal}}{1\text{hr}}$$

$$Q_e = (m)h_{fg} = \left(\frac{100\text{kg water}}{\text{hr}} \right) \left(\frac{540\text{kCal}}{\text{kg}} \right) = \frac{54,000\text{kCal}}{\text{hr}}$$

$$P_i \text{ heating} = \frac{146,000\text{kCal}}{0.4 \times 0.7} = \frac{521,428.6\text{kCal}}{\text{hr}}$$

$$P_i \text{ evaporating} = \frac{\frac{54,000\text{kCal}}{\text{hr}}}{0.4 \times 0.7} = \frac{192,857.1\text{kCal}}{\text{hr}}$$

276. There are 1000 heads of chicken in a poultry farm. Estimate the kilos of dung that can be produced by the chicken within 1 month.

Given:

Animal - poultry

No. of heads - 1000 chicken

Required:

Weight of dung produce in 1 month

Solution:

$$\text{Weight of Dung} = \left(\frac{0.075\text{kg}}{\text{day} - \text{animal}} \right) (1000 \text{ animals})(30 \text{ days}) = 2,250 \text{ kg}$$

277. A digester requires 1000 kilos of substrates per day. How many heads of hogs are needed to sustain the requirement of the digester? Assume a water to manure ratio of 1:1.

Given:

Animal- Hogs

Weight of substrate required –

1000 kilos per day

Required:

No of heads of hogs to sustain operation

Solution:

$$\text{No. of Heads} = (500 \text{ kg}) \left(\frac{1 \text{ day}}{8 \text{ kg} - \frac{\text{day}}{\text{animal}}} \right) = 62.5 \text{ hogs or use 63 heads}$$

278. Determine the size of the digester required to produce biogas for a livestock farm having 500 heads of cattle. The projected retention time is 25 days with feed material to water ratio of 1:1.5. Also estimate the amount of gas that can be produced from the digester and the number of gas burner that can be supplied by the system.

Given:

No of heads of cattle – 500

Retention time – 25 days

Feed material to water ratio – 1: 1.5

Density of substrate – 1000 kg/m³

Required:

Size of the digester and number of burner that can be supplied by the system

Solution:

$$\text{Weight of manure} = (500 \text{ heads}) \left(\frac{14 \text{ kg}}{\text{day} - \text{head}} \right) = 7,000 \text{ kg/day}$$

$$\text{Volume of manure} = \left(\frac{7,000 \text{ kg}}{\text{day}} \right) \left(\frac{1 \text{ m}^3}{1000 \text{ kg}} \right) = \frac{7 \text{ m}^3}{\text{day}}$$

$$\text{Volume of Slurry} = \frac{7 \text{ m}^3}{\text{day}} + \frac{7 \text{ m}^3}{\text{day}} (1.5) = \frac{17.5 \text{ m}^3}{\text{day}}$$

$$\text{Volume of Digester} = \left(\frac{17.5 \text{ m}^3}{\text{day}} \right) (25 \text{ days}) = 437.5 \text{ m}^3$$

$$\text{Gas produced per day} = \left(\frac{0.03 \text{ m}^3}{\text{kg}} \right) \left(\frac{7,000 \text{ kg}}{\text{day}} \right) = \frac{210 \text{ m}^3}{\text{day}}$$

$$\text{Number of Burners} = \frac{\left(\frac{0.42 \text{ m}^3}{\text{hr} - \text{burner}} \right)}{\left(\frac{210 \text{ m}^3}{\text{day}} \right) \left(\frac{1 \text{ day}}{6 \text{ hrs}} \right)} = 35 \text{ burners}$$

279. Compute the diameter of the solid shaft of a fluted mill rotating at a speed of 100 rpm and transmitting power of 1.2 hp. Assume a 6000-psi design stress for shear. Consider that the shaft is subjected to torsion only.

Given:

N – 100 rpm

S_s – 6000 psi

Hp – 1.2

Required:

Shaft diameter

Solution:

$$T = \frac{P}{2\pi N} = \frac{(1.2 \text{ hp})(33,000 \text{ ft} - \text{lb})(12 \text{ in})}{2\pi(100 \text{ rpm})} = 756.3 \text{ in} - \text{lb}$$

$$D = \sqrt[3]{\frac{16T}{\pi\sigma_d}} = \left(\frac{16 \times 756 \text{ in} - \text{lb}}{\pi \times 6000 \text{ psi}}\right)^{\frac{1}{3}} = 0.863 \text{ in. use 1in diameter shaft}$$

280. A solid steel shaft of a huller is needed to deliver 2.5 kW when rotating at 1200 rpm. Find the minimum diameter of the shaft if its allowable shear stress is 40 MN/m².

Given:

Hp – 2.5kW

Shaft speed – 1200 rpm

Allowable shear – 40 MN/m²

Required:

Minimum shaft diameter

Solution:

$$T = \frac{P}{2\pi N} = \frac{\left(\frac{2500\text{N} - \text{m}}{\text{sec}}\right)}{2\pi(1200\text{rpm})\left(\frac{1\text{min}}{60\text{sec}}\right)} = 19.89\text{N} - \text{m}$$

$$D = \sqrt[3]{\frac{16T}{\pi\sigma_d}} = \sqrt[3]{\frac{16(19.89\text{N} - \text{m})}{\pi\left(\frac{40 \times 10^6 \text{N}}{\text{m}^2}\right)}} = 0.0136\text{m or } 13.63\text{mm}$$

281. Determine the speed of a hammer mill driven by an electric motor at a speed of 1740 rpm. The motor pulley is 4 inches while the milling machine pulley is 3 inches. Assume that there is no slip on the drive system.

Given:

Motor pulley - 4 in.

Motor speed - 1740 rpm

Machine pulley - 3 in.

Required: Machine speed

Solution:

$$N_1 D_1 = N_2 D_2$$

$$N_1 = \frac{N_2 D_2}{D_1} = \frac{(1740\text{rpm})(4\text{in})}{3\text{in}} = 2320\text{rpm}$$

282. A five-row reaper, with a spacing of 30 cm per row, is harvesting at a speed of 2kph. The field efficiency of the reaper is 70%. If the specific draft of the machine is 2 kg per row, what is the power output of the reaper?

Given:

No. of row - 5

Row spacing - 30 cm

Speed of harvesting - 2kph

Field efficiency - 70%

Specific draft - 2 kg/row

Required:

Power of the reaper and effective field capacity

Solution:

$$P = \left(\frac{2\text{kg}}{\text{row}}\right)(5\text{rows})\left(\frac{2000\text{m}}{\text{hr}}\right)\left(\frac{1\text{hr}}{3600\text{sec}}\right) = \frac{5.56\text{kg} - \text{m}}{\text{sec}} \text{ or } 5.56\text{Watts or } 0.00556\text{kW}$$

$$TFC = \frac{SW}{10} = \frac{(2\text{kph})(0.3 \times 5)}{10} = \frac{0.30\text{ha}}{\text{hr}}$$

$$EFC = \frac{SWEff}{10} = \frac{(2\text{kph})(0.3 \times 5)(0.70)}{10} = \frac{0.21\text{ha}}{\text{hr}}$$

283. A 6-meter wide harvester is harvesting corn at a rate of 100 kg per minute. The effective width of the machine is 90% of its width. If it runs at a speed of 2.5kph, what is the effective field capacity and harvesting rate of the machine in tons per hectare? Assume a machine field efficiency of 70%.

Given:

Width of harvester - 6 meters
 Effective width - 90% of machine width
 Harvesting rate - 100 kg per minute
 Speed of harvesting - 2.5kph
 Field efficiency - 70%

Required:

Effective field capacity and harvesting rate in tons per hectare

Solution:

$$EFC = \frac{SWEff}{10} = \frac{(2.5\text{kph})(6\text{m} \times 0.90)(0.70)}{10} = \frac{0.945\text{ha}}{\text{hr}}$$

$$\text{Harvesting Rate} = \frac{\left(\frac{100\text{kg}}{\text{minute}}\right)\left(\frac{1\text{ton}}{1000\text{kg}}\right)\left(\frac{60\text{min}}{\text{hr}}\right)}{\frac{0.945\text{ha}}{\text{hr}}} = \frac{6.35\text{ton}}{\text{ha}}$$

284. Compute the apparent air velocity in a 1-ton-capacity flat-bed dryer having a dimension of 2.4 m x 2.4 m square bin. The specific airflow rate of the dryer is 50 m³ per minute of air per m³ of grain. Assume a density of grain of 500 kg/m³.

Given:

Total airflow - 50 m³ /min
 Area - 2.4 m x 2.4 m
 Grain drying - 500 kg/m³

Required:

Apparent air velocity

Solution:

$$\text{Volume} = (1\text{ton})\left(\frac{1000\text{kg}}{\text{ton}}\right)\left(\frac{\text{m}^3}{500\text{kg}}\right) = 2\text{m}^3$$

$$Q = \left(\frac{50\text{m}^3}{\text{min} - \text{m}^3}\right)(2\text{m}^3) = 100\text{m}^3/\text{min}$$

$$V_{\text{app}} = \frac{\left(\frac{100\text{m}^3}{\text{min}}\right)}{2.4\text{m} \times 2.4\text{m}} = 17.36\text{m}/\text{min}$$

285. A solar dryer is used to reduce the moisture content of four tons of paddy rice from 25% to 15% wet basis. What is the amount of moisture that can be removed from the paddy?

Given:

MC initial - 25%
 MC final - 14%
 Initial weight - 4 tons

Required:

Weight of moisture removed

Solution:

$$W = 4\text{tons} - \frac{4\text{tons}(1 - 0.25)}{1 - 0.14} = 0.512\text{ton or } 512\text{kg}$$

286. A single-phase 220-V electric motor has a current input of 4.5 amp. If the power factor of the motor is estimated to be at 0.98, what is the power input of the motor?

Given:

Input voltage - 220 volt

Input current - 4.5 amp

Power factor - 0.98

Required:

Power input

Solution:

$$P = VI\cos\theta = (220\text{V})(4.5\text{amp})(0.98) = 970.2\text{Watts}$$

287. The drive shaft of an engine is driving the rice mill line shaft that drives a 24-inches flat pulley in a crossed-drive assembly. The drive shaft pulley has a diameter of 6 inches. The distance between pulleys measured from the center of the shaft is 20 ft. What would be the length of the flat belt needed for the drive? Add 1ft from the computed belt length for splicing purposes.

Given:

D1 – 6 in.

D2 – 24 in.

C = 20ft

Required:

Length of belt

Solution:

$$L = 2C + 1.57(D1 + D2) + \frac{(D1 + D2)^2}{4C}$$
$$L = 2(20\text{ft}) + 1.57(0.5\text{ft} + 2\text{ft}) + \frac{(0.5\text{ft} + 2\text{ft})^2}{4(20\text{ft})} = 44\text{ft or } 45\text{ft}$$

288. If a NSK 6200 single-row deep-groove ball bearing will be used to drive a pulley that delivers 119.8kgf, how many revolution does the bearing last? If the bearing runs at 1800 rpm, how long it will last?

Given:

Bearing - 6200 single-row deep groove

Actual load - 119.8kgf

N - 1800 rpm

Required:

Fatigue life in rev and number of hours the bearing will last

Solution: From bearing catalogue, basic dynamic load of 6200 ball bearing is 400kgf

$$L_r = \left(\frac{C}{P}\right)^3 = \left(\frac{400\text{kgf}}{119.8\text{kgf}}\right)^3 = 37.22 \times 10^6 \text{rev}$$

$$L_h = \left(\frac{33.33}{1800}\right)^3 (37,220,000 \text{ rev}) = 997.86 \text{ hours}$$

289. Rice husk fuel will be used as a replacement for LPG fuel in providing heat to a poultry farm. If two 50-kg LPG tanks are needed per 24-hour operation, what would be the amount of rice husk fuel (in kilogram) that will be consumed per hour? Assume a heating value for the LPG of 11,000 kcal/kg and a 60% conversion efficiency for the system. Rice husk heating value is 3,000 kcal/kg.

Given:

Replacement fuel - rice husk

Fuel to be replaced - LPG

Weight of LPG - two 50-kg

Time of use - 24hrs

LPG heating value - 11,000 kcal/kg

System conversion efficiency - 60%

Required: Weight of rice husk fuel needed per hour

Solution:

$$\text{Energy consumption using LPG} = \frac{(2\text{tanks}) \left(\frac{50\text{kg}}{\text{tank}} \right) \left(\frac{11,000\text{kcal}}{\text{kg}} \right)}{24\text{hrs}} = \frac{45,833.33\text{kcal}}{\text{hr}}$$

$$\text{Energy required to replace LPG} = \frac{\frac{45,833.33\text{kcal}}{\text{hr}}}{0.6} = \frac{76,388.89\text{kcal}}{\text{hr}}$$

$$\text{Weight of rice husk needed per hour} = \frac{\frac{76,388.89\text{kcal}}{\text{hr}}}{\frac{3,000\text{kcal}}{\text{kg}}} = \frac{25.46\text{kg}}{\text{hr}}$$

290. The result of the ultimate analysis for coal are as follows: carbon content - 81.3% by weight; hydrogen content - 5.3%; oxygen content - 9.8%; and sulphur content - 1.9%. What is the stoichiometric air of coal?

Given:

C - 81.3%

H - 5.3%

O - 9.8%

S - 1.9%

Required: Stoichiometric Air

Solution:

$$\text{Stoichiometric Air} = \frac{[2.67 (0.813) + 8 (0.053) + 1 (0.019) - 0.098]}{0.23} = \frac{10.9\text{kg air}}{\text{kg of fuel}}$$

291. If a mechanical dryer consumes 9 liters per hour of diesel fuel during drying operation, what is the equivalent amount of rice husk (in kg) per hour needed to replace the diesel? Assume a 70% efficiency for diesel burner and a 40% efficiency for rice husk furnace. The heating value of diesel is 9,000 kcal/liter at 0.9 specific gravity while the energy content of rice husk is 3,000 kcal/kg.

Given:

Dryer consumption - 9 li of diesel per hr.

Diesel burner efficiency - 70 %

Rice husk furnace efficiency - 40%

Required: Rice Husk Consumption Rate

Solution:

$$\text{fuel energy production rate} = \left(\frac{9\text{li}}{\text{hr}}\right) \left(\frac{9,000\text{kcal}}{\text{li}}\right) = 81,000 \text{ kcal/hr}$$

$$\text{fuel replacement rate} = \frac{\left[\frac{(81,000\text{kcal})}{\text{hr}}\right]}{\frac{3,000\text{kcal}}{\text{kg}}} \div \frac{0.7 \times 0.4}{1} = 96.43 \text{ kg/hr}$$

292. The thermal efficiency of a 1.2-kW biomass stove is 15%. If wood, with a heating value of 4,000 kcal per kg, is used as fuel, what is the amount of wood consumption per hour?

Given:

Po - 1.2 kW

Eff_{th} - 15%

HVF - 4,000 kcal/kg

Required: Fuel Consumption Rate

Solution:

$$\text{FCR} = \frac{1.2\text{kW}}{\frac{0.0012\text{kW}}{\frac{\text{kcal}}{\text{hr}}} \left(\left(\frac{4,000\text{kcal}}{\text{kg}} \right) (0.15) \right)} = \frac{1.7\text{kg}}{\text{hr}}$$

293. What is the heating value of fuel with ultimate analysis result as follows: carbon - 38.9%; hydrogen - 4.81%; sulphur - 0.10%; and oxygen - 34.4%?

Given:

C - 38.9 %

H - 4.81 %

S - 0.10 %

O - 34.4 %

Required: Heating value of fuel

Solution:

$$\text{HV} \left(\frac{\text{kJ}}{\text{kg}} \right) = 33,823(0.389) + 144,250 \left(\frac{0.0481 - 0.344}{8} \right) + 9,419(0.001) = \frac{13902.245\text{kJ}}{\text{kg}}$$

294. What is the lower heating value of a biomass fuel with 14.5 MJ/kg heating value obtained from a bomb calorimeter? The moisture content of fuel is 9% wet basis.

Given:

HHV - 14.5 MJ/kg

MC - 9 % wb

Required: Lower Heating Value

Solution:

$$\text{LHV} = 14.5(1 - 0.09) - 2.477(0.09) = \frac{12.97\text{MJ}}{\text{kg}}$$

295. Calculate the indicated horsepower of a 6-cylinder engine with 4-in. bore and 4-in. stroke. It is running at a speed of 1500 rpm with 80 psi mean effective pressure.

Given:

No. of cylinders - 6

Bore, D - 4 in.

Stroke, L - 4 in.

RPM, N - 1500 rpm

Pressure, P - 80 psi

Required: Indicated Horsepower

Solution:

$$\text{IHP} = \text{PLAN} = \frac{\left(\frac{80\text{lb}}{\text{in}^2}\right)(4\text{in})\left(\frac{\pi(4\text{in})^2}{4}\right)(1500\text{rpm})(6)\left(\frac{1\text{ft}}{12\text{in}}\right)}{\frac{33,000\text{ft} - \text{lb}}{\text{min} - \text{hp}}(2)} = 45.70\text{hp}$$

296. An engine was tested on a brake dynamometer at a speed of 2000 rpm. The brake arm of the machine is 1.2 meters and it registers a load of 5 kg. What is its shaft power?

Given:

Engine speed - 2000 rpm

Brake arm length - 1.2 m

Force registered - 5 kg

Required: Engine Shaft Power

Solution:

$$P = 2\pi FN\alpha = \frac{2\pi(5\text{kg})(2000\text{rpm})(1.2\text{m})}{\left(\frac{76.2\text{kg} - \text{m}}{\text{sec}}\right)(60\text{sec})} = 16.50\text{hp}$$

297. What is the thermal efficiency of an engine that uses 0.6-lb of fuel per horsepower-hr? The fuel contains 20,000 BTU/lb.

Given:

Specific fuel consumption - 0.6lb of fuel/hp.-hr.

Heating value - 20,000 BTU/lb.

Required: Brake Thermal Efficiency

Solution:

$$\text{Thermal Efficiency} = \frac{\left(\frac{2545\text{BTU}}{\text{hr} - \text{hp}}\right)}{\left(\frac{0.6\text{lb of fuel}}{\text{hp} - \text{hr}}\right)\left(\frac{20,000\text{BTU}}{\text{lb}}\right)} = 21.2\%$$

298. The indicated power of an engine is 69 horsepower. The brake horsepower is 54. What is the mechanical or engine efficiency?

Given:

IHP - 69hp

BHP - 54hp

Required: Mechanical Efficiency

Solution:

$$\text{M. E.} = \frac{54\text{hp}}{69\text{hp}} = 78.26\%$$

299. A two-cylinder, four-stroke cycle engine has 100-mm piston diameter and 115-mm stroke. The engine runs at 1500 rpm with 120 psi effective mean pressure. What is the indicated power of the engine, in kilowatt? If the engine has 12 compression ratio, what is its clearance volume?

Given:

No. Cylinder - 2

No. of Stroke - 4

Piston Diameter - 100 mm

Length of Stroke - 115 mm
 Engine Speed - 1500 rpm
 Mean Effective pressure - 120 psi
 Compression Ratio - 12

Required:

Indicated Power, in kW
 Clearance Volume

Solution:

$$\text{IHP} = \text{PLANn} = \frac{\left(\frac{120\text{lb}}{\text{in}^2}\right) \left(\frac{115\text{mm}}{25.4\text{mm}}\right) \left(\frac{1\text{ft}}{12\text{in}}\right) \left(\frac{\pi \left(\frac{100}{25.4}\right)^2}{4}\right) (1500\text{rpm})(2)}{(2) \left(\frac{33,000\text{ft} - \text{lb}}{\text{min}}\right)} = 18.80\text{kW}$$

$$\text{CR} = \frac{\text{PD} + \text{CV}}{\text{CV}}$$

$$\text{CV} = \frac{\text{PD}}{\text{CR} - 1} = \frac{\frac{\pi(100\text{mm})^2(115\text{mm})}{4}}{12 - 1} = 82109.81\text{mm}^3 \text{ or } 82.1\text{cm}^3$$

300. A 6-cylinder, 4-stroke-cycle gas engine has 123-mm piston diameter and 124-mm stroke. It has a 12 compression ratio with 95% volumetric efficiency. The maximum power is 340hp at 2200 rpm while the maximum torque is 160kgf-m at 1260 rpm. What is the volume of gas the engine can receive in m³ per hour if it is operated at maximum power? Consider an air requirement of 20% for the combustion of fuel. What is the engine input power in kW if the gas has 1,500 kcal/m³ heating value? If the brake thermal efficiency of the engine is 25%, what is its brake horsepower?

Given:

No. of Cylinder - 6
 No. of Stroke - 4
 Piston Diameter - 123 mm
 Piston Stroke - 124 mm
 Compression Ratio - 12
 Volumetric Efficiency - 95%
 Maximum Power – 340hp
 Engine Speed @ max power - 2200 rpm
 Maximum Torque – 160kgf-m
 Engine Speed @ max torque - 1260 rpm

Required:

Volume of Gas (in m³/hr.) at maximum engine power at 20% air requirement
 Engine Power Input (in kW) at 1,500 kcal/m³ of air
 Brake Power at brake thermal efficiency of 25%

Solution:

$$\text{PD} = \frac{\pi(12.3\text{cm})^2(12.4\text{cm})}{4} = 1473.40\text{cm}^3$$

$$\text{CV} = \frac{\text{PD}}{\text{CR} - 1} = \frac{1473.40\text{cm}^3}{12 - 1} = 134\text{cm}^3$$

$$\text{Total Volume} = 1607.34\text{cm}^3$$

For 6 Cylinders = $(1607.34\text{cm}^3)(6) = 9644.07\text{cm}^3$ or 9.644liters

$$\text{Volume of gas} = \frac{(9.644\text{liters})(2200\text{rpm})(0.95)}{2} = \frac{10077.98\text{li}}{\text{min}} \text{ or } \frac{604.67\text{m}^3}{\text{hr}}$$

$$\text{Power} = \left(\frac{604.67\text{m}^3}{\text{hr}}\right)\left(\frac{1500\text{kcal}}{\text{m}^3}\right)(0.80) = \left(\frac{725604\text{kcal}}{\text{hr}}\right)\left(\frac{0.0012\text{kW} - \text{hr}}{1\text{kcal}}\right) = 870.72\text{kW}$$

$$\text{BHP} = (870.72\text{kW})(0.25) = 217.68\text{kW} \text{ or } 291.7\text{hp}$$

301. A 2-stroke, single-cylinder engine with 80-mm bore and 85-mm stroke develops a 30 bar pressure. What is the indicated power of the engine if it runs at a speed of 2000 rpm? What is the brake power of the engine if it has 80% mechanical efficiency? What is the amount of gasoline fuel it consumes if it has 25% thermal efficiency? Assume an 11,200 kcal/kg heating value of gasoline and 0.76 specific gravity.

Given:

No. of Stroke - 2

No. of Cylinder - 1

Engine Bore - 80 mm

Piston Stroke - 85 mm

Pressure - 30 bar

Engine Speed - 2000 rpm

Fuel Heating Value - 11,200 kcal/kg

Specific Gravity - 0.76

Required:

Indicated Power of the engine

Brake Power at 80% mechanical efficiency

Amount of Gasoline Consumed at 25% brake thermal efficiency

Solution:

$$\text{IHP} = \frac{\left(\frac{30 \times 14.5\text{lb}}{\text{in}^2}\right)\left(\frac{85\text{mm}}{25.4\text{mm}}\right)\left(\frac{\pi\left(\frac{80}{25.4}\right)^2}{4}\right)(2000\text{rpm})\left(\frac{1\text{ft}}{12\text{in}}\right)}{\frac{33,000\text{ft} - \text{lb}}{\text{min}}} = 57.28\text{hp}$$

$$\text{BHP} = \text{IHP} \times \text{M. Eff} = (57.28\text{hp})(0.80) = 45.82\text{hp}$$

$$\text{fuel consumed} = \frac{(45.82\text{hp})\left(\frac{2545\text{BTU}}{\text{hr} - \text{hp}}\right)\left(\frac{0.252\text{kcal}}{\text{BTU}}\right)}{\left(\frac{11,200\text{kcal}}{\text{kg}}\right)\left(\frac{0.76\text{kg}}{\text{lit}}\right)(0.25)} = \frac{13.82\text{lit}}{\text{hr}}$$

302. An 11.1-liter, 6-cylinder, 4 stroke engine has 120-mm bore and 125-mm stroke. If the compression ratio of the engine is 16, what is its clearance volume? What is the engine displacement per minute if it runs at 2,000 rpm? If biogas with an energy content of 4,500 kcal/m³ will be fueled into the engine, what would be its power output in kW considering a 10% brake thermal efficiency?

Given:

Engine Displacement - 11.1 li

No. of Cylinders - 6 cylinders

No. of Stroke - 4

Engine Bore - 120 mm

Length of Stroke - 125 mm

Compression Ratio - 16

Required:

Clearance Volume

Engine Displacement per min at 2,000 rpm

Power Output using biogas as fuel with 4,500 kcal/m³ heating value and 10% brake thermal efficiency

Solution:

$$CV = \frac{11.1 \text{ liters}}{16 - 1} = 0.69 \text{ lit}$$

$$\text{Engine Displacement per minute} = \frac{(11.1 \text{ liter})(2000 \text{ rpm})}{2} = \frac{11,100 \text{ liters}}{\text{min}}$$

$$P_{\text{Output}} = \left(\frac{11,100 \text{ liters}}{\text{min}} \right) \left(\frac{1 \text{ m}^3}{1000 \text{ lit}} \right) \left(\frac{60 \text{ min}}{\text{hr}} \right) \left(\frac{4500 \text{ kcal}}{1 \text{ m}^3} \right) (0.10) \left(\frac{0.0012 \text{ kW} - \text{hr}}{\text{kcal}} \right) = 482 \text{ hp}$$

303. The specific fuel consumption of a 200-hp diesel engine is 250 g/kW-hr. How much would be the fuel cost if it is to be operated for 10 hours? Assume a 0.79 kg per liter specific gravity of fuel and P47 cost per liter of diesel.

Given:

SFC - 300 g diesel/kW-hr

To - 10hrs

BHP - 250hp

SG_{fuel} - 0.79 kg/li

Price - P47 per li

Required:

Cost of Fuel, in 10 hours

Solution:

$$\text{BHP} = (250 \text{ hp})(0.746 \text{ kW}) = 186.5 \text{ kW}$$

$$\text{Fuel Cost} = (186.5 \text{ kW}) \left(\frac{300 \text{ g}}{\text{kW} - \text{hr}} \right) (10 \text{ hrs}) \left(\frac{1 \text{ kg}}{1000 \text{ g}} \right) \left(\frac{1 \text{ lit}}{0.79 \text{ kg}} \right) = 708.23 \text{ liters} \left(\frac{\text{P}47}{\text{liter}} \right) \\ = \text{P}33,286.71$$

304. The indicated horsepower of the tractor engine is 300 hp. If the specific fuel consumption of the tractor engine during plowing operation is 350 g of diesel/kW-hr, how much diesel the tractor engine will consume in 8 hours operation? If the cost of diesel is P42 per liter, how much is the total cost of fuel needed by the tractor?

Given:

IHP - 300hp

SFC - 350 g diesel/kW-hr

SG_{fuel} - 0.76 kg/li

T O - 8hr

Required: Fuel Consumption and Cost

Solution:

$$\text{FC} = \frac{\left(\frac{350 \text{ g}}{\text{kW} - \text{hr}} \right) \left(\frac{1 \text{ kg}}{1000 \text{ g}} \right) (300 \text{ hp})(0.746 \text{ kW})(8 \text{ hrs})}{\frac{0.76 \text{ kg}}{\text{lit}}} = 824.53 \text{ liters}$$

$$\text{Cost} = (824.53 \text{ liters}) \left(\frac{\text{P}42}{\text{liter}} \right) = \text{P}34,630$$

305. A four-stroke cycle diesel engine, with three cylinder, has 89-mm cylinder bore and 130-mm stroke. The compression ratio is 16:1 and it is running at 2000 rpm speed. What is the piston displacement of the engine per cylinder? What is the total power stroke per minute of the piston and the total displacement volume per min of the engine? What is the piston speed of the engine? Also, calculate the stroke-to-bore ratio.

Given:

Engine - 4 stroke diesel
 No. of Cylinders - 3
 Bore - 89 mm
 Stroke - 130 mm
 Compression Ratio - 16:1
 Speed - 2,000 rpm

Required:

Area of the Piston
 Power Stroke per min
 Total Volume Displacement
 Piston Speed
 Stroke-to-Bore Ratio

Solution:

$$A_{\text{piston}} = \frac{\pi(8.9\text{cm})^2}{4} = 62.21\text{cm}^2$$

$$\text{Volume} = (62.21\text{cm}^2)(13\text{cm}) = 808.73\text{cm}^3$$

$$\text{Power Stroke per minute} = (2000\text{rpm}) \left(\frac{1\text{cylinder}}{2\text{rev}} \right) (3\text{cylinders}) = \frac{3000 \text{ power strokes}}{\text{min}}$$

$$\text{Total Volume Displacement} = \left(\frac{808.73\text{cm}^3}{\text{cylinder}} \right) (3 \text{ cylinders}) = 2426.2\text{cm}^3$$

$$\text{Piston Speed} = \left(\frac{2000\text{rev}}{\text{min}} \right) \left(0.13\text{m} \times \frac{2}{\text{rev}} \right) = \frac{520\text{m}}{\text{min}}$$

$$\text{Stroke - to - Bore Ratio} = \frac{130\text{mm}}{89\text{mm}} = 1.46:1$$

306. A four-cylinder gas engine has 20-cm cylinder diameter and 40-cm stroke. The cylinders run at a speed of 150 rpm. The engine is a four-stroke cycle with 8 kg/cm² mean effective pressure. What is the area in cm² and length of stroke of the engine piston in meter? Also, compute the power stroke per min per cylinder and the total power stroke of the 4 cylinders. Compute the indicated horsepower and the brake horsepower of the engine. And, what is the stroke-to-bore ratio of the engine?

Given:

Engine - 4 cylinders gas fueled
 Piston diameter - 20 cm
 Stroke - 40 cm
 Speed - 1500 rpm
 Mean effective pressure - 8 kg/cm²
 Mechanical .Eff. - 80%

Required:

Area and Length of Piston

Power Stroke per cylinder

Total Power Stroke of the 4 cylinders

Indicated Horsepower

Stroke-to-Bore Ratio

Solution:

$$A_{\text{piston}} = \frac{\pi(40\text{cm})^2}{4} = 1256.64\text{cm}^2$$

$$\text{Piston length} = 0.4\text{m}$$

$$\text{Power Stroke per cylinder} = \frac{\left(\frac{1500\text{rev}}{\text{min}}\right)}{2} = 750 \frac{\text{strokes}}{\text{min}}$$

$$\text{Power Stroke per cylinder} = \left(\frac{750\text{strokes}}{\text{min}}\right)(4) = \frac{3000\text{strokes}}{\text{min}}$$

$$\text{IHP} = \text{PLANn} = \frac{\left(\frac{8\text{kg}}{\text{cm}^2}\right)(0.40\text{m})\left(\frac{\pi(20\text{cm})^2}{4}\right)(1500\text{rpm})\left(\frac{1\text{min}}{60\text{sec}}\right)(4)}{\frac{76.2\text{kg} - \text{m}}{\text{sec}}(2)} = 660\text{hp}$$

$$\text{BHP} = (660\text{hp})(0.8) = 528\text{hp}$$

$$\text{Stroke} - \text{to} - \text{Bore Ratio} = \frac{40\text{cm}}{20\text{cm}} = 2:1$$

307. What is the brake constant value of a dynamometer with length of arm equal to 1 m? If the dynamometer registers 100-kg force at 2000 rpm, what is the brake horsepower of the engine?

Given:

Length of arm - 1 m

Engine speed - 2000 rpm

Force - 100 kg

Required: Brake Constant and Brake Horsepower

Solution:

$$\text{Brake Constant} = \frac{2\pi(1\text{m})}{\frac{76.2\text{kg} - \text{m}}{\text{sec} - \text{hp}}} = \frac{0.0824\text{sec} - \text{hp}}{\text{kg}}$$

$$\text{BHP} = \frac{2\pi(100\text{kg})\left(\frac{2000}{60}\right)(1\text{m})}{\frac{76.2\text{kg} - \text{m}}{\text{sec}}} = 274.86\text{hp}$$

308. A gasoline engine was tested on a prony brake dynamometer. The engine runs at a speed of 200 rpm and registers 100-kg load to a 0.8-m brake arm. What is the BHP of the engine?

Given:

Engine speed - 200 rpm

Force - 100 kg

Brake arm - 0.8 m

Required: Brake Horsepower

Solution:

$$\text{BHP} = 2\pi FN\alpha = \frac{2\pi(100\text{kg})\left(\frac{200}{60}\right)(0.8)}{\frac{76.2\text{kg} - \text{m}}{\text{sec}}} = 22\text{hp}$$

309. What is the brake thermal efficiency of an engine that consumes gasoline at 1.2 liters per hour and with 8-hp power output? The heating value of the fuel is 45.7 MJ/kg and its density is 0.76 kg/liter.

Given:

Fuel consumed - 1 li/hr.

Power output - 8hp

Heating value of fuel - 45.7 MJ/kg

Fuel density - 0.76 kg/li

Required: Brake Thermal Efficiency

Solution:

$$\text{BTE} = \frac{(8\text{hp})\left(\frac{0.746\text{kW}}{\text{hp}}\right)}{\left(\frac{1\text{lit}}{\text{hr}}\right)\left(\frac{45.7\text{MJ}}{\text{kg}}\right)\left(\frac{1000\text{kJ}}{\text{MJ}}\right)\left(\frac{0.76\text{kg}}{\text{lit}}\right)\left(\frac{1\text{kCal}}{4.188\text{kJ}}\right)\left(\frac{0.0012\text{kW} - \text{hr}}{1\text{kCal}}\right)} = 60\%$$

310. What is the efficiency of a $\frac{3}{4}$ -horsepower (hp) motor that requires an input of 1000 watts (W) of electric power?

Given:

Power input (Pin) - 1000 W

Power out (Pout) - $\frac{3}{4}$ hp

Required: Efficiency

Solution:

$$P = 1000\text{Watts or } 1\text{kW} \times 1.34\text{hp} = 1.34\text{hp}$$

$$\text{Eff} = \frac{0.75\text{hp}}{1.34\text{hp}} = 56\%$$

311. 10-kWe micro-hydro power generating device is to be constructed for barangay electrification project. If the total cost of materials is P120,000.00, how much is the direct cost for the project? Assume a labor cost of 40% of the material cost plus 2% rental for the facility in the fabrication of the machine. How much is the indirect cost for the project above if the OC&M is 30% and the mobilization cost is 5%? How much is the profit margin of the project above if the mark-up is 15%? What is the total Fabrication Cost? If the installation, delivery, testing, and commissioning costs of the project above are P10,000, P5,000, P1,000, and P2,000 respectively, how much is the total project cost? Assume a 12% VAT.

Given:

Micro-hydro rated capacity - 10kWe

Material cost - P120,000.00

Labor cost - 40% of material cost

Facility rental - 2% of material cost

OC & M - 30%

Mobilization cost - 5%

Mark-up - 15%

Installation - P10,000.00

Delivery - P5,000.00

Testing -P1,000.00

Commissioning - P2,000.00=

VAT - 12%

Required:

Direct Cost of the project, Indirect Cost, Profit Margin, Fabrication Cost, and Total Project Cost

Solution:

$$\text{Direct Cost} = \text{P}120,000 + (\text{P}120,000 \times 0.40) + (\text{P}120,000 \times 0.02) = \text{P}170,400$$

$$\text{Indirect Cost} = \text{P}170,400(0.30) + \text{P}170,400(0.05) = \text{P}59,640$$

$$\text{Profit Margin} = (\text{P}170,400 + \text{P}59,640)(0.15) = \text{P}34,506$$

$$\text{Fabrication Cost} = \text{P}170,400 + \text{P}59,640 + \text{P}34,506 = \text{P}264,546$$

$$\text{Total Project Cost} = (1.12)(\text{P}264,546 + \text{P}10,000 + \text{P}5,000 + \text{P}1,000 + \text{P}2,000) = \text{P}316,451.52$$

312. Ten tons of paddy is to be sundried from 20% to 14% wet basis. What is the amount of moisture that can be removed from the paddy if it will be exposed under the sun for 6 hours and what is the rate of moisture removal per hour? How much energy is required to remove moisture per hour? What is the approximate area of the drying floor needed if the collection efficiency of the drying system is 60%? Assume a 0.89 kW/m² average power density and a 1,200 kcal/kg heat of vaporization of water from the grain. If paddy has 580 kg/m³ density, what should be its approximate thickness on the drying floor?

Given:

MC content - 20% to 14 % in 6hrs

Weight of paddy - 10 tons

Heat of vaporization - 1,200 kcal/kg

Average power density - 0.89 kW/m²

Collection efficiency - 60%

Paddy density - 580 kg/m³

Required:

Weight of Moisture Removed

Rate of Moisture Removal

Energy Required

Drying Floor Area

Paddy Thickness on the drying floor

Solution:

$$W_{\text{moisture}} = 10,000\text{kg} - \frac{10,000\text{kg}(1 - 0.2)}{1 - 0.14} = 697.67\text{kg or } 0.7\text{ton}$$

$$\text{Rate of Moisture removal} = \frac{0.7\text{ton}}{6\text{hrs}} = \frac{0.12\text{ton}}{\text{hr}} \text{ or } \frac{116\text{kr}}{\text{hr}}$$

$$\text{Required Energy} = \left(\frac{116\text{kg}}{\text{hr}}\right)\left(\frac{1200\text{kCal}}{1\text{kg}}\right) = \left(\frac{139,200\text{kCal}}{\text{hr}}\right)\left(\frac{0.0012\text{kW} - \text{hr}}{1\text{kCal}}\right) = 167\text{kW}$$

$$\text{Drying Floor Area} = \frac{168\text{kW}}{\left(\frac{0.89\text{kW}}{\text{m}^2}\right)(0.60)} = 313\text{m}^2$$

$$\text{Floor Thickness} = \frac{(10\text{tons})(1000\text{kg})}{\left(\frac{580\text{kg}}{\text{m}^3}\right)(313\text{m}^2)} = 0.055\text{m}$$

313. Determine the size of the flat-plate solar collector, which is perpendicularly inclined at 11 degrees to the direction of the sunlight, required to supply thermal heat to a fruit dryer that consumes 1000 kcal per hour of heat energy. Assume a 0.89 kW per m² average power density and a 0.77 transmission factor.

Given:

Angle of inclination - 11 degrees

Heat required - 1,000 kcal/hr.

Average power density - 0.89 kW/m²

Transmission factor - 0.77

Required: Collector Area

Solution:

$$A = \frac{mC_p\Delta T \left(\frac{1,000\text{kCal}}{\text{hr}} \right) \left(\frac{0.0012\text{kW} - \text{hr}}{1\text{kCal}} \right)}{\eta R_g \cos\theta \left(\frac{0.89\text{kW}}{\text{m}^2} \right) (0.77)(\cos 11)} = 17.84\text{m}^2$$

314. Two hundred (200) liters of water is to be heated from 30° to 80°C in 1 hour. What is the area of the tube-type solar collector required for the heater if the collection efficiency is 80% directly oriented to the sunlight? The average power density of the sun is 0.89 kW/m².

Given:

Volume of water - 200 li

Temperature - 30° to 80°C

Heating time - 1hr

Collection efficiency - 80%

Average power density - 0.89 kW/m²

Required: Collector Area

Solution:

$$\begin{aligned} \text{Power required} &= mC_p\Delta T = \left(\frac{200\text{lit}}{1\text{hr}} \right) \left(\frac{1\text{kg}}{1\text{lit}} \right) \left(\frac{1\text{kCal}}{\text{kg} \cdot ^\circ\text{C}} \right) (80^\circ\text{C} - 30^\circ\text{C}) \\ &= \left(\frac{10,000\text{kCal}}{\text{hr}} \right) \left(\frac{0.0012\text{kW} - \text{hr}}{1\text{kCal}} \right) = 12\text{kW} \\ \text{Area of Collector} &= \frac{12\text{kW}}{\left(\frac{0.89\text{kW}}{\text{m}^2} \right) (0.80)} = 16.85\text{m}^2 \end{aligned}$$

315. An indirect-fired rice husk furnace is to be designed to supply heat for a 6-tonner flatbed dryer to dry paddy from 21% to 14% in 10 hours with 60% efficiency. The thermal efficiencies of the furnace and of the heat exchanger are 40% and 70%, respectively. Determine the power input and the power output of the furnace. What is the grate area and the combustion chamber volume required if the specific burning rate is 60 kg/hr-m² and the specific combustion chamber volume is 90 kg/hr-m³? Assume a heat of vaporization of water from paddy of 1020 kcal/kg.

Given:

Dryer size - 6 tons (21-14% in 10hrs)

Dryer efficiency - 60%

Furnace thermal efficiency - 40%

Heat exchanger efficiency - 70%

Specific burning rate - 60 kg/hr-m²

Spec comb chamber volume - 90 kg/hr-m³

Required:

Power Input and Output of the Furnace

Grate Area and Combustion Chamber Volume

Solution:

$$\text{Water to remove} = 6\text{ton} - \frac{6\text{ton}(1 - 0.21)}{1 - .014} = \frac{0.488\text{ton}}{10\text{hrs}} = \frac{0.0488\text{ton}}{\text{hr}} \text{ or } \frac{48.8\text{kg}}{\text{hr}}$$

$$\text{Energy required} = \left(\frac{48.8\text{kg}}{\text{hr}}\right) \left(\frac{1020\text{kCal}}{\text{kg}}\right) = \frac{49776\text{kCal}}{\text{hr}}$$

$$\text{Power input} = \frac{\left(\frac{49,776\text{kCal}}{\text{hr}}\right)}{(0.60)(0.7)(0.4)} = \left(\frac{296,286\text{kCal}}{\text{hr}}\right) \left(\frac{0.0012\text{kW} - \text{hr}}{1\text{kCal}}\right) = 355.54\text{kW}$$

$$\text{Power Output} = (355.54\text{kW})(0.4)(0.7) = 99.552\text{kW}$$

$$\text{Grate Area} = \frac{\left(\frac{296,286\text{kCal}}{\text{hr}}\right)}{\left(\frac{3,000\text{kCal}}{\text{kg}}\right) \left(\frac{60\text{kg}}{\text{hr} - \text{m}^2}\right)} = 1.65\text{m}^2$$

$$\text{Chamber Volume} = \frac{\left(\frac{296,286\text{kCal}}{\text{hr}}\right)}{\left(\frac{3,000\text{kCal}}{\text{kg}}\right) \left(\frac{90\text{kg}}{\text{hr} - \text{m}^3}\right)} = 1.097\text{m}^3$$

316. A furnace is to be designed for a poultry dressing plant to heat 1200 liters of water from 27° to 70°C within 30 minutes. What is the power input of the furnace, in kW, to heat the water to the required temperature level? What is the amount of wood fuel the furnace will consume if it is to be operated continuously for 8 hours, considering a 60% furnace overall thermal efficiency? Assume a 5,310 kcal/kg heat content of fuel.

Given:

Volume of water to be boiled - 1,200 li

Initial temperature - 27°C

Final temperature - 70°C

Time - 30 min

Furnace efficiency - 60%

Operating time - 8hrs

Required:

Power Input

Amount of Wood Fuel Required for 8 hours' operation

Solution:

$$\begin{aligned} \text{Heat required to raise temperature} &= mC_p\Delta T = (1200\text{lit}) \left(\frac{1\text{kg}}{1\text{lit}}\right) \left(\frac{1\text{kCal}}{1\text{kg} - ^\circ\text{C}}\right) (70^\circ\text{C} - 27^\circ\text{C}) \\ &= \frac{51600\text{kCal}}{0.5\text{hr}} = \left(\frac{103,200\text{kCal}}{\text{hr}}\right) \left(\frac{0.0012\text{kW} - \text{hr}}{1\text{kCal}}\right) = \frac{123.84\text{kW}}{0.6} = 206.4\text{kW} \end{aligned}$$

$$\text{power input} = \frac{\left(\frac{103,200\text{kCal}}{\text{hr}}\right)}{\frac{5,310\text{kCal}}{\text{kg}}} = \frac{32.40\text{kg}}{\text{hr}}$$

$$W_{\text{wood}} = \left(\frac{32.40\text{kg}}{\text{hr}}\right) (8\text{hrs}) = 260\text{kg}$$

317. 500-hectare coconut plantation is designed to supply coconut oil for a biodiesel plant. There are 100 trees of coconut per hectare. How many liters of biodiesel can be produced by the plant considering a 90% conversion efficiency? Each tree has a capacity to produce an average of 12 nuts per month and 1 liter of coconut oil per 10 nuts.

Given:

Area of plantation - 500 has.

Conversion - 90%

Production per tree - 12 nuts per mo.

Volume of oil - 1 li per 10 nuts

100 trees per hectare

Required:

Volume of Biodiesel Production per month

Solution:

$$\text{No. of trees} = (500\text{ha}) \left(\frac{100\text{trees}}{\text{ha}} \right) = 50,000\text{trees}$$

$$\text{Volume of oil} = (50,000\text{trees}) \left(12 \frac{\text{nuts}}{\text{month}} \right) \left(\frac{1\text{lit}}{10\text{nuts}} \right) = \frac{60,000\text{lit}}{\text{month}}$$

$$\text{Volume of biodiesel} = \left(\frac{60,000\text{lit}}{\text{month}} \right) (0.90) = 54,000\text{liters}$$

318. A 1,000-liter-capacity biodiesel plant processes coconut oil. How much potassium hydroxide and methanol will be needed to produce biodiesel?

Given:

Input capacity of plant - 1,000 li

Required:

Amount of Potassium Hydroxide Needed

Volume of Methanol Needed

Solution:

$$\text{Amount of Potassium Hydroxide} = (1000\text{lit}) \left(\frac{1.1\text{kg of KOH}}{100\text{lit}} \right) = 11\text{kg of KOH}$$

$$\text{Volume of Methanol} = (1,000\text{lit}) \left(\frac{20\text{lit Methanol}}{100\text{lit oil}} \right) = 200\text{lit}$$

319. A 10,000-liter-per-day-capacity bioethanol plant uses fermented sap from sweet sorghum with 15% alcohol content. If the plant has 90% distillation efficiency in producing anhydrous bioethanol, what is the volume of ethanol that can be derived from the plant in one day? Also, compute the volume of stillage that can be derived.

Given:

Capacity - 10,000 li per day

%Ac of feed - 15%

Distillation Efficiency- 90% - 313 ml

Required:

Volume of Bioethanol Produced

Volume of Stillage Derived

Solution:

$$\text{Volume of Bioethanol} = \left(\frac{10,000\text{lit}}{\text{day}} \right) (0.9)(0.15) = \frac{1350\text{lit}}{\text{day}}$$

$$\text{Volume of Stillage} = \frac{10,000\text{lit}}{\text{day}} - \frac{1350\text{lit}}{\text{day}} = \frac{8650\text{lit}}{\text{day}}$$

320. bioethanol distiller is filled with 200 liters of fermented sap having 12% alcohol. What is the expected volume of alcohol that can be obtained if the distiller has 80% distillation efficiency?

Given:

Capacity of distiller - 200 li

% Ac of sap - 12%

% efficiency of distiller - 80%

Required:

Expected volume of alcohol that can be derived from the distiller

Solution:

$$\text{Volume of Alcohol} = (200\text{li})(0.8)(0.12) = 19.2\text{liters}$$

321. 1000-ml sample of feedstock contains 28.1% alcohol. After distillation, the volume of distillate recovered was 313 ml with 66.9% alcohol content. What is the distillation efficiency?

Given:

V of sample - 1000 ml

%Ac of sample - 28.1%

V of distillate - 313 ml

% Ac of distillate - 66.9%

Required: Distillation Efficiency

Solution:

$$\text{Distillation Efficiency} = \frac{(313)(0.669)}{(1000)(0.281)} = 74.51\%$$

322. A rice husk gasifier with reactor diameter of 120 cm is designed to provide thermal power for a steam boiler. If the gasifier has a specific gasification rate of 150 kg/hr-m², what is the fuel consumption of the gasifier in number of sacks per hour, assuming that one sack weighs 10 kg? Also, compute the amount of gas to be produced by the gasifier in m³/hr. if the stoichiometric air requirement of rice husks is 4.7 kg air per kg of fuel, and the equivalence ratio is 0.32. What is the amount of heat energy coming out of the gasifier if the gas produced has 1200 kcal/ m³ energy content? What would be the power input and the power output of the gasifier in kW?

Given:

Reactor diameter - 120 cm

Specific gasification rate - 150 kg/hr.- m²

Weight of rice husk - 10 kg per sack

Stoichiometric air - 4.7 kg air/kg fuel

Equivalence ratio - 0.32

Gas heating value - 1,200 kcal/ m³

Required:

Fuel Consumption Rate, sacks per hr.

Gas Flow Rate, m³/hr.

Energy Output, kcal/hr.

Power Input and Output, kW

Solution:

$$\text{Fuel Consumption Rate} = \left(\frac{150\text{kg}}{\text{hr} - \text{m}^2} \right) \left(\frac{1\text{sack}}{10\text{kg}} \right) \left(\frac{\pi(1.2)^2}{4} \right) = \frac{16.96\text{sacks}}{\text{hr}}$$

$$\text{Gas Flow Rate} = \frac{\left(\frac{169.65\text{kg}}{\text{hr}} \right) \left(\frac{4.7\text{kg air}}{\text{kg fuel}} \right) (0.32)}{\frac{1.25\text{kg}}{\text{m}^3}} = \frac{204.12\text{m}^3}{\text{hr}}$$

$$\text{Energy Output} = \left(\frac{204.12\text{m}^3}{\text{hr}} \right) \left(\frac{1200\text{kCal}}{\text{m}^3} \right) = \frac{244,944\text{kCal}}{\text{hr}}$$

$$\text{Power Input} = \left(\frac{169.65\text{kg}}{\text{hr}} \right) \left(\frac{3,000\text{kCal}}{\text{kg}} \right) \left(\frac{0.0012\text{kW}}{\frac{\text{kCal}}{\text{hr}}} \right) = 610.74\text{kW}$$

$$\text{Power output} = \left(\frac{244,944\text{kCal}}{\text{hr}} \right) \left(\frac{0.0012\text{kW}}{\frac{\text{kCal}}{\text{hr}}} \right) = 294\text{kW}$$

323. A biomass furnace is designed to supply heat for drying 6 tons of shelled corn from 20% to 13% in 14 hours. If the drying efficiency of the flatbed dryer is 40% and the furnace efficiency is 55%, what would be the amount of corn cobs needed to be burned in the furnace? What is the amount of combustion air, in m³/hr., needed for the fuel and the secondary air to be supplied, in m³/hr., if 20% excess air is needed in the furnace? Assume a heat of vaporization of moisture in shelled corn is 1,120 kcal/kg of water evaporated and a heating value for fuel of 4,200 kcal/kg. Consider a stoichiometric air of 5.6 kg air/kg of fuel.

Given:

- Dryer capacity - 6 tons
- Moisture reduction - 20% to 13%
- Drying time - 14hrs
- Dryer efficiency - 40%
- Furnace efficiency - 55%
- Fuel - corn cob w/ 4,220 kcal/kg
- Heat of vaporization - 1,120 kcal/kg water
- Stoichiometric air - 5.6 kg air/kg fuel
- Excess air - 20%

Required:

- Fuel Consumption Rate
- Combustion Air Requirement
- Secondary Air Requirement

Solution:

$$W_{\text{moisture}} = 6\text{tons} - \frac{6\text{ton}(1 - 0.20)}{1 - 0.14} = 418.6\text{kg}$$

324. A four – cylinder petrol engine has a bore of 57mm and a stroke of 90mm. Its rated speed is 2800 rpm and it is tested at this speed against a brake which has a torque arm of 0.356m. The net brake load is 155N and the fuel consumption is 6.74 l/h. The specific gravity of the petrol used is 0.735 and it has a lower calorific value of; 44200 kJ/kg. The indicated load for the engine as calculated by the Morse test method is 187.3N. Calculate for this speed, the engine torque, the brake mean effective pressure, the brake thermal efficiency, the specific consumption, the mechanical efficiency and the indicated mean effective pressure.

Give:

No. of Cylinder-4
 Bore-57mm
 Stroke-90mm
 Rated speed-2800rpm
 Brake arm-0.356m
 Brake load-155N
 FCR-6.74li/h
 SG-0.735
 HV-44200kJ/kg
 Indicated load-187.3N

Solution:

$$T = (155\text{N})(0.356\text{m}) = 55.18\text{N} - \text{m}$$

$$\text{Brake power} = 2\pi TN = 2\pi(55.18\text{N} - \text{m}) \left(\frac{2800}{60\text{sec}} \right) = 16.20\text{kW}$$

$$\text{BMEP} = \frac{\text{IHP}(2)}{\text{LANn}} = \frac{(16.20\text{kW})(2)}{(0.09\text{m}) \left(\frac{\pi(0.057)^2}{4} \right) \left(\frac{2800}{60} \right) (4)} = \frac{755,781\text{N}}{\text{m}^2} \text{ or } 7.55\text{bar}$$

$$\text{BTE} = \frac{16.20\text{kW}}{\left(\frac{44200\text{kJ}}{\text{kg}} \right) \left(\frac{6.74\text{lit}}{\text{hr}} \right) \left(\frac{0.735\text{kg}}{1\text{lit}} \right)} = 26.63\%$$

$$\text{mass of fuel rate} = \left(\frac{6.74\text{lit}}{\text{hr}} \right) \left(\frac{0.735\text{kg}}{\text{lit}} \right) \left(\frac{1\text{hr}}{3600\text{s}} \right) = \frac{0.001376\text{kg}}{\text{s}}$$

$$\text{SFC} = \frac{m_f}{\text{brake power}} = \frac{\frac{0.001376\text{kg}}{\text{s}}}{16.20\text{kW}} = \frac{8.50 \times 10^{-5}\text{kg}}{\text{kW} - \text{s}} \text{ or } \frac{0.306\text{kg}}{\text{kW} - \text{hr}}$$

$$\text{ME} = \frac{\text{brake power}}{\text{indicated power}} \text{ or } \frac{\text{brake load}}{\text{indicated load}} = \frac{155\text{N}}{187.3\text{N}} = 82.75\%$$

$$\text{Indicated power} = \frac{16.20\text{kW}}{0.8275} = 19.58\text{kW}$$

$$\text{IMEP} = \frac{\text{BMEP}}{\text{ME}} = \frac{7.55\text{bar}}{0.8275} = 9.12\text{bar}$$

325. A four cylinder car engine has a bore and stroke = 79 mm and 77 mm respectively. What is the capacity of the engine in cc?

Solution:

$$\text{PD} = \frac{\pi D^2 L}{4} = \frac{\pi(7.9\text{cm})^2(7.7\text{cm})(4)}{4} = 1509.71\text{cc}$$

326. If the same engine (Ex-2) (i.e., four-stroke, 2 liters) as above produces 76 kW at 5400 rpm, find its brake mean effective pressure.

Solution:

$$P = T\omega; T = \frac{P}{\omega} = \frac{\left(\frac{76,000\text{N} - \text{m}}{\text{s}} \right)}{\frac{2\pi(5400\text{rpm})}{60}} = 134.40\text{N} - \text{m}; \omega = \frac{565\text{rad}}{\text{s}}$$

$$T = \frac{P}{2\pi N} = \frac{\left(\frac{76,000\text{N} - \text{m}}{\text{s}} \right)}{\frac{2\pi(5400\text{rpm})}{60}} = 134.40\text{N} - \text{m}$$

$$\text{BMEP} = \frac{(\text{IHP})(2)}{LANn} \text{ or } \frac{2\pi T n_{\text{stroke}}}{V} = \frac{2\pi(134.40\text{N} - \text{m})(2)}{(2\text{lit})\left(\frac{0.001\text{m}^3}{1\text{lit}}\right)} = \frac{844,460\text{N}}{\text{m}^2} \text{ or } 8.44\text{bar}$$

327. A 4-Cylinder, 2-stroke IC engine has the following particulars: engine speed = 3000 rpm, bore = 120 mm, crank radius = 60 mm, mechanical efficiency = 90% and the engine develops 75 bhp. Calculate the swept volume and mean effective pressure (MEP).

Give:

N-3000rpm
No. Cylinder-4, 2-stroke
Bore-120mm
Crank radius-60mm
ME-90%
BP-75bhp

Required:

MEP

Solution:

$$\text{IHP} = \frac{75\text{hp}}{0.90} = 83.33\text{hp}$$

$$T = \frac{P}{2\pi N\alpha} = \frac{(83.33)(0.746\text{kW})(1000)}{\frac{2\pi(3000)}{60}} = 197.87\text{N} - \text{m}$$

$$\text{BMEP} = \frac{(\text{IHP})(2)}{LANn} \text{ or } \frac{2\pi T n_{\text{stroke}}}{V} = \frac{2\pi(197.87\text{N} - \text{m})(1)}{\frac{\pi(0.12\text{m})^2(2 \times 0.06\text{m})(4)}{4}} = \frac{229,016\text{N}}{\text{m}^2} \text{ or } 2.29\text{bar}$$

328. The peak pressure of a SI engine rotating at 1500 rpm occurs 0.003s after the spark, what will be the spark timing when peak pressure is at TDC. If the inlet valve opens at 10 degrees before TDC and closes at 45 degrees after BDC, how long the inlet valve opening period is in seconds.

Solution:

$$N = \frac{1500\text{rpm}}{60\text{s}} \text{ or } \frac{25\text{rev}}{\text{sec}}$$

$$\text{No. of revolutions between spark timing and TDC} = \left(\frac{25\text{rev}}{\text{sec}}\right)(0.003\text{sec}) = 0.075\text{rev}$$

$$\text{crank angle in this period of time} = (0.075\text{rev})(360) = 27^\circ$$

$$\text{Inlet valve opening angle} = 10 + 180 + 45 = 235^\circ$$

$$\text{Inlet valve opening time in seconds} = \frac{235}{(360)(25)} = 0.026\text{sec}$$

329. In a four stroke single cylinder gas engine the indicated mean effective pressure is 0.46 MN/m², the brake power 9 kW, speed 250 rpm, mechanical efficiency, $\eta_m = 0.8$, and bore to stroke ratio = 0.66. Calculate cylinder diameter and mean piston speed.

Give:

IMEP-0.46MN/m²
BP-9kW
N-250rpm
ME-0.8
Bore to stroke ration-0.66

Required:

Cylinder diameter

Mean piston Speed

Solution:

$$\text{IHP} = \frac{PLANn}{2}$$

$$LA = \frac{2(\text{IHP})}{PNn} = \frac{2\left(\frac{9\text{kW}}{0.8}\right)}{\left(\frac{460\text{kN}}{\text{m}^2}\right)\left(\frac{250}{60}\right)(1)} = 0.01174\text{m}^3$$

$$\frac{D}{L} = 0.66; L = \frac{D}{0.66}$$

$$\left(\frac{D}{0.66}\right)\left(\frac{\pi D^2}{4}\right) = 0.01174\text{m}^3; D = 0.214\text{m}; L = 0.325\text{m}$$

$$\text{Mean Piston Speed} = \pi DN = \frac{\pi(0.214)(250)}{60} = \frac{2.8\text{m}}{\text{s}}$$

330. A four stroke petrol engine delivers 35.75kW with a mechanical efficiency of 80%, the fuel consumption of the engine is 0.4 kg per brake power hour, and the A/F ratio is 14:1. The heating value of the fuel is 41870 kJ/kg. Find: (a) indicated power (b) friction power (c) η_{BT} , (d) η_{IT} , (e) fuel consumption per hour (f) air consumption per hour.

Give:

BP-35.75kW

ME-80%

FC-0.4kg/hr.

Air fuel ration-14:1

HV-41870kJ/kg

Solution:

$$\text{IP} = \frac{\text{BP}}{\text{ME}} = \frac{35.75\text{kW}}{0.80} = 44.70\text{kW}$$

$$\text{Friction Power} = \text{IP} - \text{BP} = 44.70\text{kW} - 35.75\text{kW} = 8.95\text{kW}$$

$$\text{BTH} = \frac{3600(35.75\text{kW})}{\left(\frac{0.4\text{kg}}{\text{hr}}\right)(35.75\text{kW})\left(\frac{41,870\text{kJ}}{\text{kg}}\right)} = 21.5\%$$

$$\text{ITH} = \frac{\text{BTH}}{\text{ME}} = \frac{0.215}{0.8} = 26.87\%$$

$$\text{FCR} = \left(\frac{0.4\text{kg}}{\text{hr}}\right)(35.75) = \frac{14.3\text{kg}}{\text{hr}}$$

$$\text{Air consumption per hour} = (14)(14.3\text{kg}) = 200.5\text{kg}$$

331. The air flow to a four cylinder four – stroke engine is 2.15 m³/min. During a test on the engine the following data were recorded: Bore 10.5cm; stroke 12.5cm; engine speed 1200 rpm, torque 150 N-m, fuel consumption 5.5 kg/h, calorific value of fuel, 43124 kJ/kg, ambient temperature and pressure are 20 degree C and 1.03 bars. Calculate:

1- The brake thermal efficiency.

2- The brakes mean effective pressure.

3- The volumetric efficiency.

Give:

No. of cylinder-4

No. of stoke-4
 Air flow-2.15m³/min
 Bore-10.5cm
 Stroke-12.5cm
 N-1200rpm
 Torque-150N-m
 FC-5.5kg/hr.
 HV-43124kJ/kg
 Ambient temp-20 degrees
 Pressure-103bar

Solution:

$$P = 2\pi TN = \frac{2\pi(150\text{N} - \text{m})(1200\text{rpm})}{60} = 18.849\text{kW}$$

$$\text{BTE} = \frac{75.40\text{kW}}{\left(\frac{5.5\text{kg}}{\text{hr}}\right)\left(\frac{43124\text{kJ}}{\text{kg}}\right)} = \frac{18.849\text{kW}}{65.88\text{kW}} = 28.61\%$$

$$\text{BMEP} = \frac{\text{IHP}(2)}{\text{LANn}} = \frac{18.849\text{kW}(2)}{(0.125\text{m})\left(\frac{\pi(0.105\text{m})^2}{4}\right)\left(\frac{1200}{60}\right)(4)} = 4.35\text{bar or } 435.36\text{kPa}$$

$$\text{Volumetric Speed} = \frac{\pi D^2 L N}{4} = \frac{\pi(0.105\text{m})^2(0.125\text{m})(4)(1200)}{4(2)} = \frac{2.6\text{m}^3}{\text{min}}$$

$$\text{Volumetric Efficiency} = \frac{\frac{2.15\text{m}^3}{\text{min}}}{\frac{2.6\text{m}^3}{\text{min}}} = 82.70\%$$

332. 3liters six – cylinders SI engine operates on a four – stroke cycle and run at 3600 rpm. The compression ratio is 9.5 the length of connecting rod is 16.6cm, and the bore equal the stroke. Combustion ends at 20° after TDC calculate: (1) Cylinder bore and stroke, (2) average piston speed, (3) clearance volume of one cylinder

Give:

Capacity-3lit
 No. of cylinder-6
 No. of stoke-4
 N-3600rpm
 Compression ratio-9.5
 Length of rod-16.6cm
 Bore=stroke
 Combustion ends at 20° after TDC

Solution:

$$PD = \frac{\pi D^2 L}{4} = \frac{3\text{liters}\left(\frac{1000\text{cm}^3}{1\text{lit}}\right)}{6 \text{ cylinders}} = 500\text{cm}^3$$

$$\frac{\pi D^2 D}{4} = 500\text{cm}^3; D = 8.6\text{cm} = L$$

$$V = 2LN = 2(0.086\text{m})\left(\frac{3600}{60}\right) = \frac{10.32\text{m}}{\text{s}}$$

$$CR = \frac{PD + CV}{CV} : CV = \frac{PD}{CR - 1} = \frac{500\text{m}^3}{9.5 - 1} = 59\text{cm}^3$$

333. The engine in problem #332 is connected to a dynamometer which gives a brake output torque of 205 Nm at 3600 rpm. At this speed air enters the cylinder at 85kPa and 60°C, and the mechanical efficiency of the engine is 85%. Calculate: (1) brake power, (2) indicated power (3) BMEP, (4) IMEP, (5) FMEP, (6) FP, (7) engine specific volume.

Give:

Torque-205N-m

N-3600rpm

Pressure-85kPa

T-60°C

ME-85%

PD or LA=500m³

Solution:

$$BP = 2\pi TN = 2\pi(205\text{N} - \text{m})\left(\frac{3600\text{rpm}}{60}\right) = 77.28\text{kW}$$

$$IHP = \frac{BP}{ME} = \frac{77.28\text{kW}}{0.85} = 90.92\text{kW}$$

$$BMEP = \frac{BP(2)}{LANn} = \frac{(77.28\text{kW})(2)}{(0.0005\text{m}^3)\left(\frac{3600}{60}\right)(6)} = 858.67\text{kPa}$$

$$IMEP = \frac{BMEP}{ME} = \frac{858.67\text{kPa}}{0.85} = 1010.20\text{kPa}$$

$$FMEP = 1010.20\text{kPa} - 858.67\text{kPa} = 151.53\text{kPa}$$

$$\text{Friction Power} = FMEPLANn = \frac{(151.53\text{kPa})((0.0005\text{m}^3)\left(\frac{3600}{60}\right)(6))}{2} = 13.64\text{kW}$$

$$\text{Engine Specific Volume} = \frac{\text{Swept volume}}{BP} = \frac{3\text{lit}}{77.28\text{kW}} = \frac{0.0388\text{Lit}}{\text{Kw}}$$

334. The engine in problem #333 is running with A/F ratio =15, a fuel of heating value; 44000kJ/kg and a combustion efficiency of 97% calculate: (1) the rate of fuel flow. (2) η_{BT} , (3) η_{IT} , (4) and brake specific consumption.

Solution:

$$CV = 59\text{cm}^3$$

$$\text{Fuel Mass } (m_f) = \frac{P(PD + CV)}{RT} = \frac{(85\text{kPa})(0.0005\text{cm}^3 + 0.000059\text{cm}^3)}{\left(\frac{0.287\text{kJ}}{\text{kgK}}\right)(333\text{K})} = 0.0005\text{kg}$$

$$\text{Fuel Mass per cycle} = \frac{0.0005\text{kg}}{15} = 0.00033\text{kg of fuel per cylincer per cycle}$$

$$\text{Mass fuel flow} = \left(\frac{0.00033\text{kg}}{\text{cycle} - \text{cylinder}}\right)(6\text{cylinders})\left(\frac{3600\text{rev}}{60\text{s}}\right) = \frac{0.006\text{kg}}{\text{s}}$$

$$BTE = \frac{BP}{(m_f)HV.CE} = \frac{77.28\text{kW}}{\left(\frac{0.006\text{kg}}{\text{s}}\right)\left(\frac{44,000\text{kJ}}{\text{kg}}\right)(0.97)} = 30.18\%$$

$$\text{ITE} = \frac{\text{BTE}}{\text{ME}} = \frac{30.18}{0.85} = 35.50\%$$

$$\text{BSFC} = \frac{m_f}{\text{BP}} = \frac{\frac{0.006\text{kg}}{\text{s}}}{77.28\text{kW}} = \frac{0.28\text{kg}}{\text{kW} - \text{sec}}$$

335. A six-cylinder 4-stroke cycle petrol engine is to be designed to develop 300 kW of (BP) at 2500 rpm the bore / stroke ratio is to be 1:1.25. Assuming ME=83% and an indicated mean effective pressure of 9.5 bar, determine the required bore and stroke. If the compression ratio of the engine is to be 6.5 to 1, determine consumption of petrol in kg/h and in kg/bp.hr. Take the ratio of the indicated thermal efficiency of the engine to that of the constant volume air standard cycle as 0.55 and the calorific value of the petrol as; 44770kJ/kg

Give:

No. of cylinder-6
 No. of stroke-4
 Brake power-300kW
 N-2500rpm
 Bore-stroke ratio-1:1.25
 ME-83%
 IMEP-9.5bar
 HV-44,770kJ/kg
 CR-6.5:1

Required:

Bore and stroke

Solution:

$$\text{ME} = \frac{\text{BP}}{\text{IP}}; \text{IP} = \frac{\text{BP}}{\text{ME}} = \frac{300\text{kW}}{0.83} = 361.44\text{kW}$$

$$\text{IP} = \frac{\text{PLANn}}{2}$$

$$361.44\text{kW} = \frac{\left(\frac{950,000\text{N}}{\text{m}^2}\right)(\text{LA})(2500\text{rpm})(6)}{2}; \text{LA} = 0.00304\text{m}^3$$

$$\frac{B}{L} = \frac{1}{1.25}; L = 1.25B$$

$$0.00304\text{m}^3 = \frac{\pi D^2 L}{4} = \frac{1.25\pi D^3}{4}; D = 0.146\text{m}; L = 1.25(0.146\text{m}) = 0.1825\text{m}$$

336. A person pays interest on a loan semiannually at a nominal interest rate of 16%. What is most nearly the effective annual interest rate?

Solution:

$$i = \left(1 + \frac{0.16}{2}\right)^2 - 1 = 0.1664 \text{ or } 16.64\%$$

337. A bank is advertising 9.5% accounts that yield 9.84% annually. How often is the interest compounded?

Solution:

$$0.0984 = \left(1 + \frac{9.5}{m}\right)^m - 1$$

$$\text{trial 1: } m = 2; \left(1 + \frac{0.095}{2}\right)^2 - 1 = 0.0972 \text{ or } 9.72\%$$

$$\text{trial 2: } m = 3; \left(1 + \frac{0.095}{3}\right)^3 - 1 = 0.0980 \text{ or } 9.80\%$$

$$\text{trial 3: } m = 4; \left(1 + \frac{0.095}{4}\right)^4 - 1 = 0.0984 \text{ or } 9.84\%$$

Answer: $m = 4$ or Quarterly